Image Data Management and Stewardship

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[1] University of Dundee, [2] EMBL-EBI

Bristol, 20 May 2026

Building a national imaging data stewardship skills base

Building a national imaging data stewardship skills base

- Direct training of research technical professionals
- Develop a comprehensive training curriculum
- Train-the-trainer events
- Support role recognition and career pathways

Training and support

- Training events: combining direct training with train-the-trainer
  - Liverpool, Cambridge, London, Bristol, Glasgow
- Consultation sessions
- Online training materials
  - https://github.com/BioImagingUK/Training
- Join our new community slack channel

Agenda

- 09:00 Arrival
- 09:30 - 10:45 Morning session 1
  - Data Stewardship
  - FAIR data
  - Data Life Cycle and planning
- 10:45 - 11:00 Coffee break
- 11:00 - 12:30 Morning session 2
  - Metadata standardisation
  - Tools for Data Management
  - OMERO
- 12:30 - 13:10 Lunch
- 13:10 - 14:30 Afternoon session 1
  - Data management: successful examples and end user tips
  - Storage and sharing strategies and tools
  - Data standardisation: bioformats and OME-NGFF/OME-Zarr
  - Visualisation strategies
- 14:30 - 14:45 Coffee break
- 14:45 - 16:45 Afternoon session 2
  - Preserving raw and processed data, qc checks
  - What do public repositories offer: Success stories
  - Submitting data to public repositories
- 16:45 Moving forward and feedback
- 17:00 End

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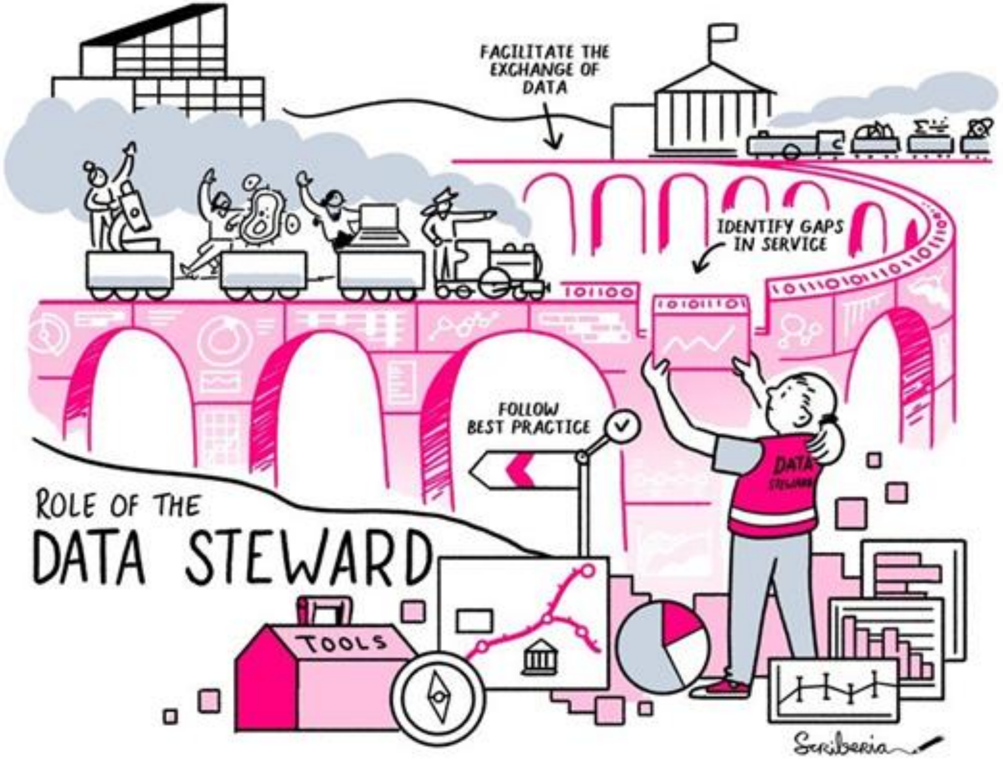
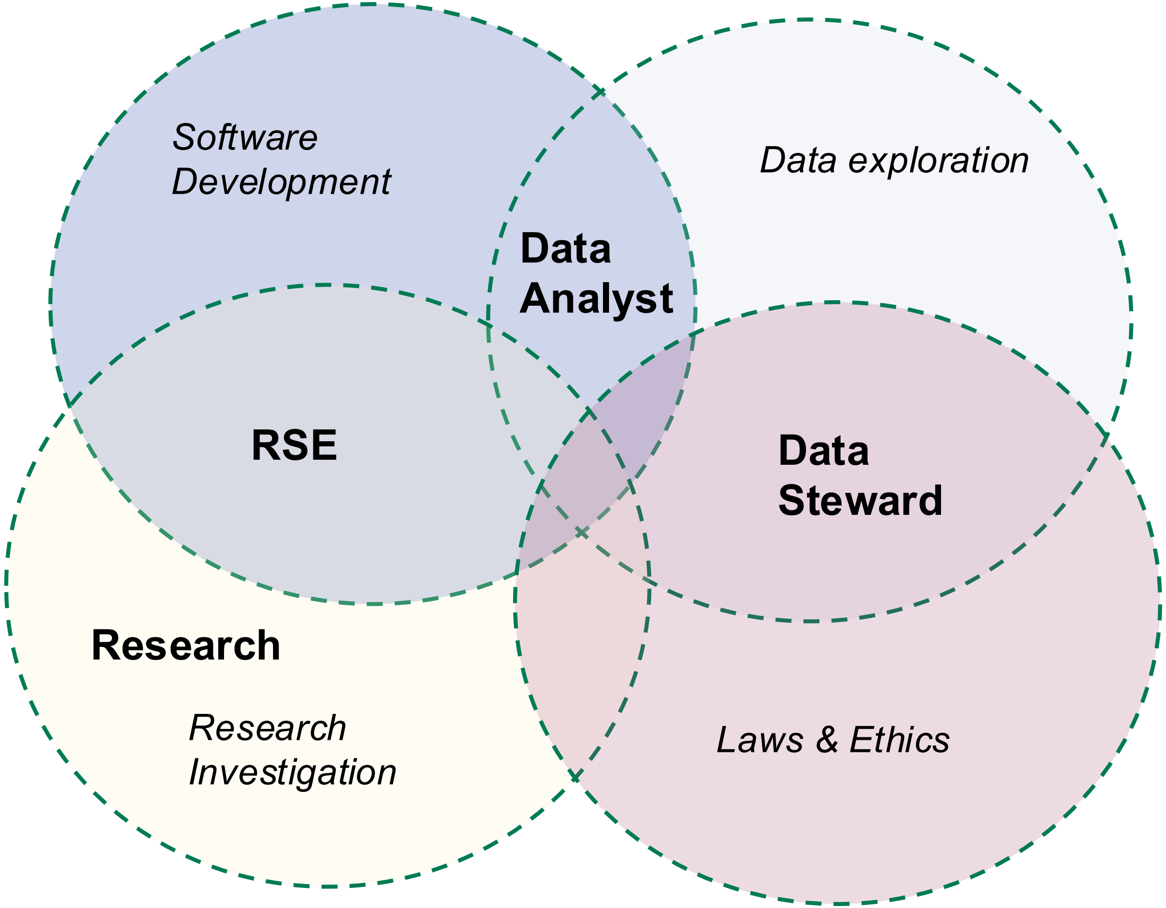
Data Steward... What does it entail?



- Complex role
- Interacts with many parts
- Tie together many aspects

Data Steward... Where does it fit?



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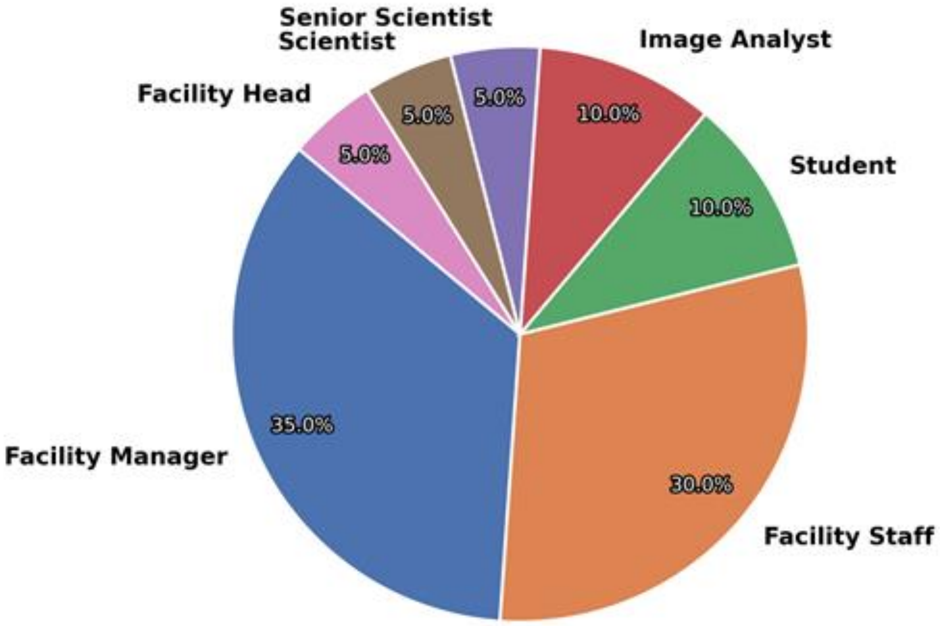
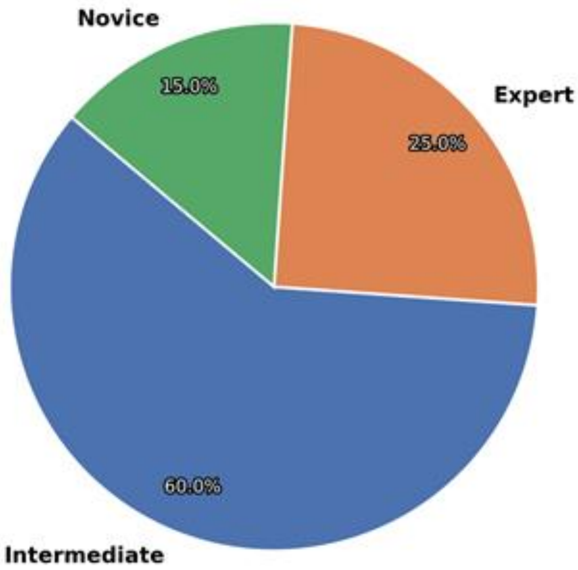
Participants Overview

Role



Level of Research Management



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Why do we need image data management?

- **Data is big and complex, or many!**
  - **Not possible to remember everything about the experiment**; sample preparation, imaging parameters, other variables (antibodies, genes, chemicals, dyes etc.).
  - **Not possible to remember where all the images are** for all of the experiments. Sometimes one person doesn't have all the images.
  - A single person's mind cannot be your data management system.

Exercise (3 minutes)

What do you need to know?

What do you need to know to be able to do your responsibilities?

menti.com

3770 5336

Why do we need image data management?

- **Facility managers:** Need to know where user's data are, what they are about (handling samples fit for facility/instrument use), if the instruments are up and running producing quality images, which instrument is allocated to which user/experiment when.
- **Researchers:** Need to know what the experiments are about and what they are looking for. Therefore need to know how the sample was prepared, how the experiment was conducted (acquisition parameters, conditions etc). If there are different sets they should compare (different acquisition parameters, different samples, different preparation methods, different variables such as genes, drugs, time etc).

  They don't always work on a single experiment at a time; you may be writing the paper of an experiment you did a year ago, while collecting data of another experiment.
- **Image Analysts:** Need to know what the experiment is about and what they are looking for. Need to know the image size, what they're running and whether the analysis is working as expected.

Why do we need image data management?

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  - **Not possible to remember where all the images are** for all of the experiments. Sometimes one person doesn't have all the images.
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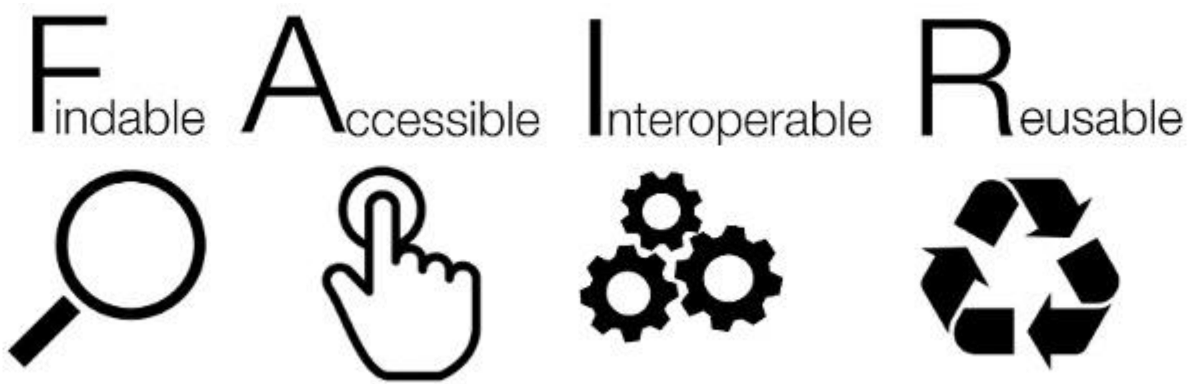
- **Science is collaborative and therefore different people will need to access and understand the data.**

- **Using software/code or AI for automating the processing or analysis requires standardised metadata and data.**



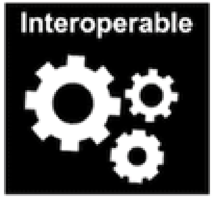
The FAIR Guiding Principles for scientific data management and stewardship

https://doi.org/10.1038/sdata.2016.18

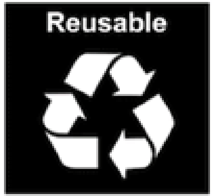
FAIR: “What does it mean for me?”

Where is my data? Can I search and find it?

Can I access it? Can I download it? *Is it on a disk or server I no longer have access? Is it password protected and I forgot the password?*



Can I read it/open and view it? *Or is the format no longer supported?* Are the data and metadata machine readable?



Can my collaborators analyse it if I give the data file to them? What other information do they need? *Where is that information?* Can they reproduce the images if they do the experiment?

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FAIR data vs Open data vs FAIR open data

- FAIR data: Data is FAIR but not open - not shared publicly, or “available upon request”!

- Open data:
    - Find my data on google drive or Cloud
    - Here’s my data as zip file on Zenodo (doi)

- FAIR open data: Here’s my data at BioImage Archive, EMPIAR or IDR (accession id, doi)

Where can I find Images?

FAIR Bioimage Repositories

BioImage Archive

BioImage Archive is a free, publicly available resource for life sciences images from any imaging modality, at any scale. It supports deposition, search, access and reuse of data that are either associated with a peer-reviewed publication, or of value beyond a single experiment.

EMPIAR
Electron Microscopy Public Image Archive

EMPIAR (Electron Microscopy Public Image Archive), is a public resource for raw images underpinning 3D cryo-EM maps and tomograms, 3D images obtained with volume EM techniques, and soft and hard X-ray tomography imaging.

IDR

IDR (Image Data Resource) is a public repository of reference image datasets from published scientific studies, enabling search, access and reuse of these highly annotated datasets.

What about analytical workflows?

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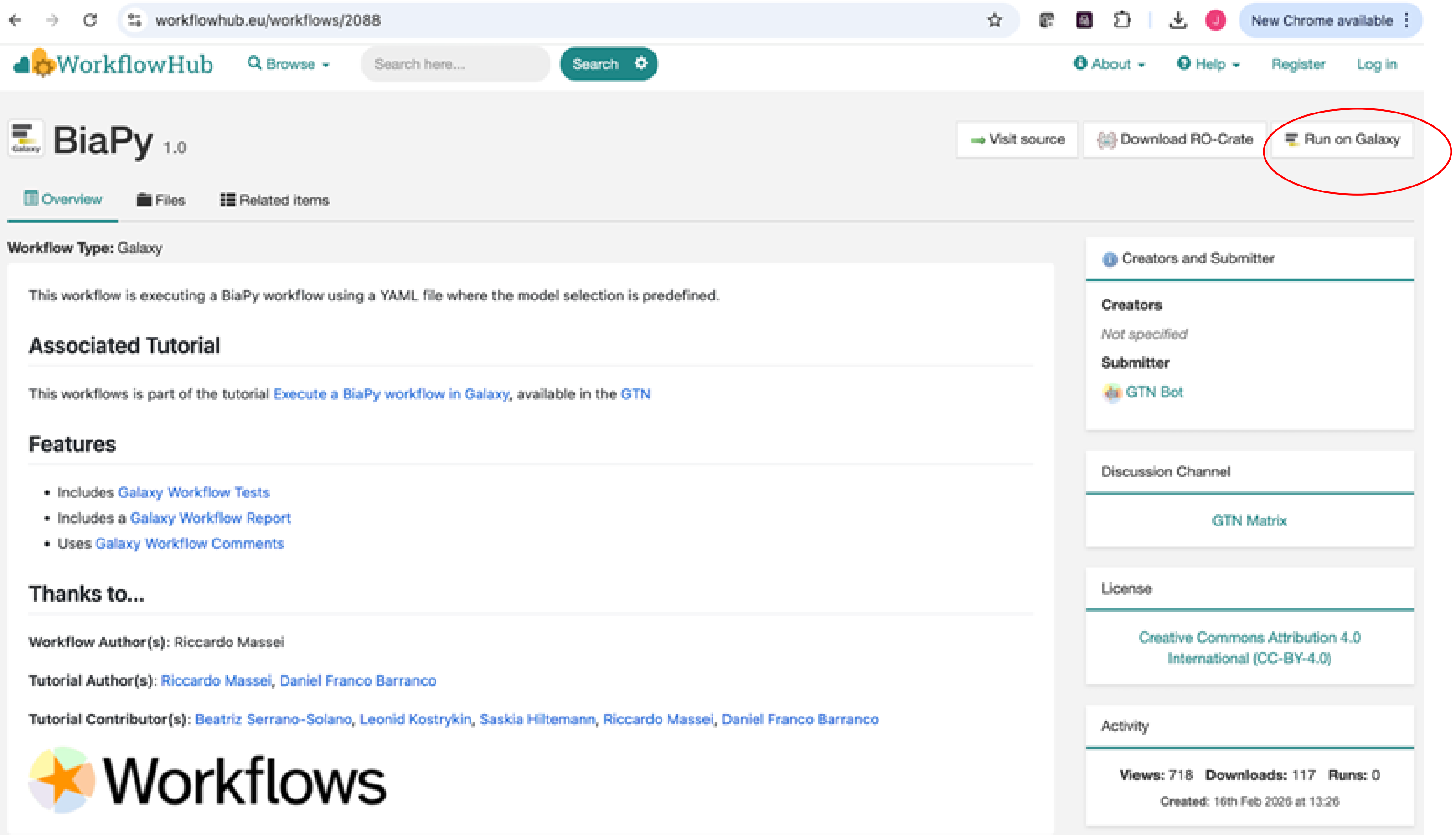
WorkflowHub

https://workflowhub.eu/

A registry for workflows

- Standardised
- Searchable
- Citable
- Interoperable
- Rich metadata



https://workflowhub.eu/workflows/2088

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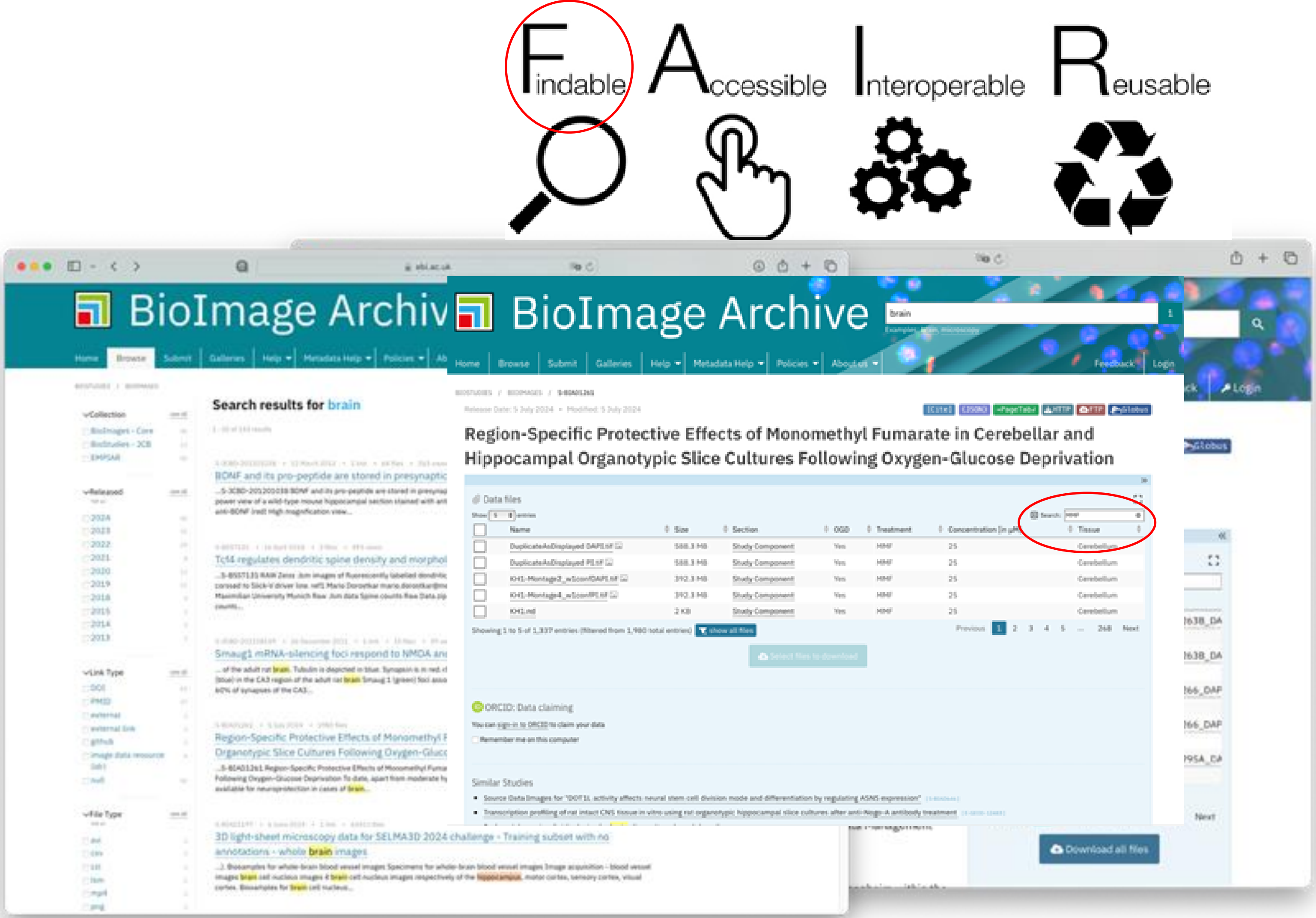
Findable
Accessible
Interoperable
Reusable
BioImage Archive
BioImage Archive
Search results for brain
Region-Specific Protective Effects of Monomethyl Fumarate in Cerebellar and Hippocampal Organotypic Slice Cultures Following Oxygen-Glucose Deprivation
Data files
ORCID: Data claiming
Similar Studies
Download all files

ebi.ac.uk/empiar/EMPIAR-11313/

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EMPIAR-11313

Notice:
Due to changes in Aspera, please use user emp_ext2 instead of emp_ext3 for any command line (ascp) downloads

Principles of mitoribosomal small subunit assembly in eukaryotes

Entry authors: Harper NJ, Burnside C, Klinge S

Publication: Principles of mitoribosomal small subunit assembly in eukaryotes
Harper NJ, Burnside C, Klinge S
Nature **614** 175-181 (2022)
PMID: 36482135
DOI: 10.1038/s41586-022-05621-0

Experiment type: EMDB

Related PDB entries: 8d8k, 8d8j, 8csr, 8css, 8d8l, 8cst, 8csp, 8csu, 8csq

Related EMDB entries: EMD-27250, EMD-27249, EMD-26968, EMD-26969, EMD-27251, EMD-26970, EMD-26966, EMD-26971, EMD-26967

Deposited: 2022-10-21

Released: 2023-02-17

Last modified: 2023-02-17

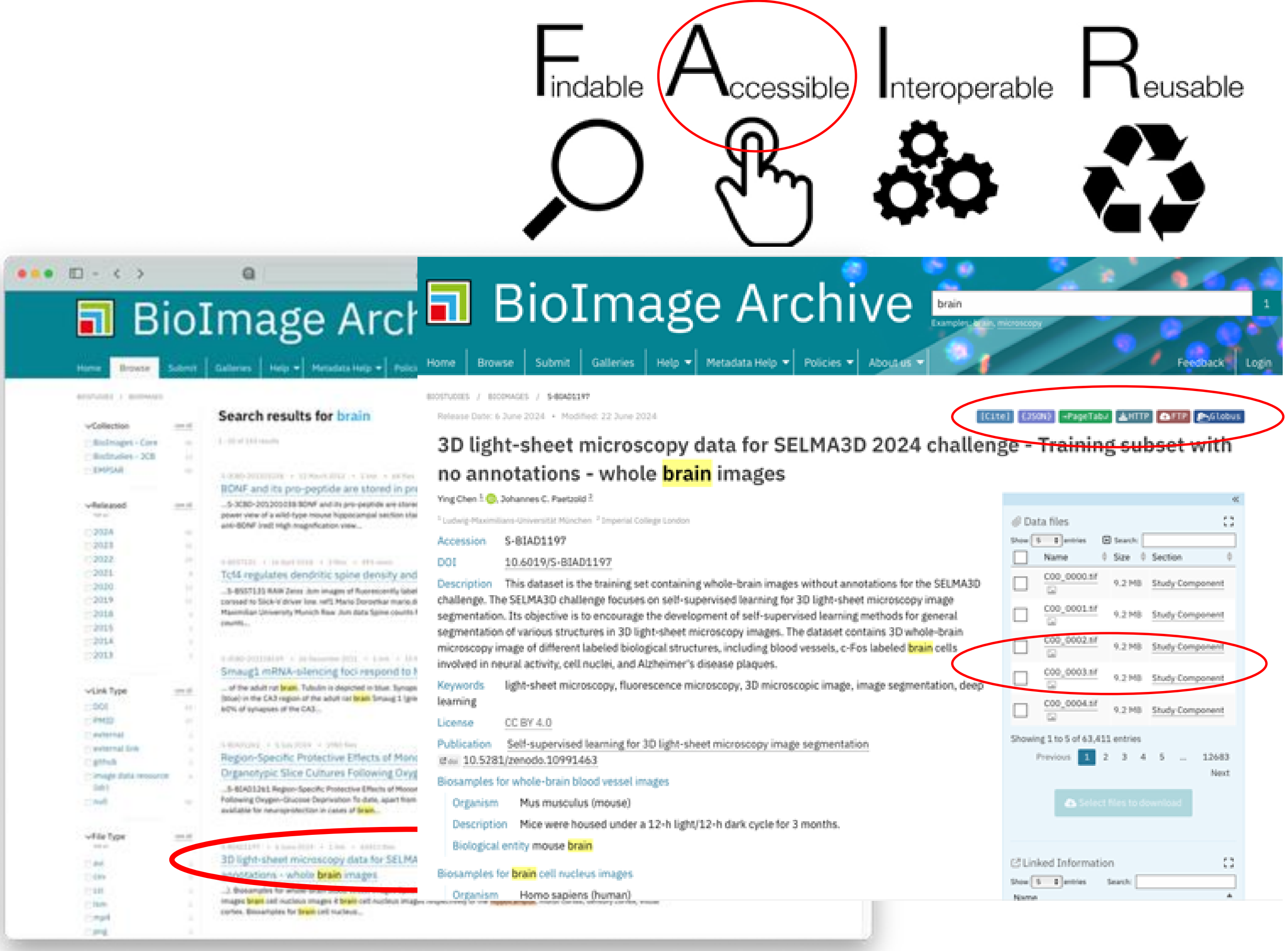
Dataset size: 73.6 TB

Dataset DOI: 10.6019/EMPIAR-11313

Experimental metadata: Download xml

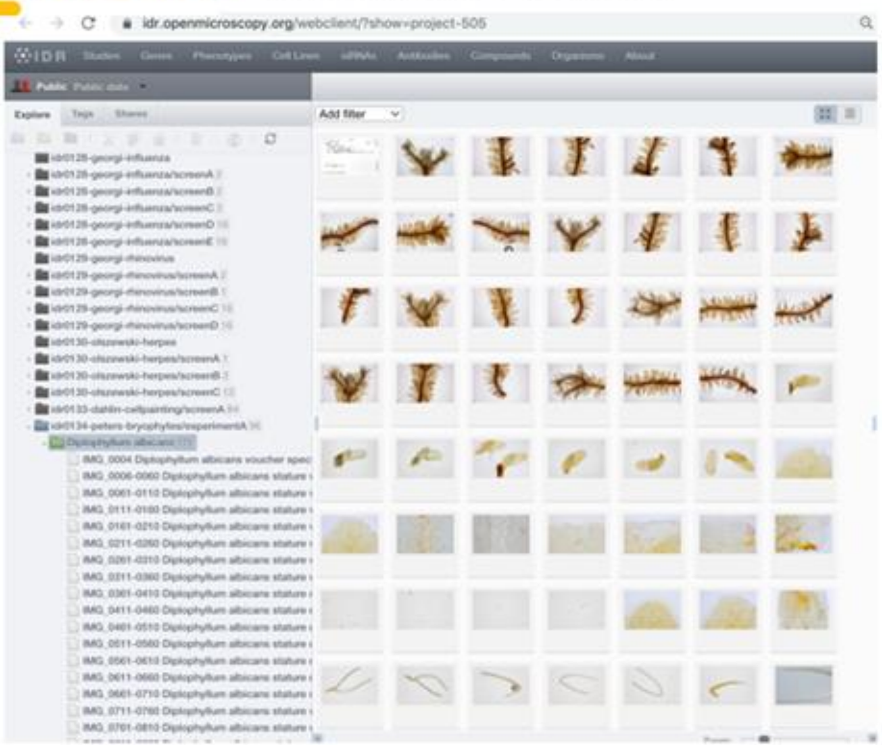
Contains:

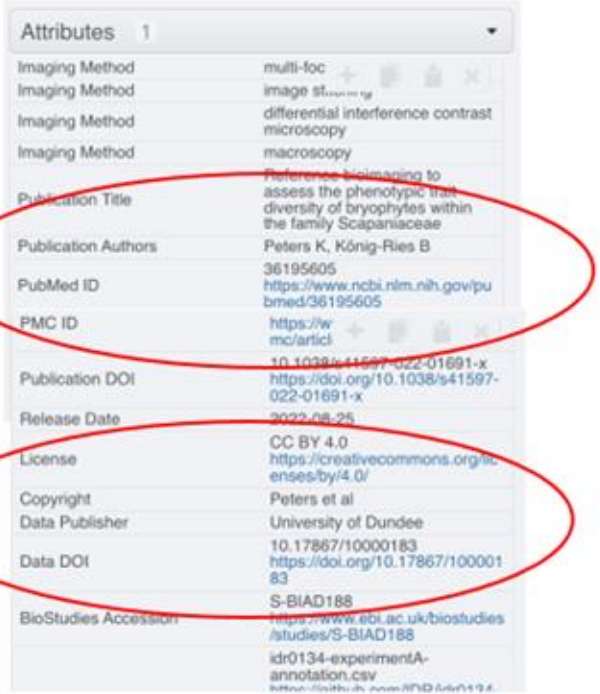
micrographs



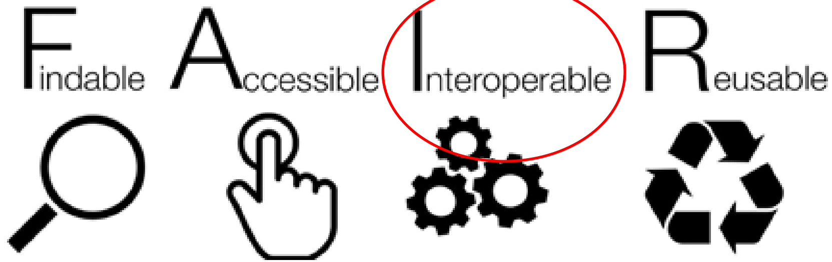
Findable
Accessible
Interoperable
Reusable
BioImage Archive
brain
Home
Browse
Submit
Galleries
Help
Metadata Help
Policies
About us
Feedback
Login
Search results for brain
Cite
JSON
PageTab
HTTP
FTP
Globus
3D light-sheet microscopy data for SELMA3D 2024 challenge - Training subset with no annotations - whole brain images
Ying Chen, Johannes C. Paetzold
Accession S-BIAD1197
DOI 10.6019/S-BIAD1197
Description This dataset is the training set containing whole-brain images without annotations for the SELMA3D challenge. The SELMA3D challenge focuses on self-supervised learning for 3D light-sheet microscopy image segmentation. Its objective is to encourage the development of self-supervised learning methods for general segmentation of various structures in 3D light-sheet microscopy images. The dataset contains 3D whole-brain microscopy image of different labeled biological structures, including blood vessels, c-Fos labeled brain cells involved in neural activity, cell nuclei, and Alzheimer's disease plaques.
Keywords light-sheet microscopy, fluorescence microscopy, 3D microscopic image, image segmentation, deep learning
License CC BY 4.0
Publication Self-supervised learning for 3D light-sheet microscopy image segmentation
10.5281/zenodo.10991463
Biosamples for whole-brain blood vessel images
Organism Mus musculus (mouse)
Description Mice were housed under a 12-h light/12-h dark cycle for 3 months.
Biological entity mouse brain
Biosamples for brain cell nucleus images
Organism Homo sapiens (human)
Data files
Name
Size
Section
C00_0000.tif
C00_0001.tif
C00_0002.tif
C00_0003.tif
C00_0004.tif
9.2 MB
Study Component
Showing 1 to 5 of 63,411 entries
Previous
Next
Linked Information

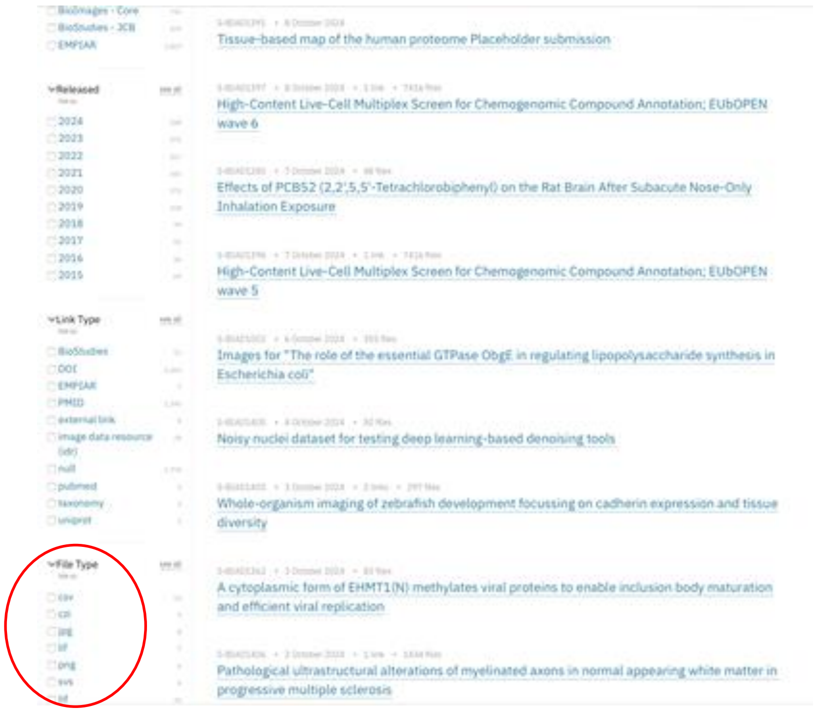
IDR Browse Data





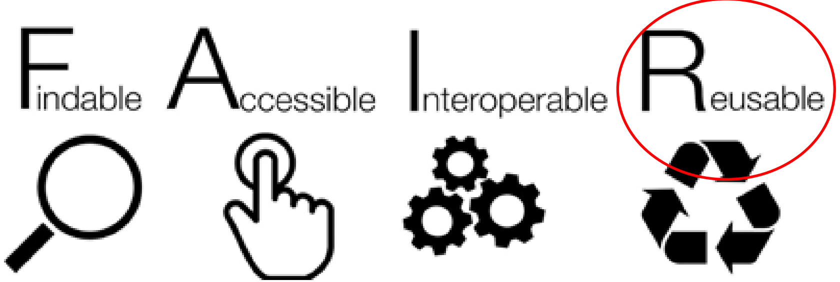
Idr0134 Peters K et al Reference bioimaging to assess the phenotypic trait diversity of bryophytes within the family Scapaniaceae [Scientific Data]

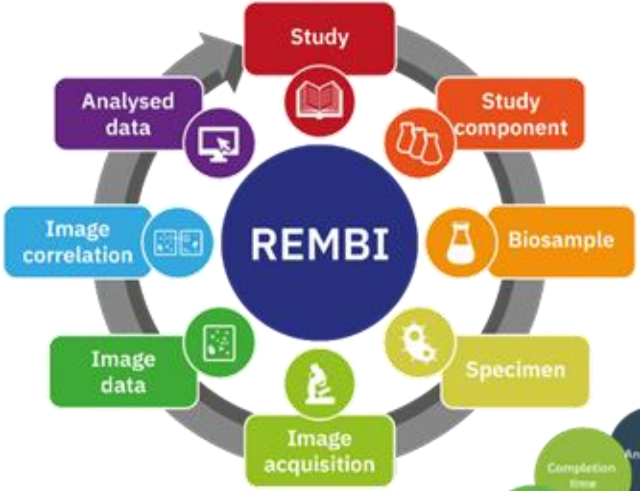




Problems:

- Lots of different data formats, including many proprietary formats
- Not possible to visualise the images without downloading





Metadata Schemas, Ontologies and controlled vocabularies

Image Data Management is not always FAIR

"Clara shares" by Henning Falk, ©2022 NumFOCUS, is used under a CC BY 4.0 license.
https://github.com/zarr-developers/zarr-illustrations-falk-2022#clara-shares

Image: Simon Li
https://twitter.com/penguinoops/status/643812318699225088

Data Life Cycle

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NFDI4 BIOIMAGE
Gosh, what a nightmare. I really need to think about data management MUCH earlier next time!
I will remember the laser power – I usually take the same value anyway.
WHAT? They are asking for my data for their analysis? But the naming is kind of not intuitive ...
PLANNING, DATA SEARCH + REUSE
IMAGE ACQUISITION
I just leave my images on the microscope PC. Nobody would delete anything on there, right?
FILE STORAGE + ACCESS
Most of the data is still on the microscope PC and I have some on this stick and there was an external drive ...
ARCHIVING + LONG-TERM STORAGE
PROCESSING + BIOIMAGE ANALYSIS
(DATA-) PUBLICATION
SHIT, how did I do this measurement?! I'll just write 'as previously done'.
We always use this old Python script this one postdoc once wrote for analysis.
BEEN THERE, DONE THAT.
HENNING FALK

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Ideal image data life cycle/management scenario?

The one where you need to do the minimal effort?

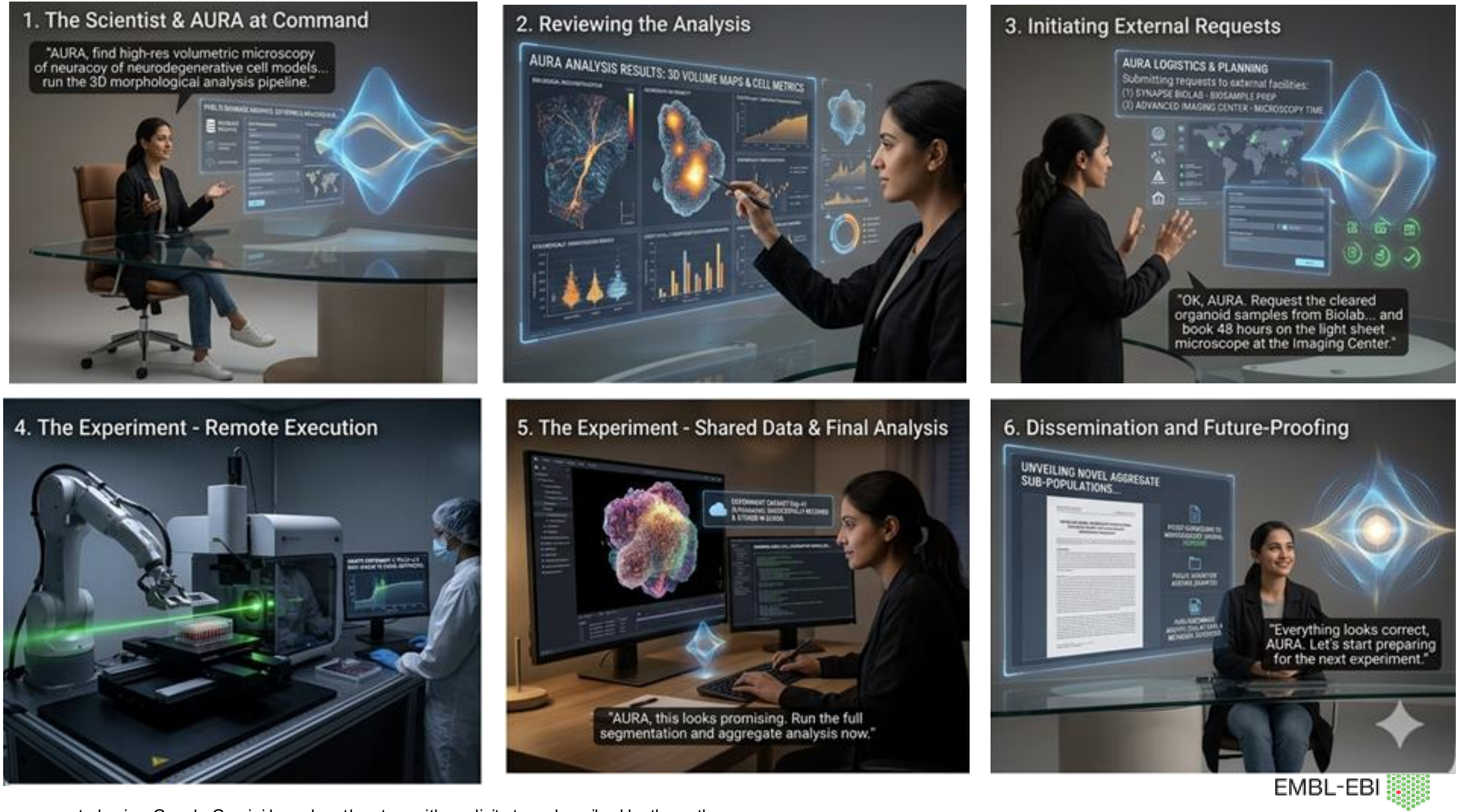
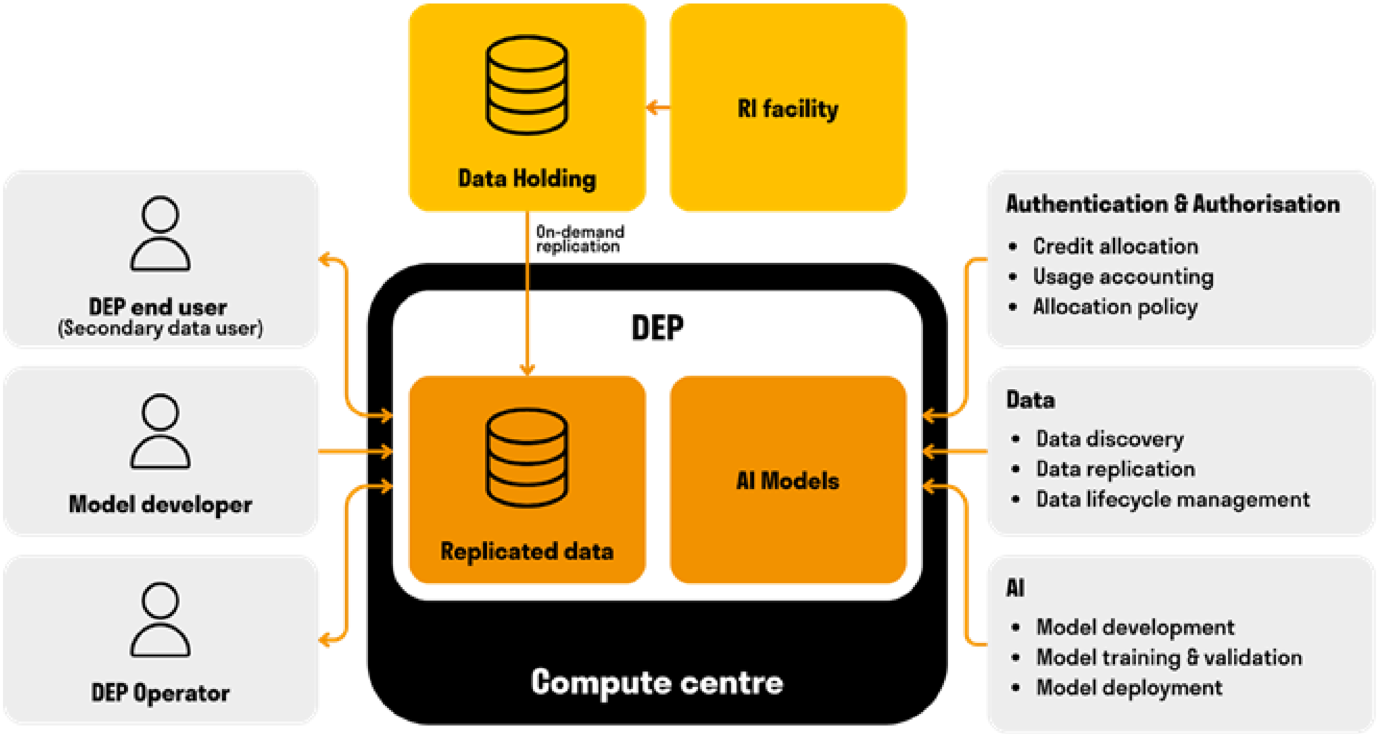


Image created using Google Gemini based on the story with explicit steps described by the authors.

A scalable, open platform for scientific data valorisation



https://www.riscale.eu/

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Ideal image data life cycle/management scenario?

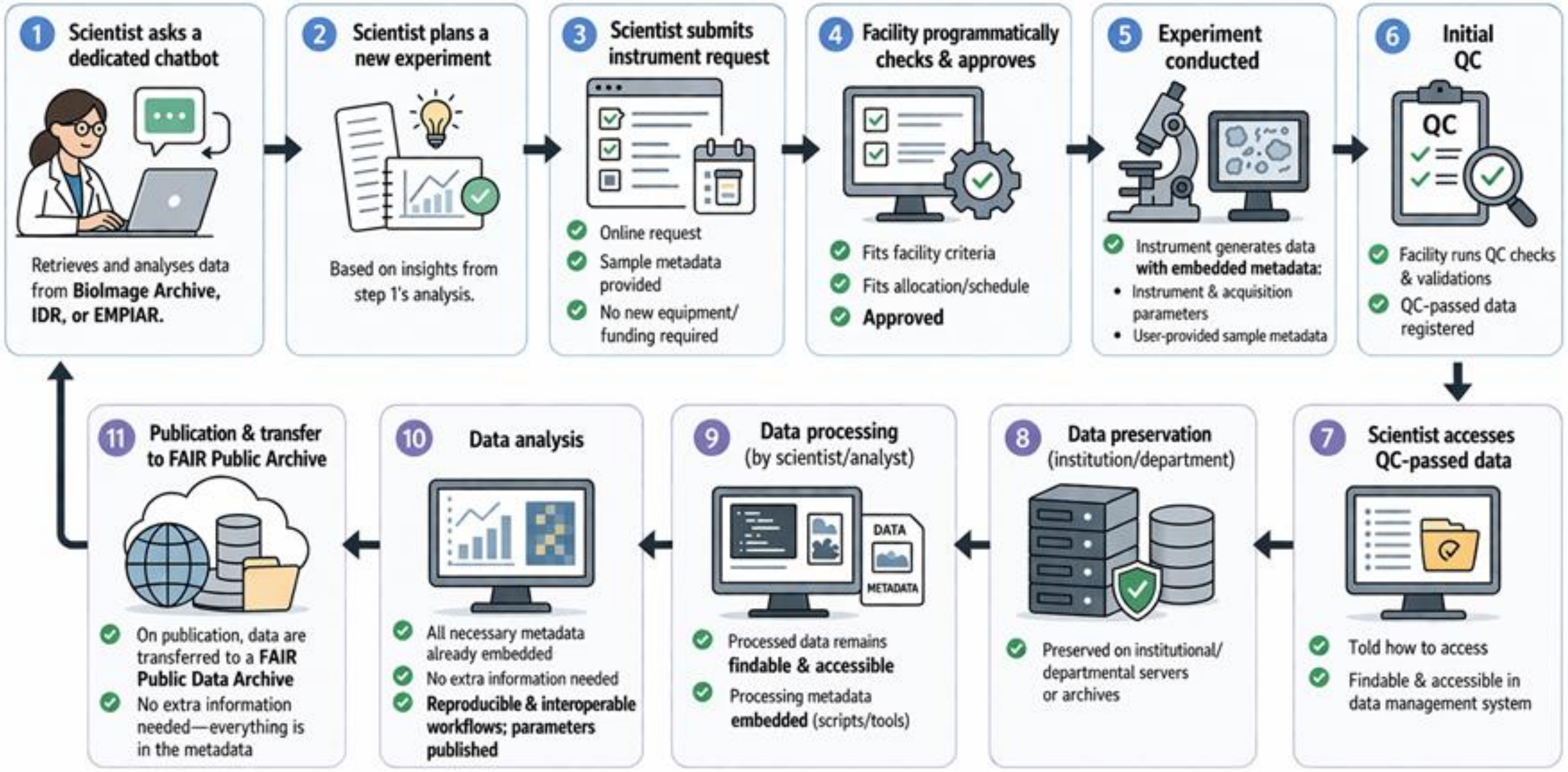

Figure created using ChatGPT based on explicit steps described by the authors.

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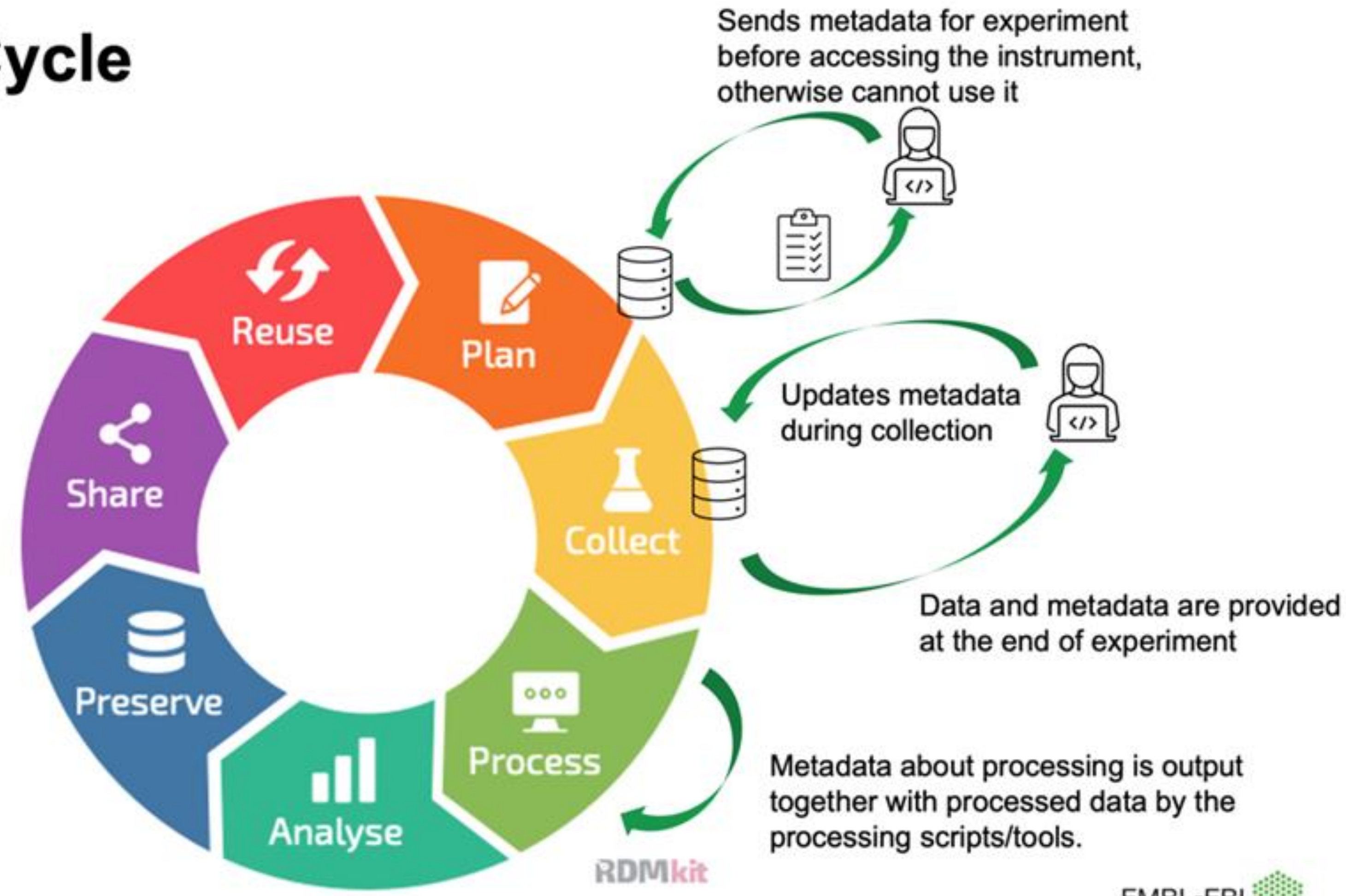
Data Life Cycle



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Group Discussion Exercise (20 minutes)

Describe the data life cycle journey of your imaging project

Write/Think about the journey of your image data

1. Which of the points in the ideal scenario works well in your setting and how do you do it?

1. What can be done to achieve the ideal scenario?

Google doc for note taking: https://bit.ly/4vJyhqr

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Data Management Plans

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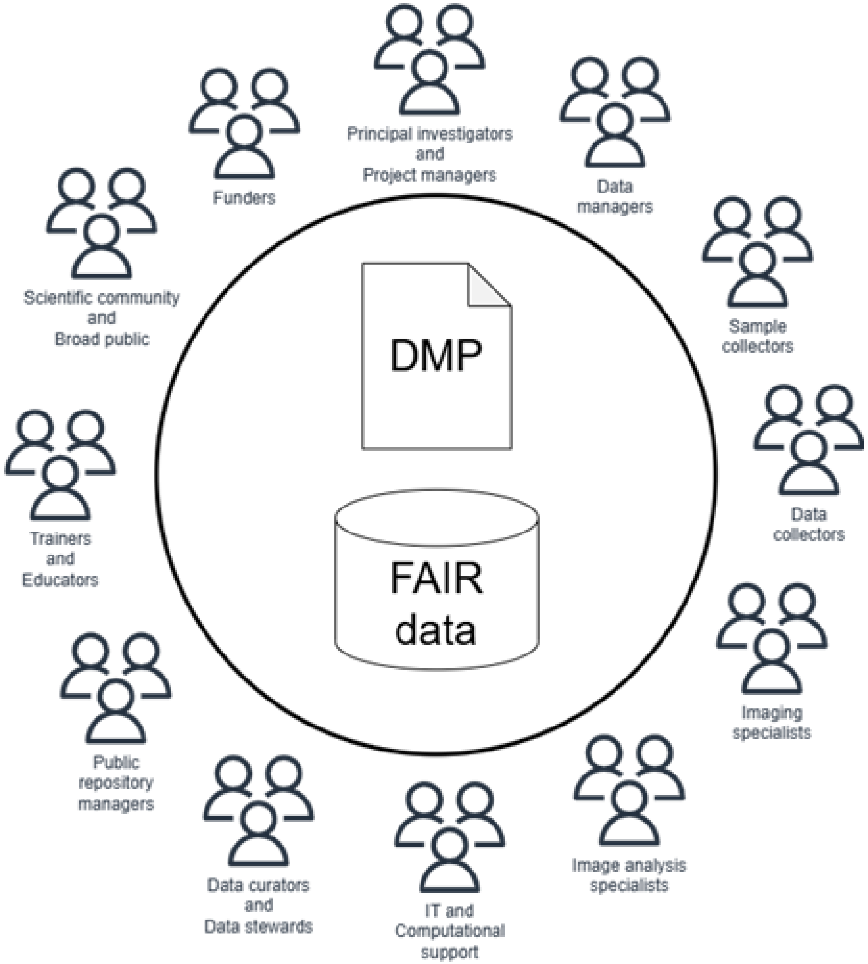
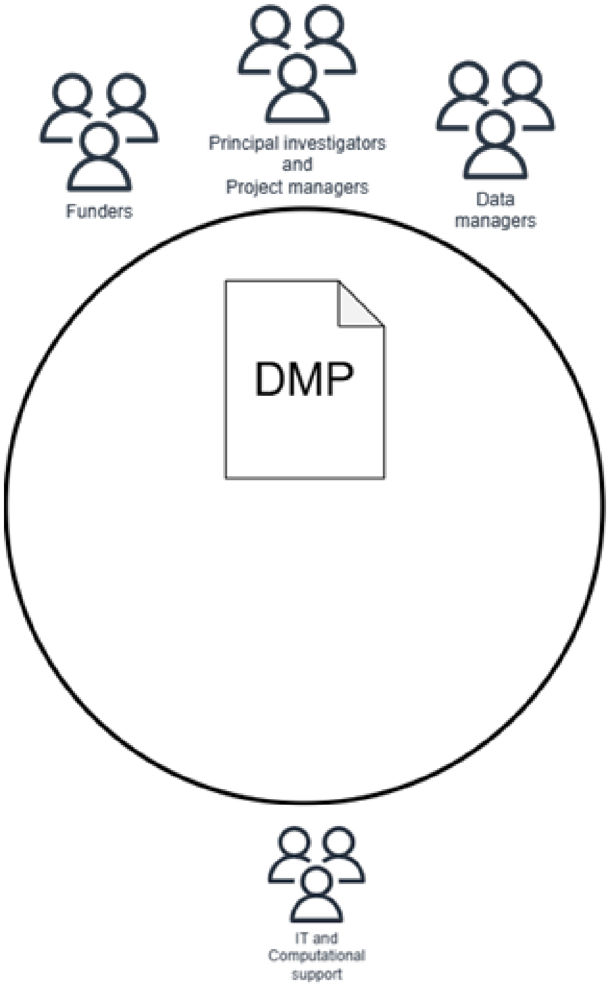
Data Management Plan

- **Strategy:** Define the strategy for managing the data and documents in your project
- **Cost:** You can avoid unexpected situations and costs See https://rdmkit.elixir-europe.org/costs_data_management
- **Why:**
  - Requirements from funders: Guidance to follow
  - It can help to determine potential risks and ways to mitigate them
  - It is a way to be more FAIR as it makes it easier to archive, share and reuse data
- **When:**
  - In early stages of a project, but it is a "**living document**" so it should be constantly revisited



Ensure that all stakeholders

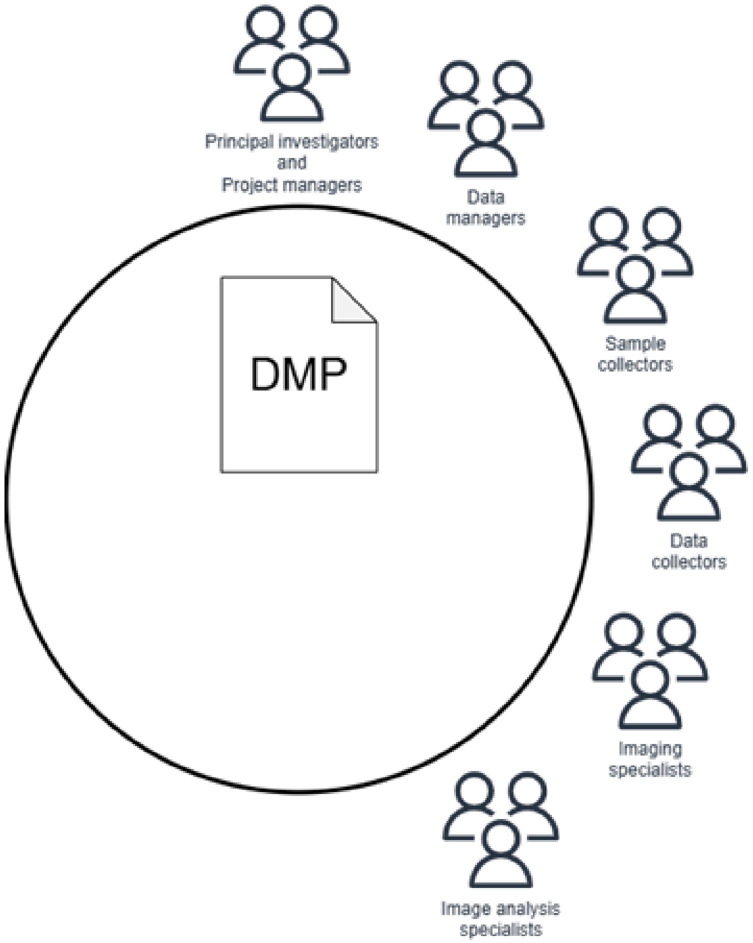
- understand data management practices
- know how to apply them,
- are aware of where to access training if they need additional knowledge or skills.



Principal Investigators and **Project Managers** bring **Data Managers** to draft the first version of the DMP in line with funder requirements.

Essential for Data Managers to understand the types of data that will be generated, review relevant ethical considerations, and identify suitable public repositories for data submission.

Who is missing?



Congratulations!

At kickoff, **schedule a data management planning session**

Data Manager to lead.

Alexandra Zakieva, Ajay Mishra et al. 2026, Managing Multi-Modal Imaging Data to Meet FAIR Principles (to bu published in EBI training collection)

Data Stewardship Wizard

Open-source data management tool for writing and managing a DMP in a collaborative format

https://ds-wizard.org/

It is distributed as a series of Docker images

You set up SMART questionnaires that guide you through building a DMP

DMPonline

Online tool to create DMPs

https://dmponline.dcc.ac.uk/

You can find funder's templates

There are publicly shared user's DMPs

Example template from Euro-BioImaging for Research Data Management plan

1-General Project Description

Project Name

Project ID

Project Leader

Project description

Project Start Date

Project End Date

Funding

https://doi.org/10.5281/zenodo.11473803

See also https://gerbi-gmb.de/i3dbio/i3dbio-rdm/dmp/

Data Management Plan (tools)

- Use the tool suggested by your funding agency or institution.
- Choose one of the following online DMP tools (ordered alphabetically).
  - Argos : the joint effort of OpenAIRE and EUDAT to deliver an open platform for Data Management Planning.
  - Data Stewardship Wizard : publicly available open-source tool to collaboratively compose data management plans through smart and customisable questionnaires with FAIRness evaluation.
  - DAMAP : tool for machine actionable Data Management Plans.
  - DMPonline : tool widely used in Europe and many universities or institutes provide a DMPonline instance to researchers.
  - DMPTool : widely used tool and many universities or institutes provide a DMPTool instance to researchers.
  - DMPRoadmap : DMP Roadmap is a Data Management Planning tool. Management and development of DMP Roadmap is jointly provided by the Digital Curation Centre (DCC), http://www.dcc.ac.uk/, and the University of California Curation Center (UC3), http://www.cdlib.org/services/uc3/. The DMPTool and DMPonline sites are both now running from the joint DMPRoadmap codebase.
  - easy.DMP : tool provided by the pan-European network EUDAT.
- Examples of useful resources for writing and implementing a DMP.
  - FAIR Implementation Profile and FIP Wizard are effective instruments for clearly defining and explaining the particular implementation choices required to effectively enact FAIR principles during and after the course of a research project.
  - Research Data Management Organiser : tool that supports the systematic planning, organisation and implementation of research data management throughout the course of a project.
  - Webinars explaining what a data management plan is and when you might need one, such as the Research Management Plan webinar produced in collaboration with Dr. Rob Hooft, Technical Coordinator of the Dutch TechCentre for Life Sciences, could be useful instruments to learn more about the topic.

https://rdmkit.elixir-europe.org/data_management_plan

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Day-to-day data management

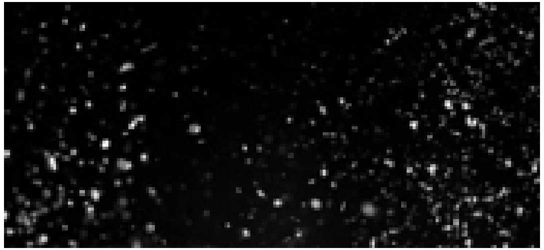
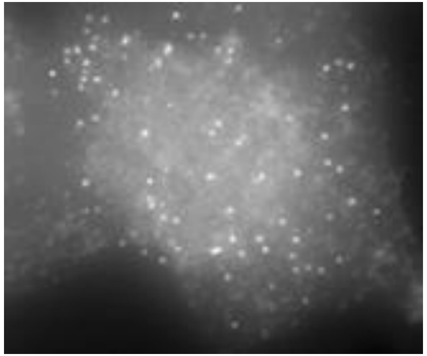
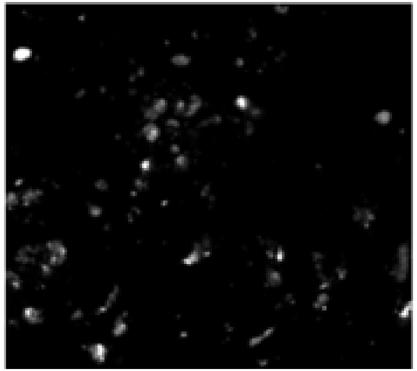
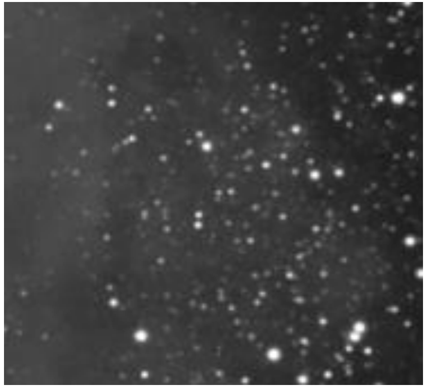
guidance on practical dos and don’ts

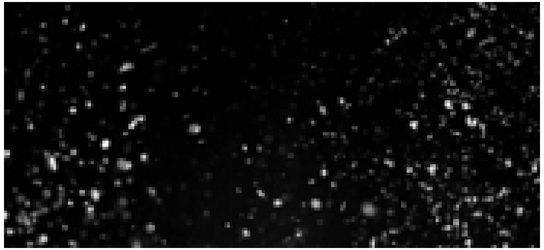
https://www.ebi.ac.uk/training/events/data-management-bioimage-informatics-data-flow/

https://www.ebi.ac.uk/training/events/principles-research-data-management

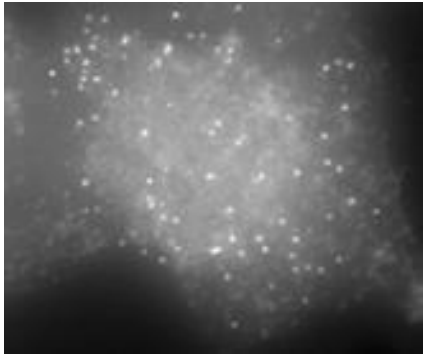
https://gerbi-gmb.de/i3dbio/i3dbio-rdm/data-management-platforms/

Metadata

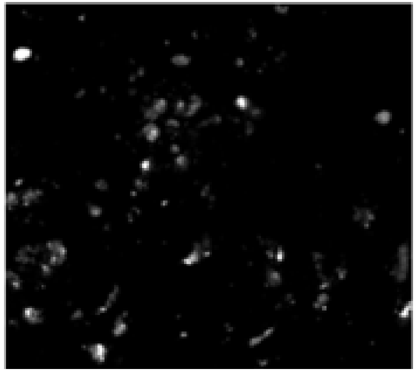







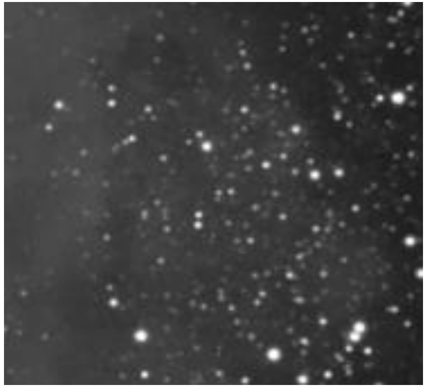
B&W photo of dust



smFISH on mouse cells

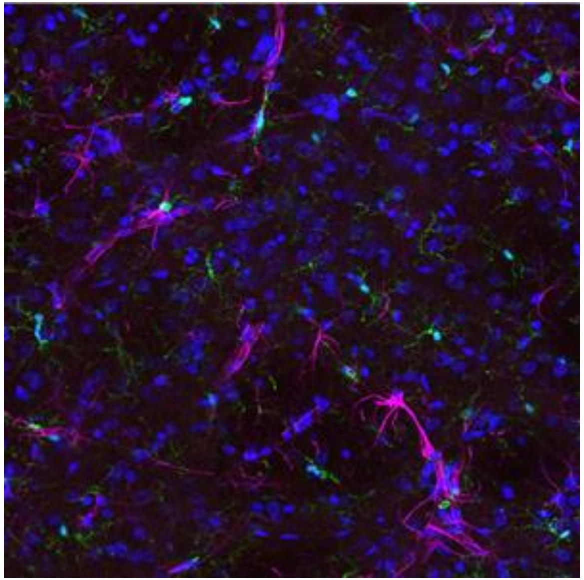


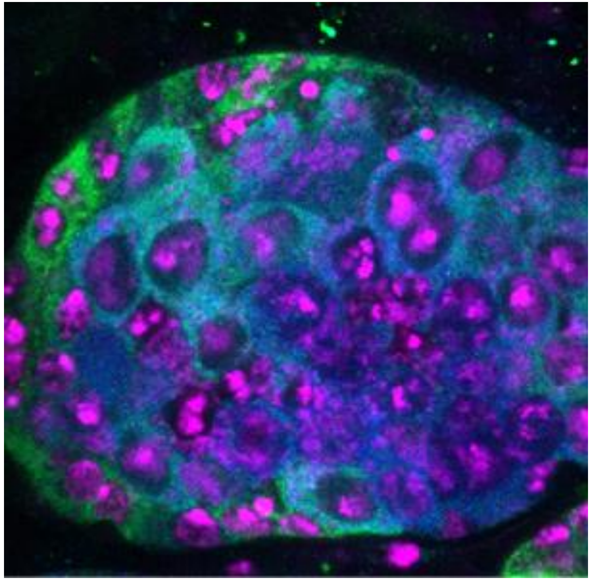
Fluorescence microscopy of mice tissue

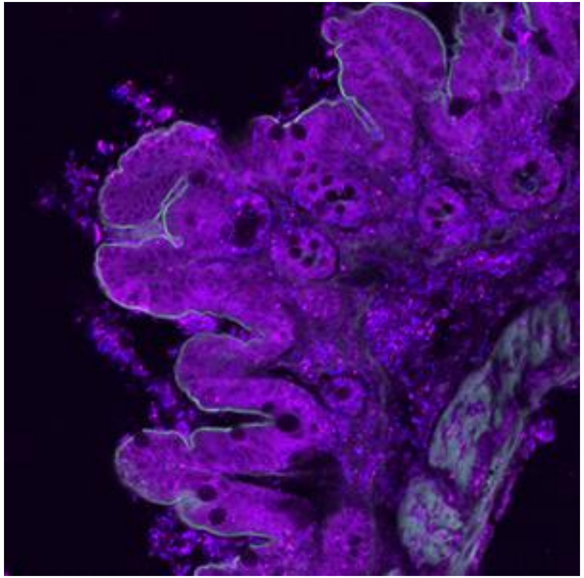


A section of M8 (The Lagoon) Nebula
(stars and interstellar dust and gas)

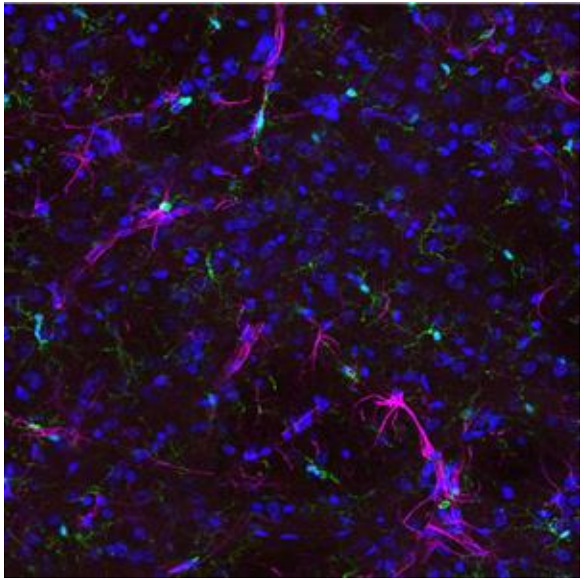
Which one of these spatial experiments was performed in a human sample?



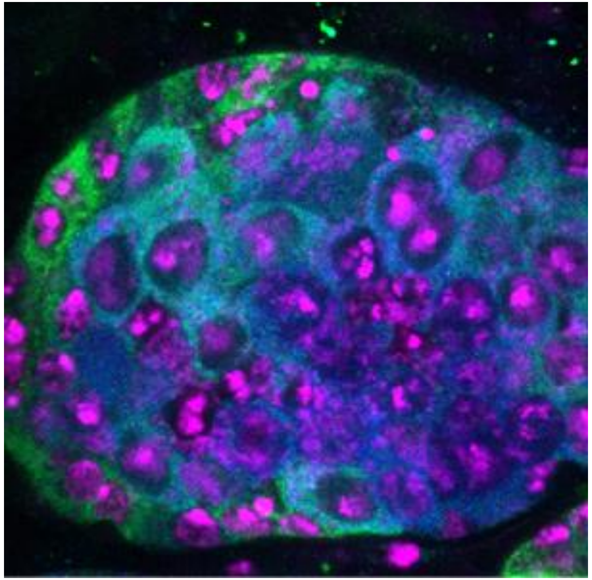




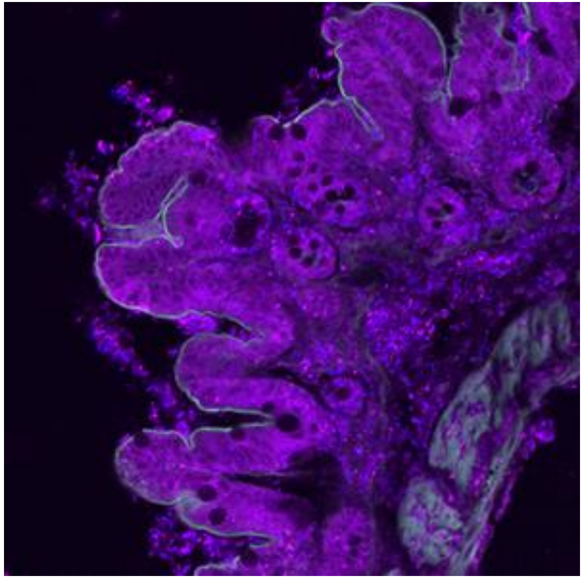
None! Experimental data without context is not useful



Rat (S-BIAD1280)



Fruit fly (S-BIAD1009)



Mouse (S-BIAD922)

Metadata

It’s data about our data → It helps to contextualise the data

What?

How?

When?

Who?

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Exercise (5 minutes)

Asses the following metadata recording

Can you re-use this or replicate this experiment? What information is missing?

https://zenodo.org/records/16618585

https://zenodo.org/records/5140334

Exercise (5 minutes)

What metadata needs to be there?

1. What metadata should be there for you to be able to do your responsibilities?

1. What metadata should be there if one needs to repeat the experiment?

1. What metadata should be there if it was someone else’s data you’d like to re-use?

menti.com

3770 5336

https://www.menti.com/alhcwchdasgg

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Exercise (for you to use for training)

The Case of the Missing Metadata

The Scenario:

Dr. X submitted the following dataset to the BIA:

Study Title: "Mitochondrial dynamics in neurons"

Description: We imaged neurons to look at mitochondria. The cells were stained and imaged on the confocal microscope. Images were processed in Fiji.

Files submitted: image_001.tif, image_002.tif, image_003.tif, processed_data.zip

Organism: Mouse (mus musculus)

Imaging Method: Confocal microscopy

Exercise

The Case of the Missing Metadata

What's Missing:

1. **Cell type specificity** - What type of neurons? (cortical, hippocampal, motor neurons?)
2. **Mouse strain, age, sex** - Critical for reproducibility
3. **Treatment/conditions** - Control vs experimental? What were they testing?
4. **Sample preparation** - Fixed or live? If fixed, what fixation method?
5. **Microscope details** - Which microscope? Which objective (magnification, NA)?
6. **Fluorophores/stains** - What was used to label mitochondria? (MitoTracker? Immunofluorescence? Genetically encoded marker?)
7. **Acquisition settings** - Laser power, detector gain, pixel size, z-step size
8. **Channel information** - How many channels? What does each show?
9. **Time information** - Single timepoint or time-lapse? If time-lapse, what intervals?
10. **File naming convention** - Which image is which condition/replicate/field of view? (though this should be in the metadata!)
11. **Image dimensions** - What are the physical dimensions (μm)? XY pixel size? Z-spacing?
12. **Raw vs processed** - Which files are raw data? What's in processed_data.zip?
13. **Analysis Information:** What specific steps in Fiji? (background subtraction, deconvolution, segmentation?)
14. **Software versions** - Which version of Fiji? Any plugins used? Is that enough?
15. **Analysis parameters** - Thresholds, filters applied?
16. **Biological replicates** - How many? Which images belong to which replicate?
17. **Technical replicates** - Multiple fields of view per sample

Metadata through the data life cycle

What to collect?

- Study-related:
  - Funder
  - Grant id
  - Researchers
  - Experimental design
- **Data-related:**
  - Sample and preparation information
  - Image acquisition protocols and parameters
  - Processing and analysis protocols

When to collect?

- Initially define in the Data Management Plan
- Before you even begin to collect any data
- During the whole data life cycle

How to collect?

- See next sections on metadata standardisation, data management tools and data standardisation

Points to consider

Metadata throughout the data life cycle

- What metadata to collect when during the data life cycle?
- What metadata is usually missing in your facilities?
- How could we prevent this at the point of data acquisition?

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Do’s and Don’ts for Metadata

Do’s:

- Collect the metadata at the beginning, during and after the study is ended
- Use metadata schemas and standard vocabulary to describe your (meta)data
- It should make the data clear for another user or even for your future self
- Keep the data and metadata together

Don'ts:

- Don’t leave the metadata gathering to when you start are writing the manuscript

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Metadata standardisation: schemas, ontologies and controlled vocabularies

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Metadata schema

- A structured way to describe a study
- It has specific components and attributes
- It will provide guidelines on how to organise, and format your data
- It helps with programmatic use

OME DATA MODEL

```xml
<OME xmlns="http://www.openmicroscopy.org/Schemas/OME/2016-06"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://www.openmicroscopy.org/Schemas/OME/2016-06
 http://www.openmicroscopy.org/Schemas/OME/2016-06/ome.xsd">
  <Image ID="Image:0" Name="6x6x1x8-swatch.tif">
    <AcquisitionDate>2010-02-23T12:51:30</AcquisitionDate>
    <Pixels DimensionOrder="XYCZT" ID="Pixels:0:0" PhysicalSizeX="10000.0"
    PhysicalSizeY="10000.0" Type="uint8" SizeC="1" SizeT="1" SizeX="6"
    SizeY="4" SizeZ="1">...</Pixels>
    <AnnotationRef ID="Annotation:1"/>
    <AnnotationRef ID="Annotation:2"/>
  </Image>
  <StructuredAnnotations>
    <!-- First Tag -->
    <MapAnnotation ID="Annotation:1">
      <Description>This is the description of the sample map A</Description>
      <Value>
        <M K="SampleKeyA">SampleValueA</M>
      </Value>
    </MapAnnotation>
    <!-- Second Tag -->
    <MapAnnotation ID="Annotation:2">
      <Description>This is the description of the sample map B</Description>
      <Value>
        <M K="SampleKeyB-1">SampleValueB-1</M>
        <M K="SampleKeyB-2">SampleValueB-2</M>
      </Value>
    </MapAnnotation>
  </StructuredAnnotations>
</OME>
```

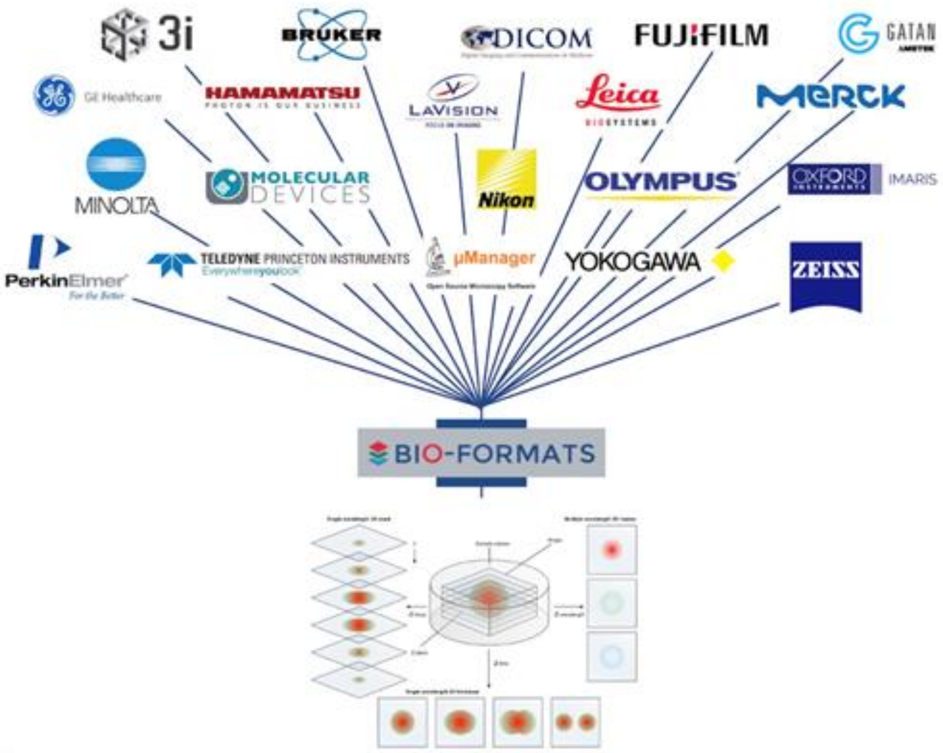
Image			Describes the actual image and its metadata.
	AcquiredDate		The acquisition date of the Image.
	Description		A multi-line description for the image.
	Name		A short description for the image. This would be used to, for example, select the image from a list.
	ImagingEnvironment		Describes the environment that the biological sample was in during the experiment.
		AirPressure	AirPressure in millibars[mbar].
		CO2Percent	%CO2 as a percent-fractions from 0.0 to 1.0 [%].
		Humidity	Humidity as a percent-fraction from 0.0 to 1.0 [%].
		Temperature	Temperature [degrees Celsius].
	ObjectiveSettings		Describes any settings on or around the objective
		CorrectionColor	An adjustable ring on the objective that corrects for changes in immersion medium refractive index. Arbitrary scale and unitless.
		Medium	A description of a Medium used for the lens. e.g., Oil, Water, WaterDipping, Air, Multi, Glycerol, Other
		RefractiveIndex	Refractive index is that of the immersion medium.
	Pixels		Defines the location and parameter sof the Pixels, the actual binary image data
		DimensionOrder	The order in which the individual planes of data are interleaved. e.g., XYZCT, XYZTC, XYCTZ, XYCZT, XYTCZ, XYTZC
		PhysicalSizeX	Physical size in x, y, z of a pixel in microns[µm]
		PhysicalSizeY	
		PhysicalSizeZ	
		SizeC	Dimensional size x, y, z, c, t of pixel data array
		SizeT	
		SizeX	
		SizeY	
		SizeZ	
		TimeIncrement	Used for time series that have a global timing specification instead of per-timepoint timing info, e.g., a video stream. [s].
		Type	The variable type used to represent each pixel in the image. e.g., int8, int16, int32, uint8, uint16, uint32, float, bit, double, complex, double-complex

Goldberg, et al. (2005) ***The Open Microscopy Environment (OME) Data Model and XML File: Open Tools for Informatics and Quantitative Analysis in Biological Imaging.*** *Genome Biol. 6:R47.*

DOI: 10.1186/gb-2005-6-5-r47

https://bit.ly/citing-ome

BIO-FORMATS



https://www.openmicroscopy.org/bio-formats/

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Bio-Formats

1. **Core metadata** includes:
   a. image resolution; number of focal planes, time points, channels, and other dimensional axes; byte order; dimension order;
   b. color arrangement (RGB, indexed color or separate channels); and thumbnail resolution.
2. **Original metadata** is information specific to a particular file format. Nomenclature often differs between formats, as each vendor is free to use their own terminology.

OME metadata is information from #1 and #2 converted by Bio-Formats into the OME data model.

https://bio-formats.readthedocs.io/en/stable/about/index.html

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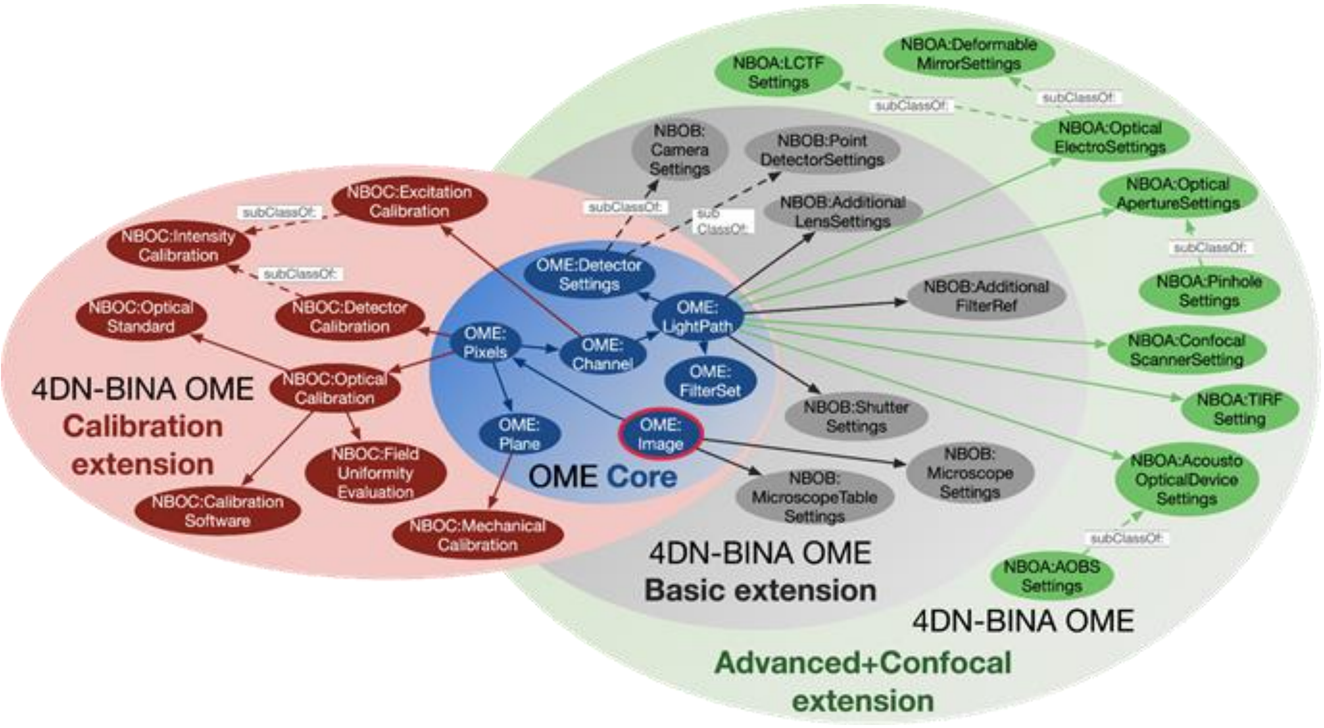
Examples tools

- Read data using Bio-Formats
  - https://github.com/BioImagingUK/Training/blob/main/notebooks/BioFormats.ipynb
- Read data using Python libraries
  - https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_Image_Using_Python.ipynb

Extending a Model??



4D-Nucleome / BINA / OME / QUAREP-LiMi (2018-2019)

https://github.com/WU-BIMAC/NBOMicroscopyMetadataSpecs

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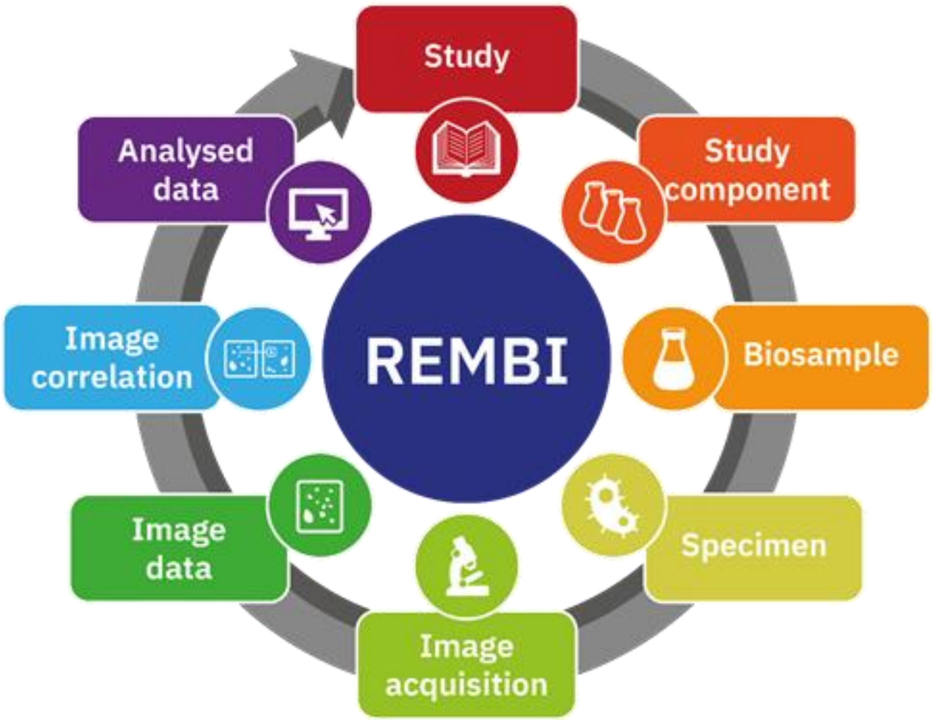
REMBI: REcommended Metadata for Biological Images



Sarkans, U., Chiu, W., Collinson, L. *et al*. REMBI: Recommended Metadata for Biological Images—enabling reuse of microscopy data in biology. *Nat Methods* 18, 1418–1422 (2021). https://doi.org/10.1038/s41592-021-01166-8

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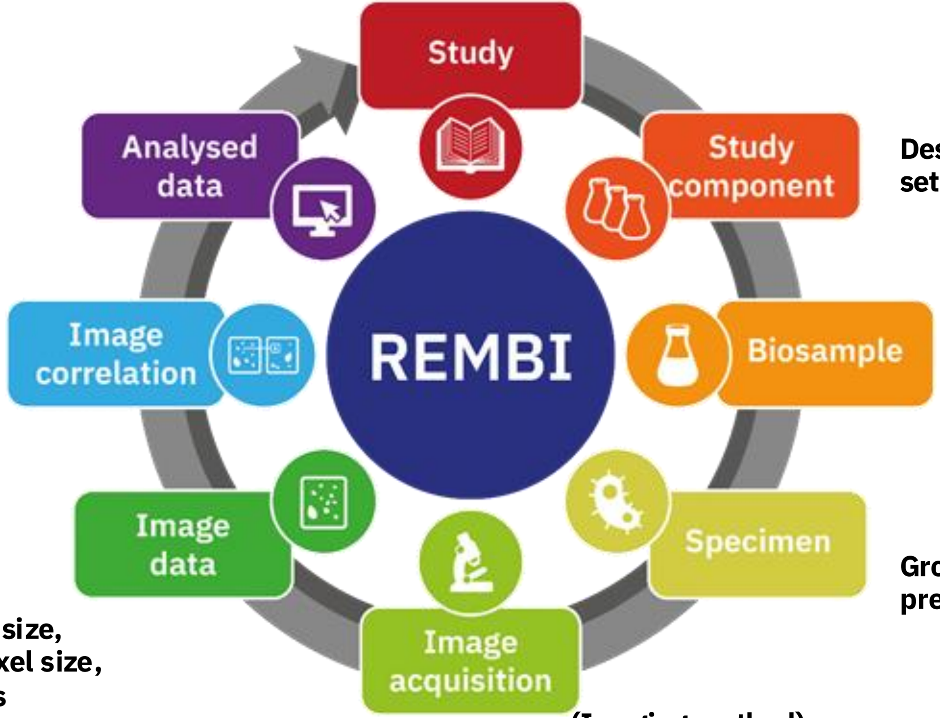
REMBI

Project description, funding and publications.

Analysis methods, workflow

Description of the data set

How you correlated images from different modalities or experiments



Species, tissue/cell/organism, genetic or extrinsic variables

Growth and preparation protocols

Physical size, Pixel/voxel size, Channels

(Imaging method), instrument, acquisition parameters

This is implemented as the File List for BioImage Archive submissions, instead of a separate component in the submission form.

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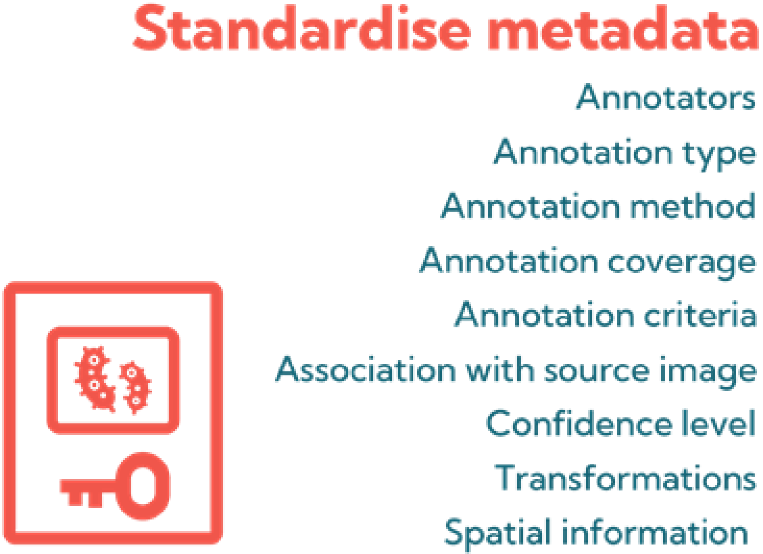
https://www.ebi.ac.uk/bioimage-archive/rembi-model-reference/

Seeing REMBI on a BIA submission

https://beta.bioimagearchive.org/bioimage-archive/study/S-BIAD1392

MIFA: Metadata, Incentives, Formats and Accessibility



MIFA guidelines are meant to improve the reuse of bioimages and annotations for AI applications.

Zulueta-Coarasa, T., Jug, F., Mathur, A. *et al.* MIFA: Metadata, Incentives, Formats and Accessibility guidelines to improve the reuse of AI datasets for bioimage analysis. *Nat Methods* 22, 2245–2252 (2025). https://doi.org/10.1038/s41592-025-02835-8

Example of BIA submission with annotations

BETA **S-BIAD1392**

PlantSeg core datasets: Ovules of Arabidopsis thaliana

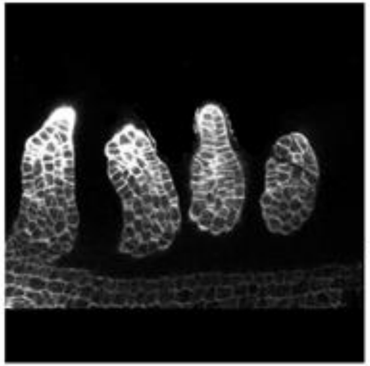
Released: 2024-10-01

Rachele Tofanelli[1], Athul Vijayan[1], Kay Schneitz[1], Adrian Wolny[2], Anna Kreshuk[2], Teresa Zulueta-Coarasa[3]

[1]School of Life Sciences Weihenstephan, Technical University of Munich, Freising, Germany [2]EMBL, Heidelberg, Germany [3]European Bioinformatics Institute

On this page

- Curated Study Information
- Models Related To This Work
- Annotated Datasets
- Viewable Annotated Images



Example image for this dataset

Example annotation for this dataset

View in BioStudies

Browse all files

https://beta.bioimagearchive.org/bioimage-archive/galleries/ai/ai-ready-study/S-BIAD1392#all_images

Controlled vocabularies

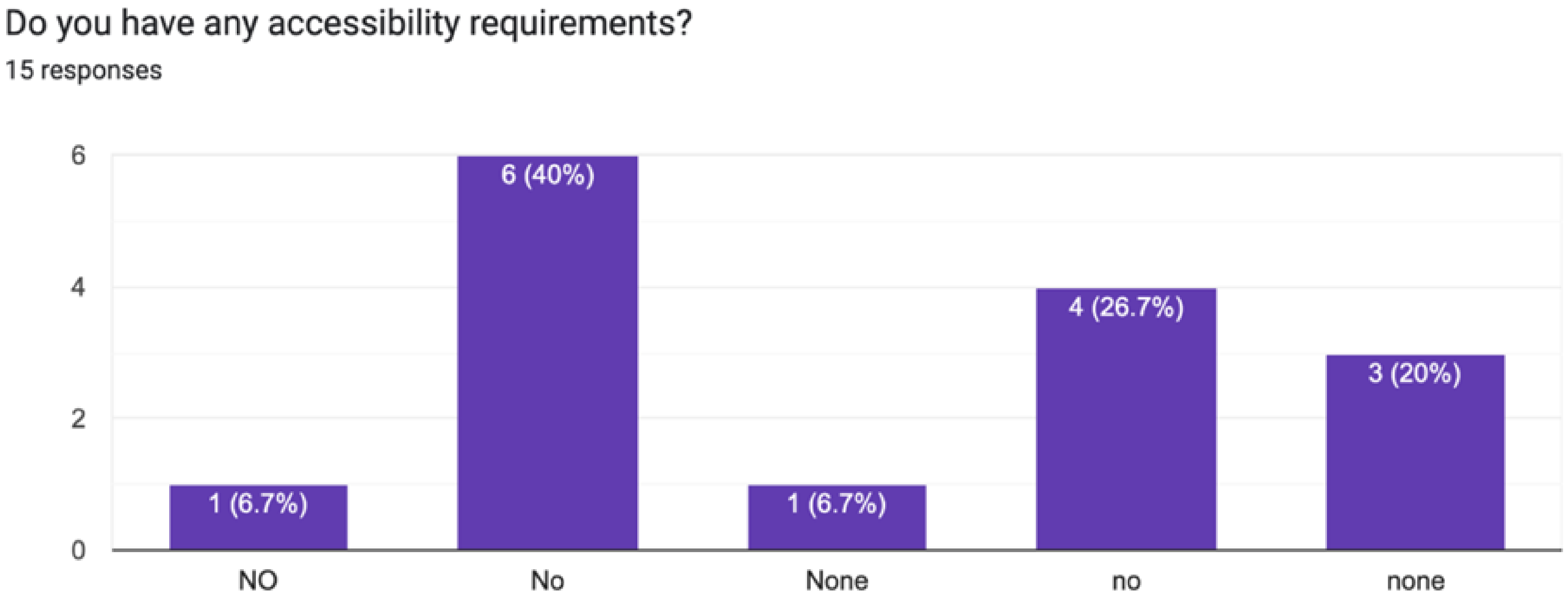
- Gives clear definitions for terms
- Avoids issues like different spelling or even different languages

Standard vocabulary:

- List of terms
- Gives a more precise way of entering data

Taxonomy:

- List of terms but with a hierarchy
  - E.g. NCBI Taxonomy


Do you have any accessibility requirements?
15 responses
6
4
2
0
6 (40%)
4 (26.7%)
3 (20%)
1 (6.7%)
1 (6.7%)
NO
No
None
no
none
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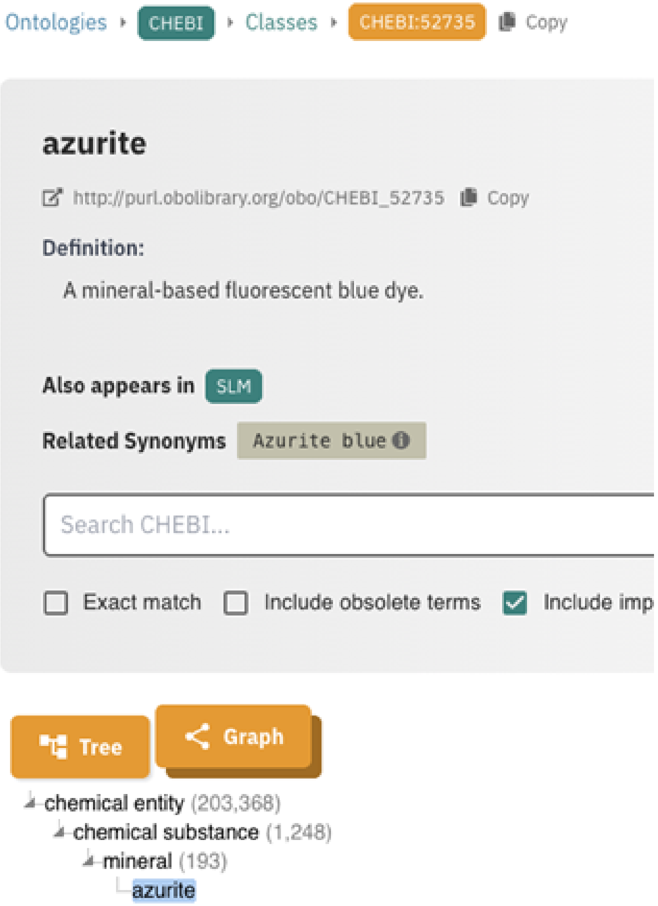
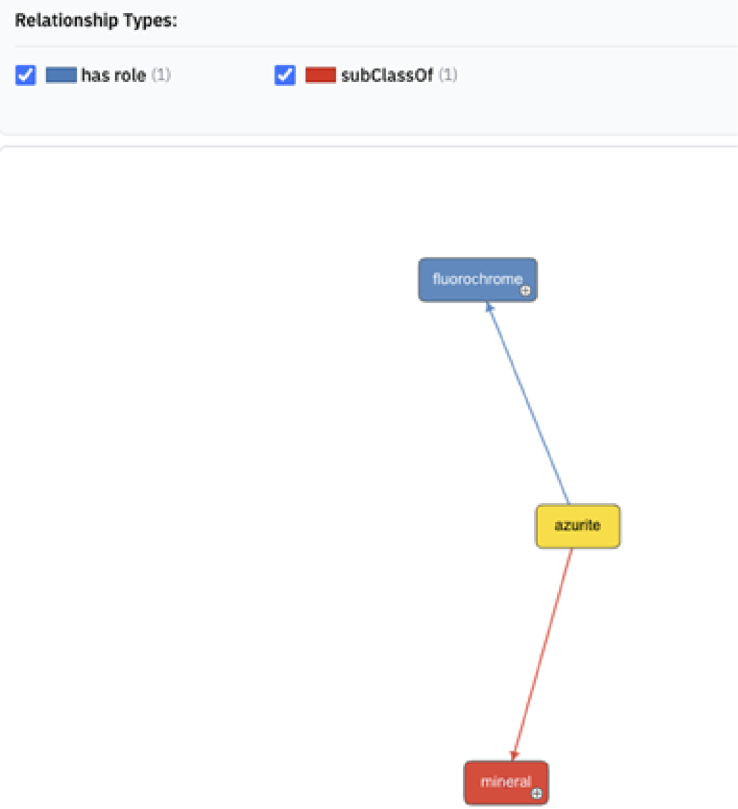
Ontologies

- A taxonomy but with added relationships between the terms
- It allows to make inferences
- Can describe properties
- Terms have a persistent identifier
- E.g. CHEBI (Chemical Entities of Biological Interest)

Ontologies





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Exercise

Define ROI

Region of Interest or Return-on-investment

Adding a controlled vocabulary or an ontology term would avoid confusion

obo/NCIT_C85402

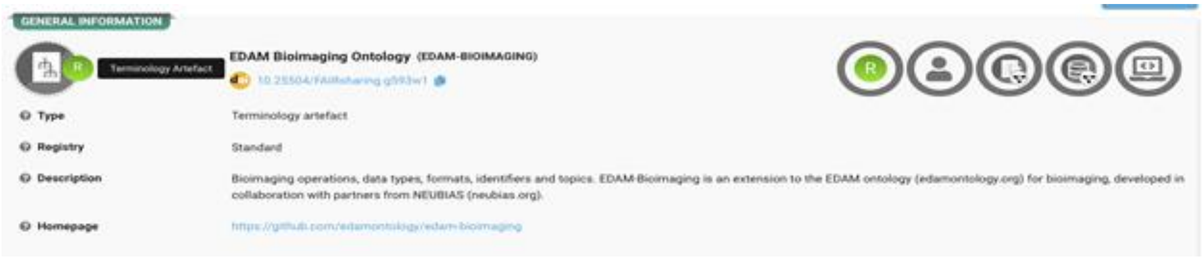
Definition:
A specific area of interest defined by a sequence of image overlays or a sequence of contours described as a single point (for a point ROI) or more than one point (representing and open or closed polygon).

Where to look for Ontologies

- Ontology Lookup Service (OLS) https://www.ebi.ac.uk/ols4/ontologies

- Ontobee https://ontobee.org/

Where to check if an ontology is maintained or how widely it's been adopted



https://fairsharing.org/

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Example from the BIA

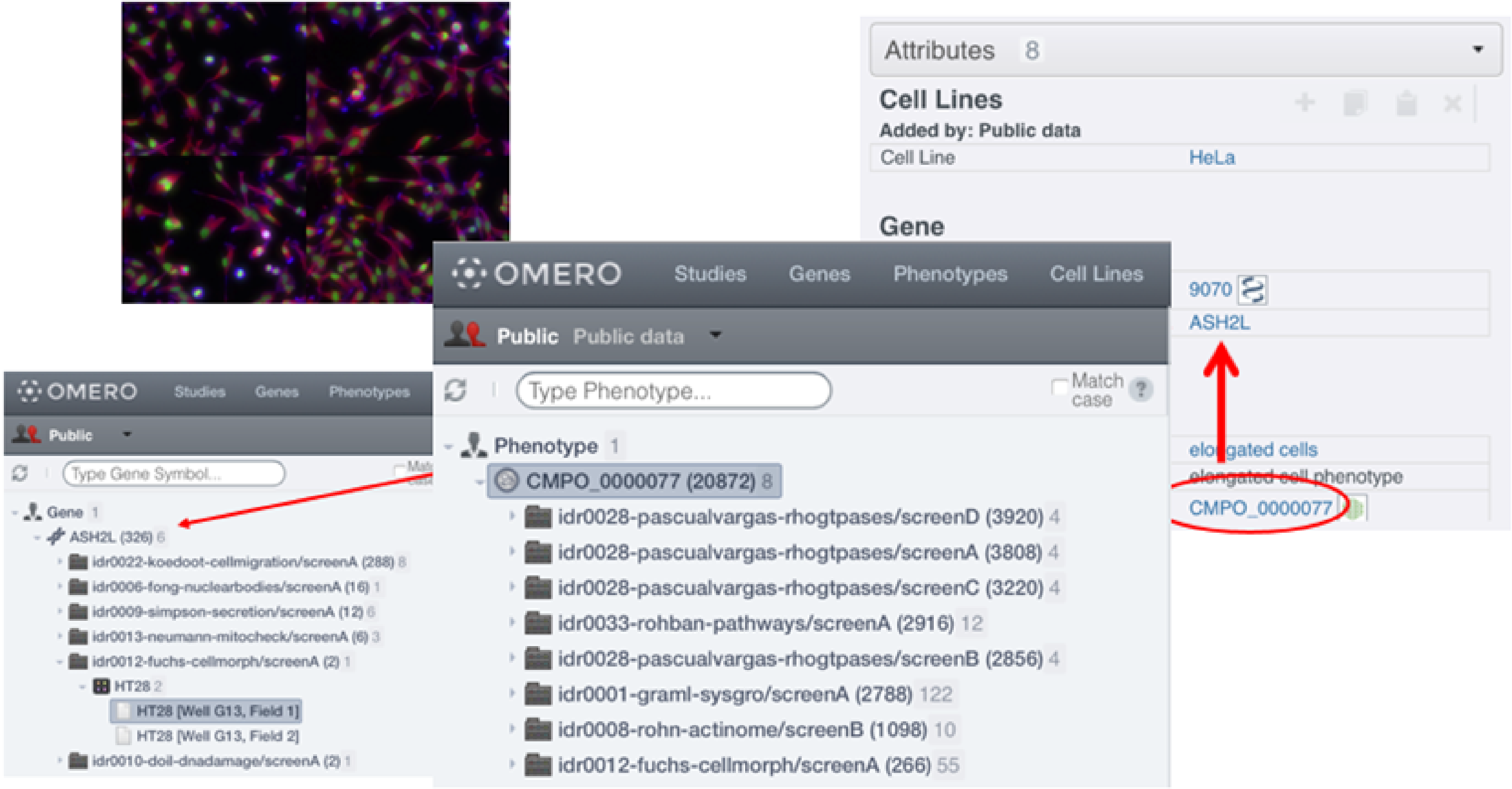
Experimental Factor Ontology (EFO)

Provides a systematic description of many experimental variables available in EBI databases

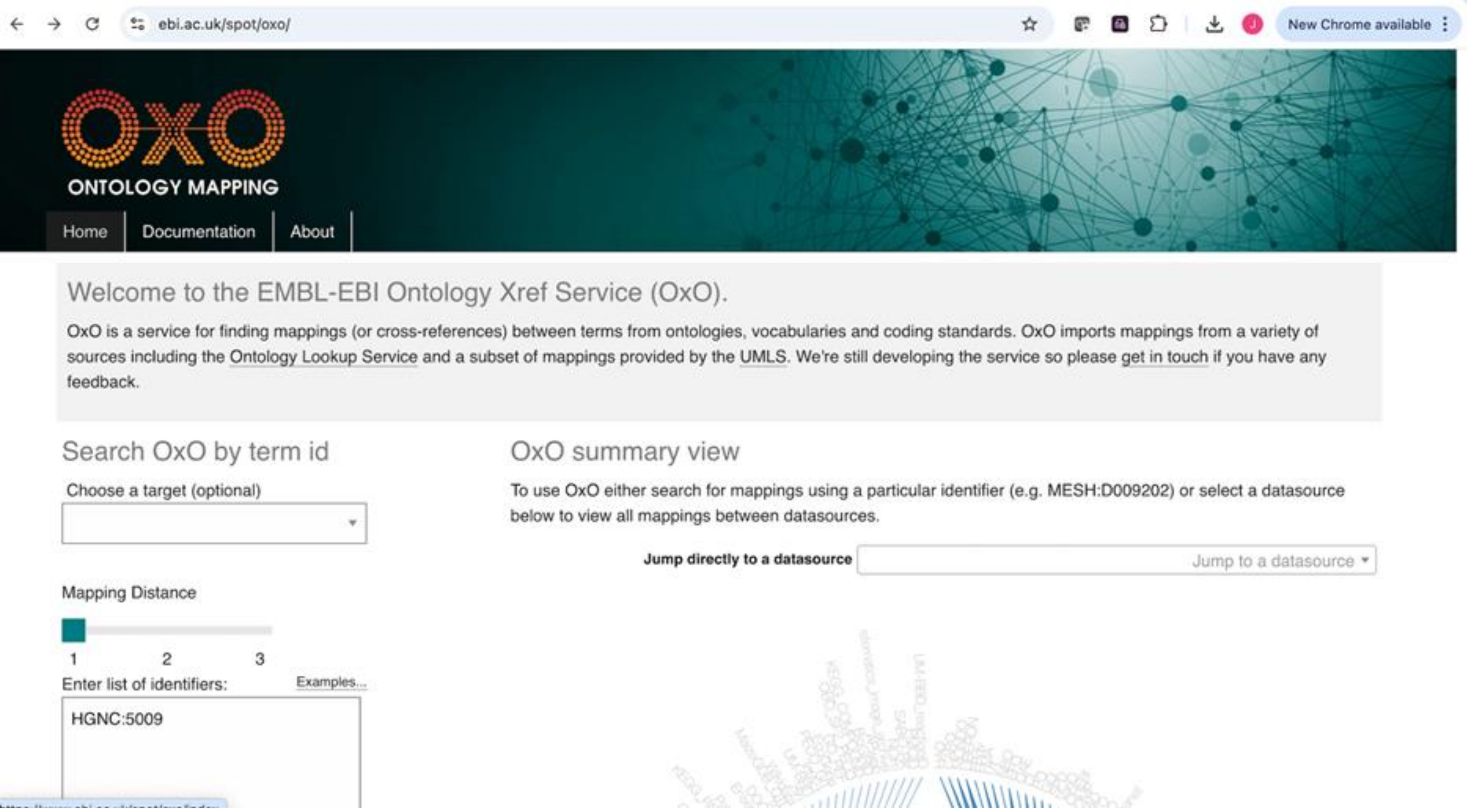
It combines several ontologies like CHEBI, FBBI, NCBI taxon

IDR

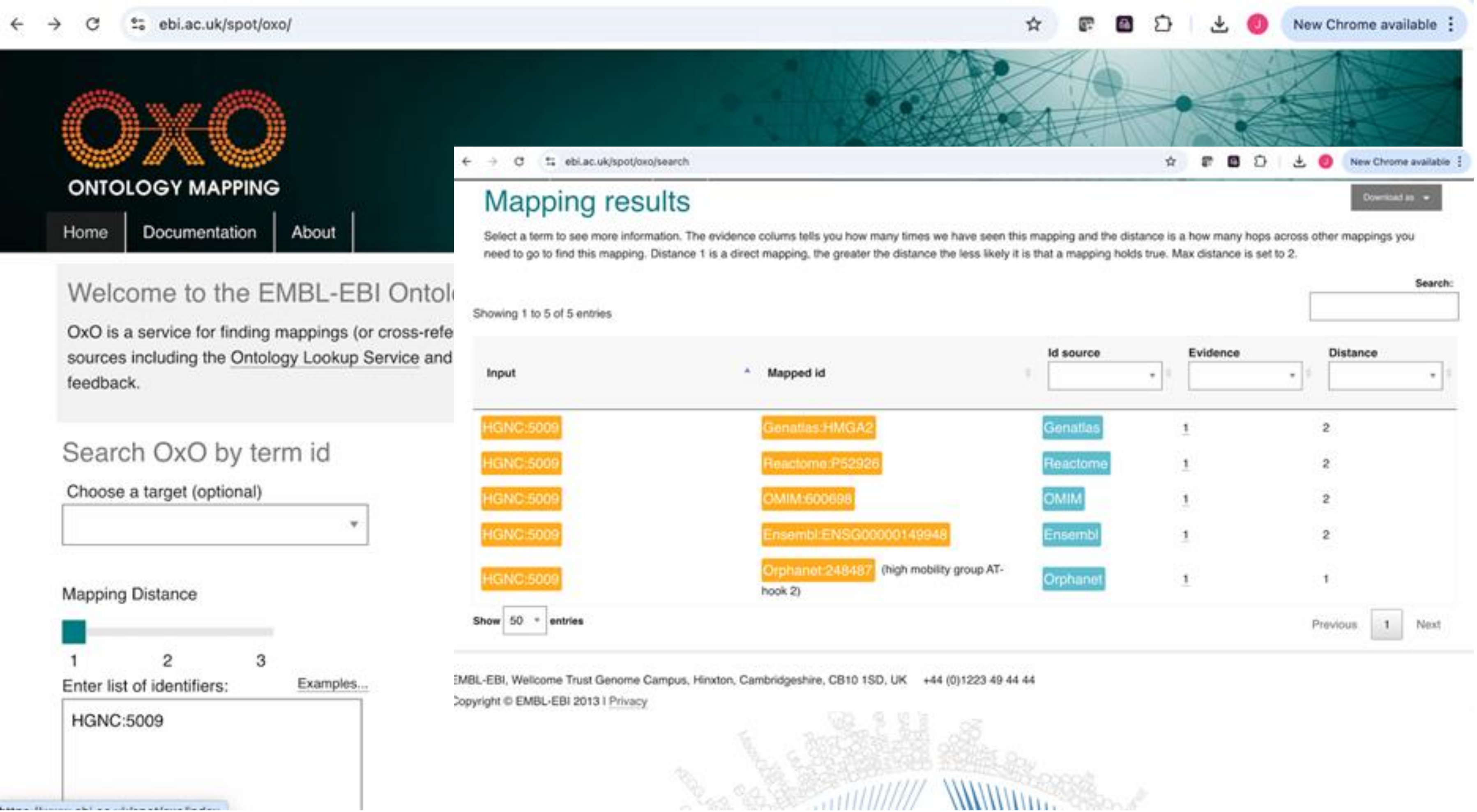
Linked Metadata



idr0012 Fuchs F et al Molecular Systems Biology [10.1038/msb.2010.25](10.1038/msb.2010.25)



https://github.com/BioImagingUK/Training/blob/main/notebooks/OXO_IDR.ipynb



https://github.com/BioImagingUK/Training/blob/main/notebooks/OXO_IDR.ipynb

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FoundingGIDE: Founding a Global Image Data Ecosystem

Bringing together image data resource owners and research infrastructures from Europe, Australia and Japan to develop the basis of **interoperable image data repositories**.

➔ common ontologies and metadata frameworks
➔ data interoperability

Biological Imaging

- D2.1 – Landscape analysis of existing ontologies and recommendations on a set of imaging ontologies
- D6.1 – Report in metadata model overlap and gaps

Preclinical Imaging

- D4.1 – Report on the recommended ontologies for preclinical image datasets and technologies

https://founding-gide.eurobioimaging.eu/

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Recommended ontologies by FoundingGIDE

Ontologies used by BioImage Archive, IDR and SSBD

Category	Ontology/Controlled vocabulary	Acronym
Imaging Method	Biological Imaging Methods Ontology	FBbi
Taxonomy	NCBI Organismal Classification	NCBITaxon
Cell Line	Cell Line Ontology	CLO
Cell type	Cell Ontology	CL
Gene	Ensembl	ENSG
	Entrez Gene	NCBI_Gene
Protein	Universal Protein Resource	UniProt
Chemical Compound	PubChem	PubChem
	Chemical Entities of Biological Interest Ontology	ChEBI
Experimental Condition	Experimental Factor Ontology	EFO
	Ontology for Biomedical Investigation	OBI
Unit	Units of Measurement Ontology	UO
Disease	Systematized Nomenclature of Medicine-Clinical Terms	SNOMED CT
Cellular Phenotype	Cellular Microscopy Phenotype Ontology	CMPO
Anatomy	Uberon Multi-species Anatomy Ontology	UBERON
Biological Process	Gene Ontology	GO
Cellular Component		
Molecular Function		
Controlled Vocabulary	Medical Subject Headings	MeSH

Imaging technique/method
We recommend use of either the Biological Imaging Methods Ontology (FBbi) or EDAM ontologies.

If some subject domain is not captured in this list, we recommend looking at the ontologies in use in EFO.

Onami, S., Kyoda, K., Yamagata, Y., & Itoga, H. (2025). foundingGIDE Deliverable D6.1: Report in metadata model overlap and gaps. Zenodo. https://doi.org/10.5281/zenodo.16794787

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Recommended ontologies by FoundingGIDE

- Using ontologies and taxons are better than not
- Some archives have specific recommendations

Recommendations for imaging facilities, data stewards, and biologists capturing imaging data.

Capture ontology IDs rather than (or as well as) text description or names. Many ontologies will have guidelines about how to reference their terms: e.g. NCBI taxon recommends NCBI:txid9606 for Homo Sapiens, and OBO ontologies recommend e.g. UBERON:0000948 when it is possible to provide context for the UBERON prefix. Preferably use ontologies that correspond to those in use of any database that may eventually store the data, but do not make that a limiting factor. It is preferable to have some IDs of a different ontology than none at all. If ontologies are missing terms, contact their maintainers to get them added.

Sherwood, F., Longo, D., & Maria, M. (2025). foundingGIDE Deliverable D2.1 Landscape analysis of existing ontologies and recommendations on a set of imaging ontologies. Zenodo. https://doi.org/10.5281/zenodo.16794948

Data management tools

Data Management Basis

Spreadsheets

- Easiest to implement
- Lightweight
- Get complicated to handle the bigger they get
- Unit of data is the cell

Electronic Lab Notebook (ELN)

- Electronic equivalent of a paper notebook
- Easier way of sharing
- Versioning
- can’t truly delete anything

Database & File Storage (e.g. OMERO)

- Better suited for more complex data
- Requires IT knowledge to set up and maintain
- Easily scalable
- Can have data models and relationships
- Can provide provenance and versioning

Packaged Files (e.g. RO-Crate)

- Metadata and data is packaged together
- Easily scalable and shareable
- Can have (relational) data models
- Can have versioning and provenance

Which one to use

- Is it for one individual only or do you want others to be able to view and edit?
- Do you want to impose a structure?
- Do you want to impose value restrictions?
- Does it need controlled access?

A spreadsheet may be fine when:

- one person is exploring a small dataset
- the work is temporary
- the data is not yet final
- the consequences of errors are low

A managed system becomes important when:

- many people depend on the data
- the data must be reused later
- permissions matter
- data quality needs to be checked
- the data links to samples, files or workflows

What you still need for image data management

Option	What it gives you	What you still need to build
Relational database	Strong structured data management	(Image viewer, file handling), metadata model, user interface, permissions, import tools
Document database	Flexible metadata storage	(Image viewer, file management), validation, search interface, permissions
Graph database	Tracks relationships and provenance	(Image viewer, file management), user interface, data entry tools
OMERO	Image repository, web viewer, metadata, annotations, sharing, permissions, image import, APIs	Some custom metadata templates, integrations

Which one to use

- Individual researcher or small group: OMERO or data management servers or ELN if you have it setup at your institution. If not, spreadsheets
- Big research groups or labs: OMERO or data management servers if you have it setup at your institution. If not, electronic lab notebooks
- Imaging centres or institutes: OMERO or custom data management servers

➔ RO-crates for data packaging and sharing.

In some cases can also be used for metadata management in combination with versioning tools (e.g. EMO BON uses ro-crates and Github for data management)

Electronic Lab Notebooks

Pros

- An almost electronic equivalent to a paper notebook but we added functionality
- Many comercial options (LabArchives, RSpace) that provide both free and paid versions
- Paid versions can be at an institutional level and have things like admins
- Easy way to transfer ownership of (meta)data when someone leaves the lab/institution

Cons

- Limited features on the free version
- You need an electronic equipment and internet connection to use them
- Trickier to use in labs with clean policies

ELN Common format: https://github.com/TheELNConsortium/TheELNFileFormat

Electronic Lab Review Notebooks

gerbi-gmb.de/i3dbio/i3dbio-rdm/electronic-lab-notebooks/

New Chrome available

Regulatory Compliance: ELNs help ensure compliance with standards.

The ELN market is diverse, with a wide range of options available. Here's a brief overview of some popular ELN's in the field of Life Sciences (as of early 2025):

	LabArchives	Benchling	SciNote	elabFTW	RSpace	openBIS
Costs	subscription	subscription	subscription	Free (self hosting)	subscription	Free (self hosting)
Open Source Code	No	No	No	Yes	No	Yes
Link	Website	Website	Website	Website	Website	Website

When using both an ELN and OMERO, scientists should be able to conveniently annotate the required metadata in either their ELN or within OMERO, without the need to duplicate annotation efforts. For this purpose, the I3D:bio team is working on an ELN-integration layer that allows bridging data entries within an electronic lab notebook and OMERO. The tool is available for testing on GitHub, but still considered under development.

Try out the OMERO-ELN-DataBridge-Tool developed by I3D:bio.

https://gerbi-gmb.de/i3dbio/i3dbio-rdm/electronic-lab-notebooks/

Exercise (5 minutes)

What can you do with this data? What needs to change?

File Name	Imaging Modality	Organism	Channel 1	Ch2	Resolution (um/pixel)	Experiment Date	Treatment
image_1.tif	Confocal	fly	actin-GFP	myosin:mCherry	0.05	2023/01/15	control
sample_B.tif	Microscopy	Drosophila	actin:GFP	myosin:mCherry	512x512	02-15-2023	drug1
sample_3.tiff		mouse	actin	myosin:mCherry	0.1	Jan 25th, 2023	drug2

Answers

- File name: Inconsistent naming of images and image formats
- Imaging modality: which microscopy? Empty values
- Organism: inconsistent naming; both scientific name and common name
- Channel 1: inconsistent formatting (actin-GFP, actin:GFP), is actin same as actin-GFP?
- Resolution: wrong values; 512x512 is not resolution but likely the image pixel size
- Experiment date: inconsistent format
- Treatment: what are drug1 and drug2?
- General: better to avoid space characters in column headers and file names, lowercase, uppercase inconsistency, inconsistent naming of column headers

Tips for spreadsheets

Do’s

Use first row as header

Use atomic values

One table per sheet

Impose formats to each cell where possible

Add one tab to work as a README

Define how you will code for missing values

Don'ts

Don’t convey information using different formats

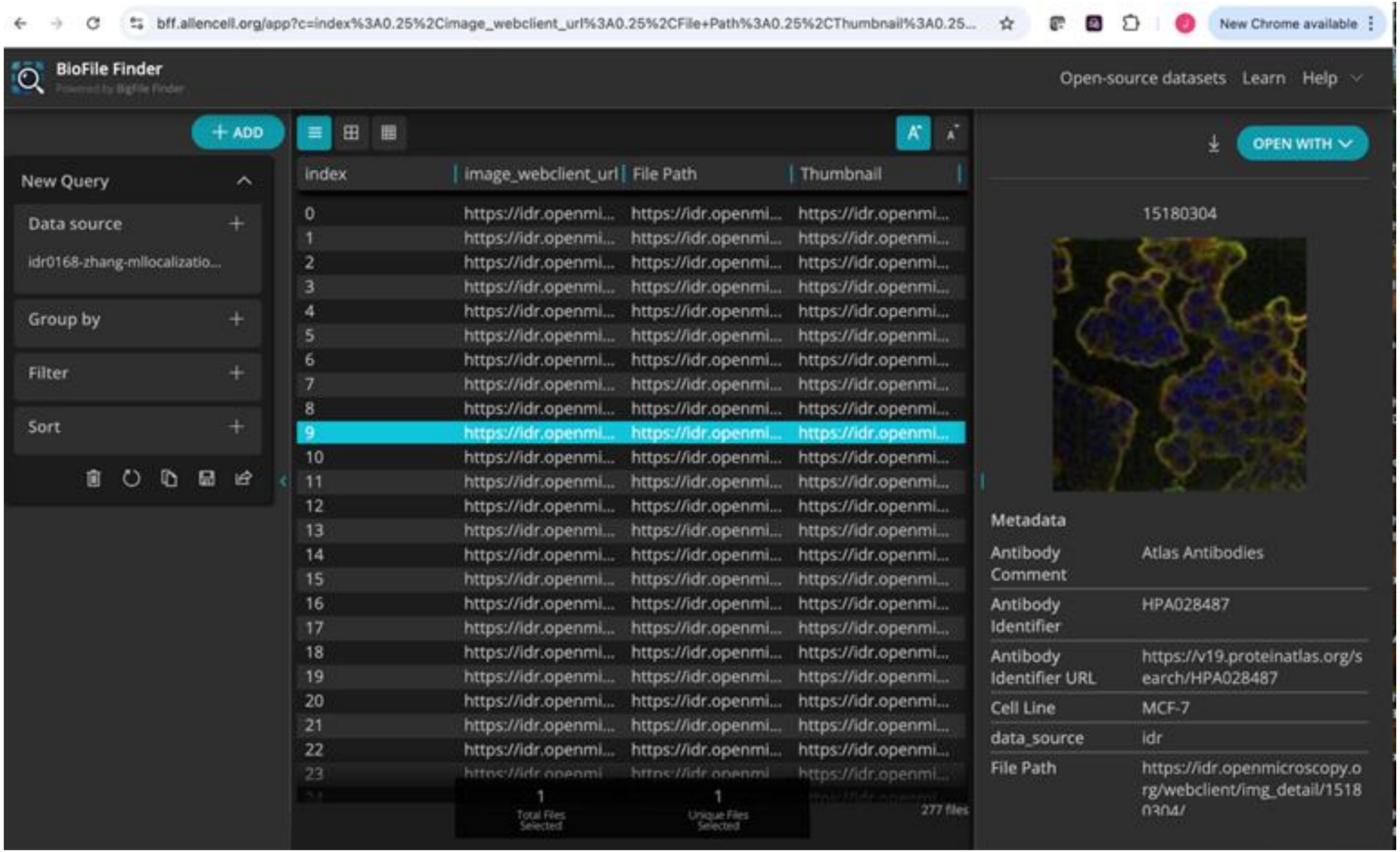
Don’t spread headings across multiple columns

Data model and metadata schema

What are the high level concepts to record, and how are they related?

For each high level concept, what information must we record, in what format, using what allowed values?

- **In a spreadsheet**: the column names, allowed values and rules.
- **In an ELN**: templates, required fields, sample links and metadata sections.
- **In a relational database**: tables, fields and relationships.
- **In a document database**: expected structure of each document, often with flexible optional fields.
- **In a graph database**: nodes and relationships



https://bff.allencell.org/

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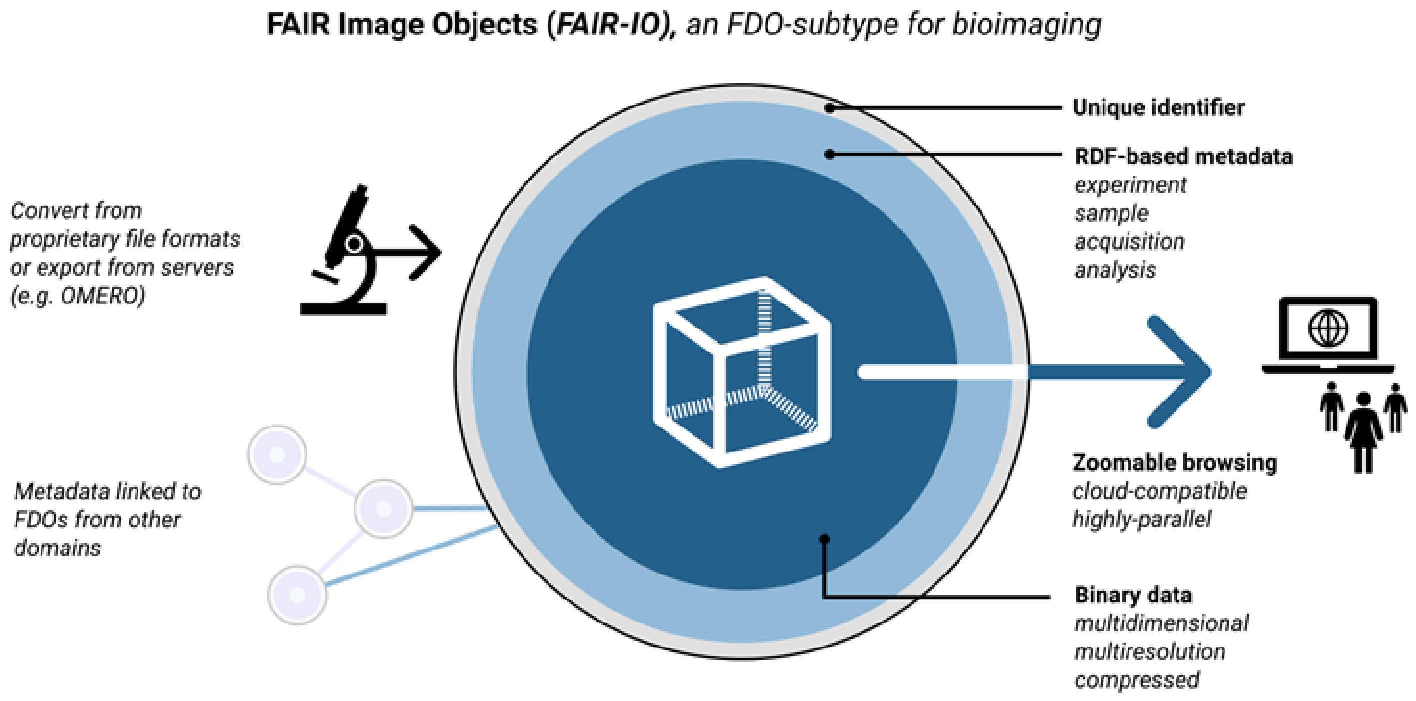
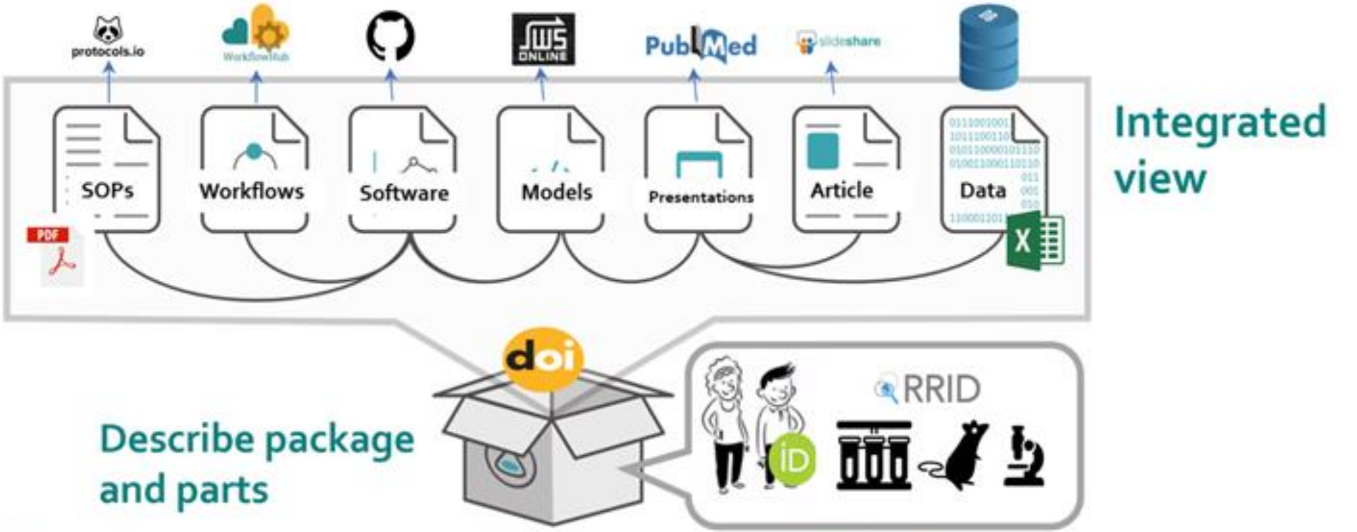
FAIR-IO (2023-2028)



https://nfdi4bioimage.github.io/FAIR-IO/specification/index.html

Research Object Crate (RO-Crate)

- A way to package data and metadata together:
  - Paper + authors + protocols + workflows + data
- It uses JSON-LD and [Schema.org](https://schema.org)
- It's a way to comply with the FAIR principles

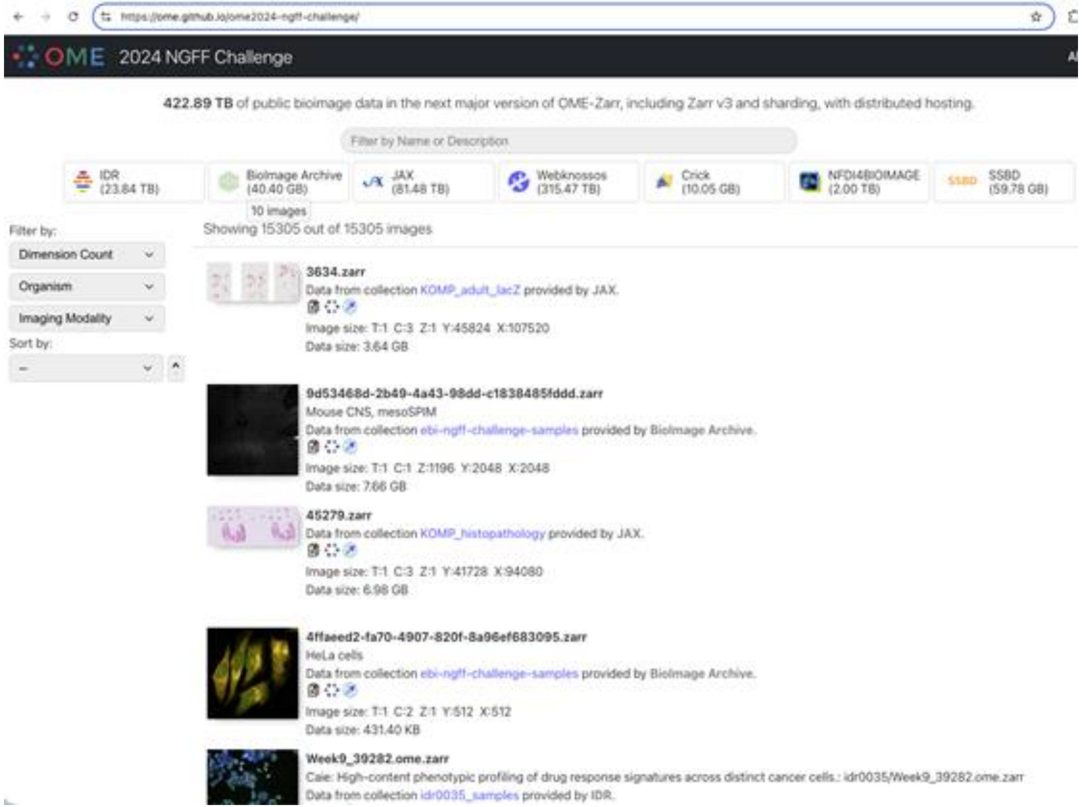


RO-Crates can link and be linked to research objects from many different sources using persistent identifiers like ORCIDs and DOIs.
Image credit: Goble, C. (2024, February 16). FAIR Digital Research Objects: Metadata Journeys. University of Auckland Seminar, Auckland. Zenodo. https://doi.org/10.5281/zenodo.10710142

Research Object Crate (RO-Crate)

Metadata and data sharing

- OME-NGFF 2024 challenge



https://ome.github.io/ome2024-ngff-challenge/

Research Object Crate (RO-Crate)

Metadata and data sharing

- OME-NGFF 2024 challenge
- FoundingGIDE search
- BIA submission (in progress)


```
{
  "@context": [
    "https://w3id.org/ro/crate/1.1/context",
    {
      "organism_classification": "https://schema.org/taxonomicRange",
      "BioChemEntity": "https://schema.org/BioChemEntity",
      "channel": "https://www.openmicroscopy.org/Schemas/Documentation/Generated/OME-
      "obo": "http://purl.obolibrary.org/obo/",
      "FBcv": "http://ontobee.org/ontology/FBcv/",
      "acquisiton_method": {
        "@reverse": "https://schema.org/result",
        "@type": "@id"
      },
      "biological_entity": "https://schema.org/about",
      "biosample": "http://purl.obolibrary.org/obo/OBI_0002648",
      "preparation_method": "https://www.wikidata.org/wiki/Property:P1537",
      "specimen": "http://purl.obolibrary.org/obo/HSO_0000308"
    }
  ],
  "@graph": [
    {
      "@id": "./",
      "@type": "Dataset",
      "name": "141-Sato-CellMorphology/Fig3a_FIB-SEM_synapse",
```

FoundingGIDE Search Portal

This search portal provides a single federated search interface over a number of biological imaging databases that contribute to gide. We currently display data from the Image Data Resource (IDR), SSBD:repository and SSBD:database, and the BioImage Archive (BIA).

https://embl-ebi.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=b2be292d-73d9-47e8-bbd2-b43200cfb63c

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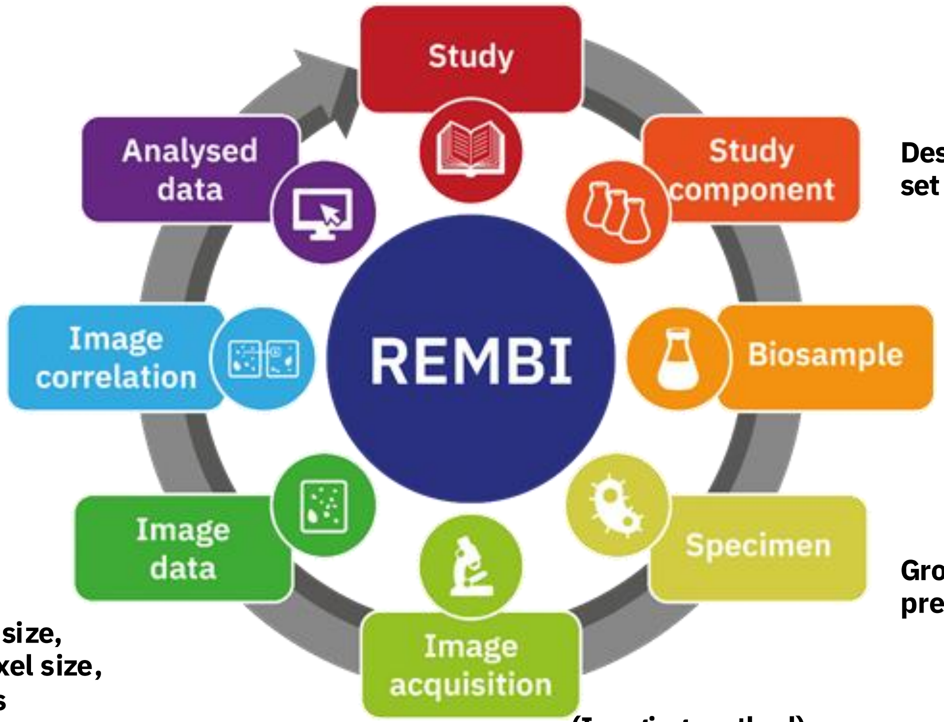
Let’s put things together: A possible solution

REMBI

Project description, funding and publications.

Analysis methods, workflow



Description of the data set

How you correlated images from different modalities or experiments

Species, tissue/cell/organism, genetic or extrinsic variables

Growth and preparation protocols

Physical size, Pixel/voxel size, Channels

(Imaging method), instrument, acquisition parameters

This is implemented as the File List for BioImage Archive submissions, instead of a separate component in the submission form.

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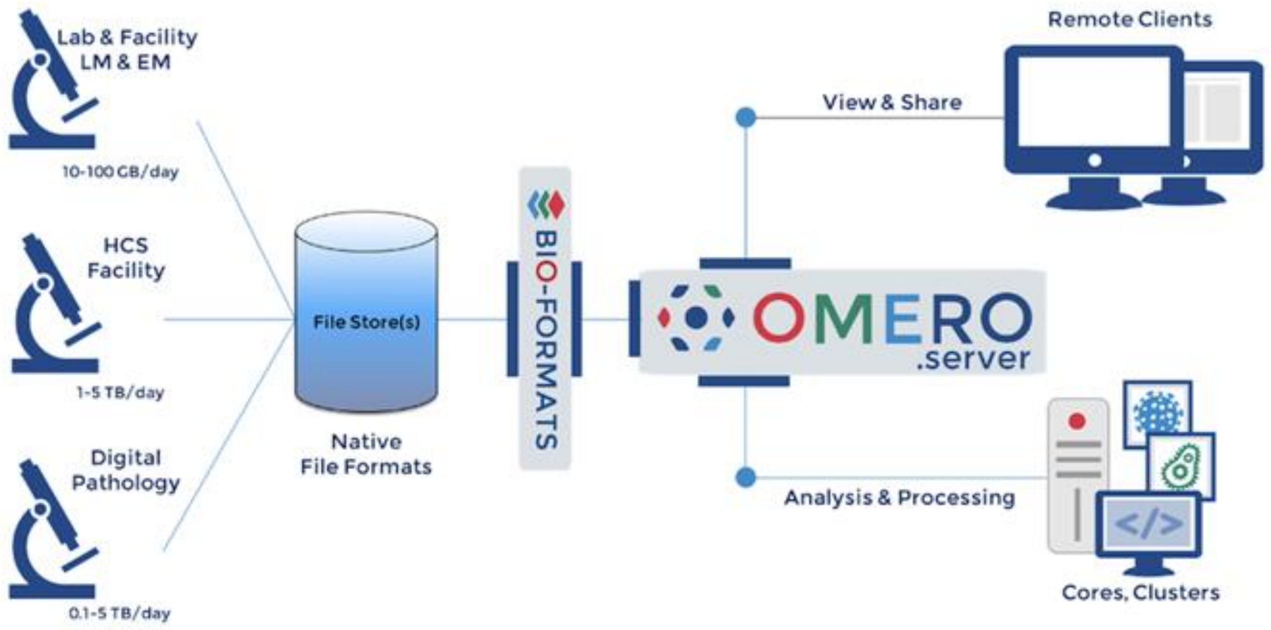
https://www.ebi.ac.uk/bioimage-archive/rembi-model-reference/

OMERO

Benefit: Full power of Database

Challenge: Custom API, bottleneck



Allan et al (2012) Nature Methods

OMERO
MATLAB
pandas
VISIOPHARM
R
learn
QuPath
scikit-image
TensorFlow
PyTorch
CellProfiler
cell image analysis software
ORBIT
KNIME
HALO

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Export and Transfer

- omero-cli-plugin: transfer object and annotations

Two objects to highlight:

`--barchive` creates a package compliant with BIA submission standards

`--rocrate` generates a RO-Crate compliant package with flat structure

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Installation

- **Manually**
  - Instruction for [RHEL 9] and [Ubuntu 24.04]

- **Docker**: containerisation
  - https://ome.github.io/training-docker/

- **Ansible**: configuration management
  - https://github.com/ome/prod-playbooks

Server Specification

- It depends on usage and type of data

- Training server:
  - 4 CPU 40GB RAM

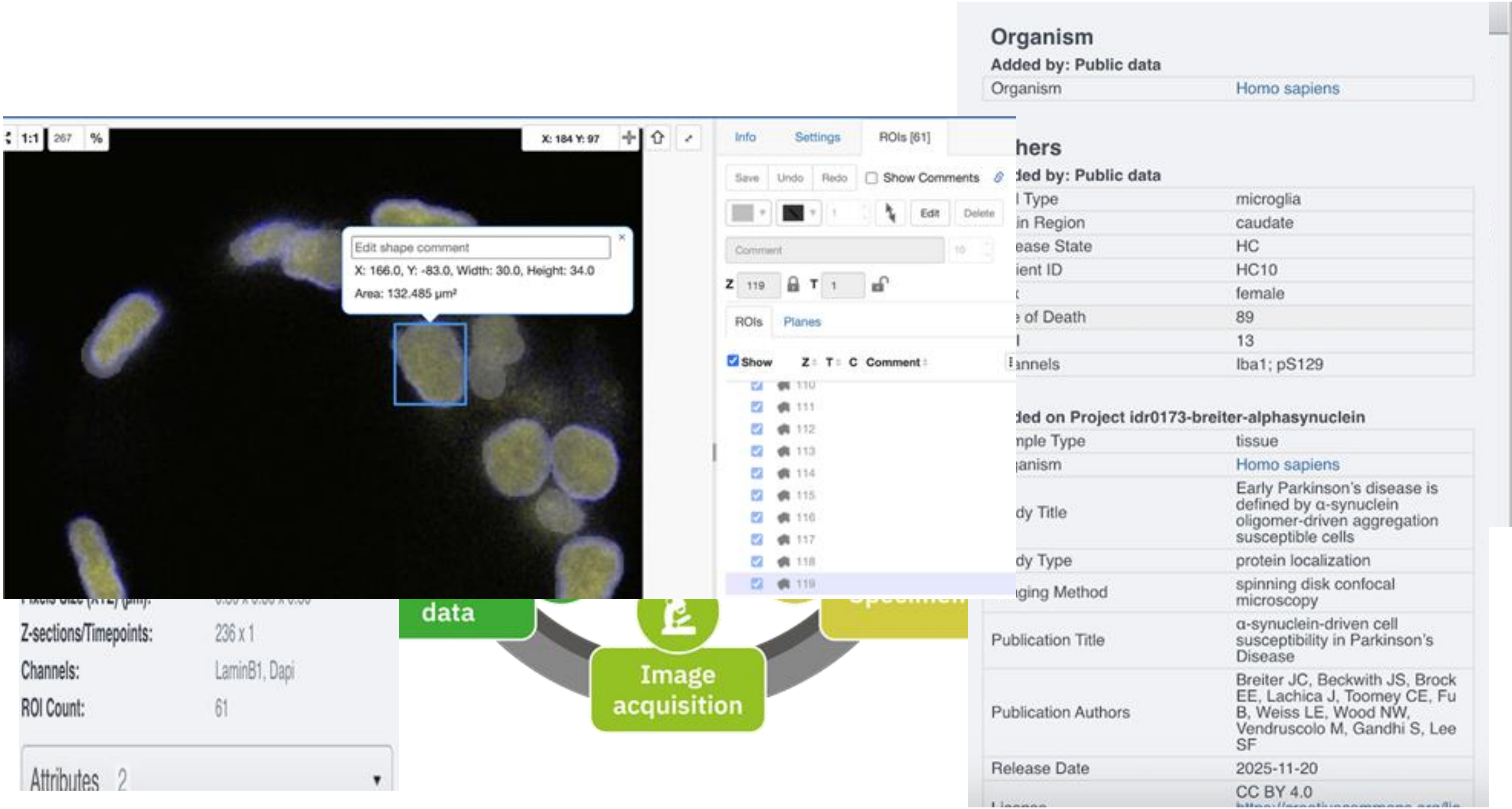
- Teaching server (Indonesia)
  - 16 CPU 100GB RAM

- Dundee server of Faculty of Life Sciences
  - 4 CPU 32GB RAM (server)
  - 2 CPU 15 GB RAM (web)

Exercise: Quick OMERO tour

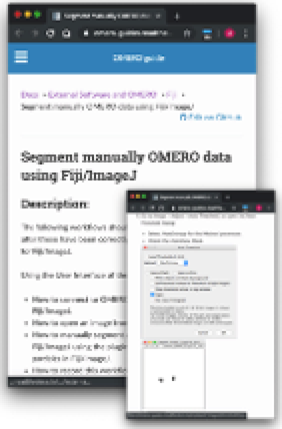
Server: https://workshop.openmicroscopy.org
Users: user-1, user-2, etc.
Password: harvard

Guide: https://omero-guides.readthedocs.io/en/latest/figure/docs/


Organism
Added by: Public data
Organism
Homo sapiens
hers
ded by: Public data
l Type
microglia
in Region
caudate
ease State
HC
ient ID
HC10
female
e of Death
89
13
annels
Iba1; pS129
ded on Project idr0173-breiter-alphasynuclein
nple Type
tissue
anism
Homo sapiens
dy Title
Early Parkinson's disease is defined by α-synuclein oligomer-driven aggregation susceptible cells
dy Type
protein localization
ging Method
spinning disk confocal microscopy
Publication Title
α-synuclein-driven cell susceptibility in Parkinson's Disease
Publication Authors
Breiter JC, Beckwith JS, Brock EE, Lachica J, Toomey CE, Fu B, Weiss LE, Wood NW, Vendruscolo M, Gandhi S, Lee SF
Release Date
2025-11-20
CC BY 4.0
1:1
%
X: 184 Y: 97
Info
Settings
ROIs [61]
Save
Undo
Redo
Show Comments
Edit
Delete
Comment
Edit shape comment
X: 166.0, Y: -83.0, Width: 30.0, Height: 34.0
Area: 132.485 µm²
Z
119
T
1
ROIs
Planes
Show
Z
T
C
Comment
111
112
113
114
115
116
117
118
119
Z-sections/Timepoints:
236 x 1
Channels:
LaminB1, Dapi
ROI Count:
61
Attributes 2
data
Image acquisition

OMERO.guides and Videos

- OMERO.guides – including community workflow contributions



https://omero-guides.readthedocs.io/en/latest/index.html

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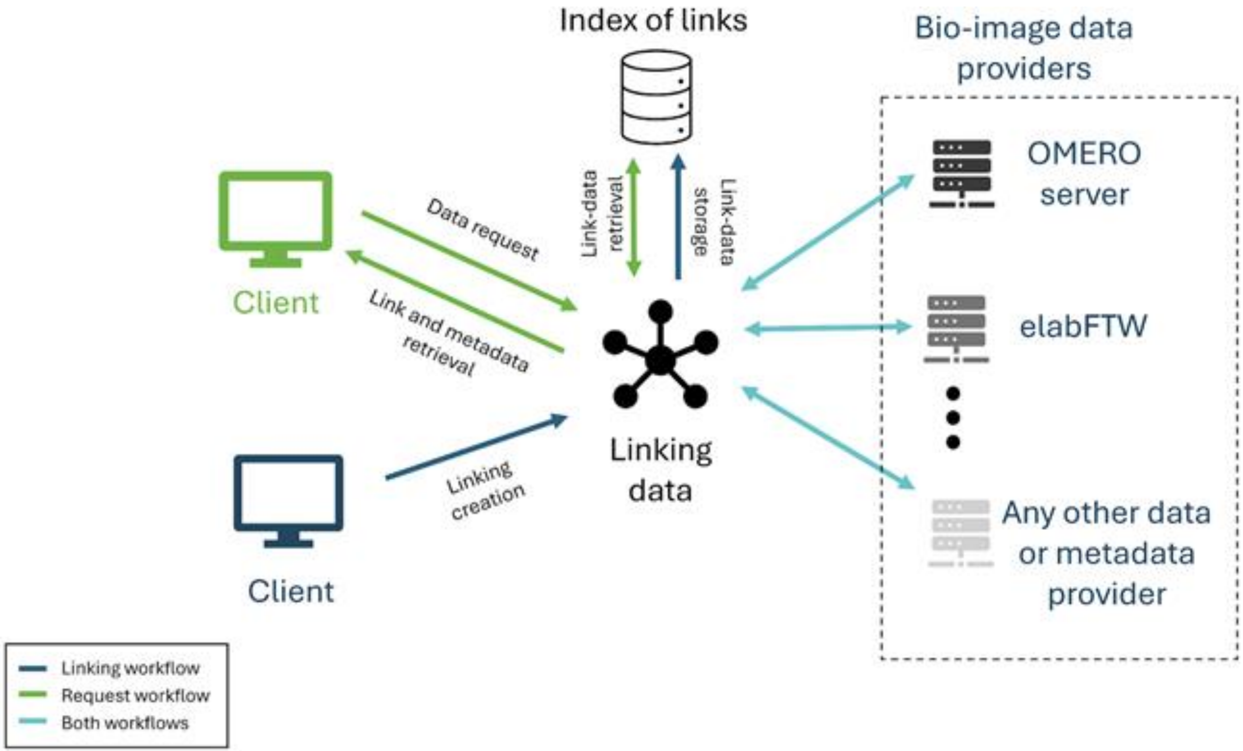
OMERO & ELN



https://github.com/NFDI4BIOIMAGE/LEO

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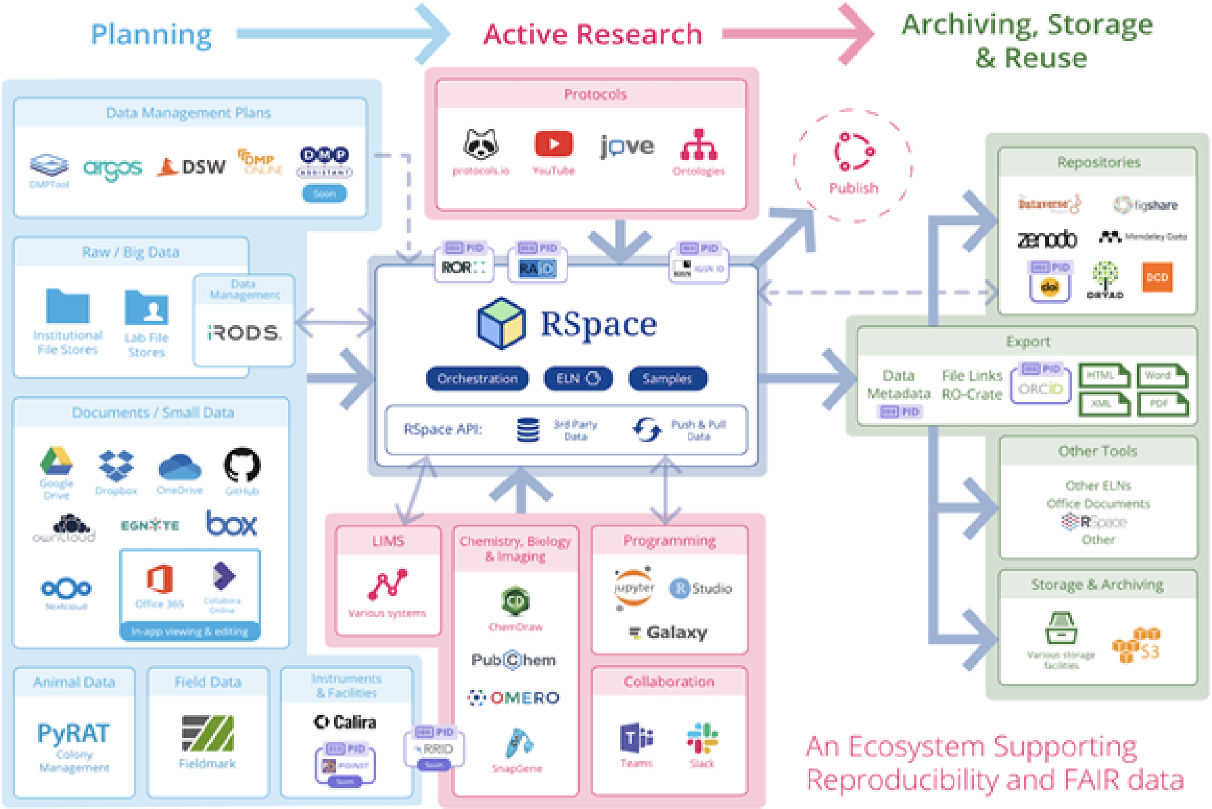
Rspace: Research Infrastructure Orchestrator



An Ecosystem Supporting Reproducibility and FAIR data

https://www.researchspace.com/

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Data management end goal: Successful real life examples

WEBINAR SERIES

How to organise and share my imaging data?

Multimodal data management for marine biologists, environmental scientists and imaging specialists

https://www.ebi.ac.uk/training/events/how-organise-and-share-my-imaging-data/

List of Webinars

1. Principles of research data management
2. Mastering data management: Go FAIR, Go Open, Go Smart — EMO BON and TREC showcases
3. Towards open and standardised imaging data: an introduction to Bio-Formats, OME-TIFF, and OME-Zarr
4. Data management in a bioimage informatics data flow
5. A FAIR and scalable workflow for plankton phenogenomics at single cell level
6. Open FAIR data: the role of public data archives
7. A journey to FAIR bioimage data

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A journey to FAIR bioimage data

Stefanie Weidtkamp-Peters, Heinrich-Heine-Universität Düsseldorf

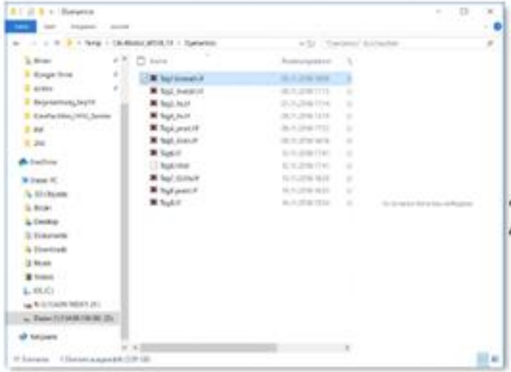
Tom Boissonnet, Heinrich-Heine-Universität Düsseldorf

Christian Schmidt, DKFZ Heidelberg

Info and recording: https://www.ebi.ac.uk/training/events/journey-fair-bioimage-data

Presentation slides: https://zenodo.org/records/15796252

What happens in between ...in reality?



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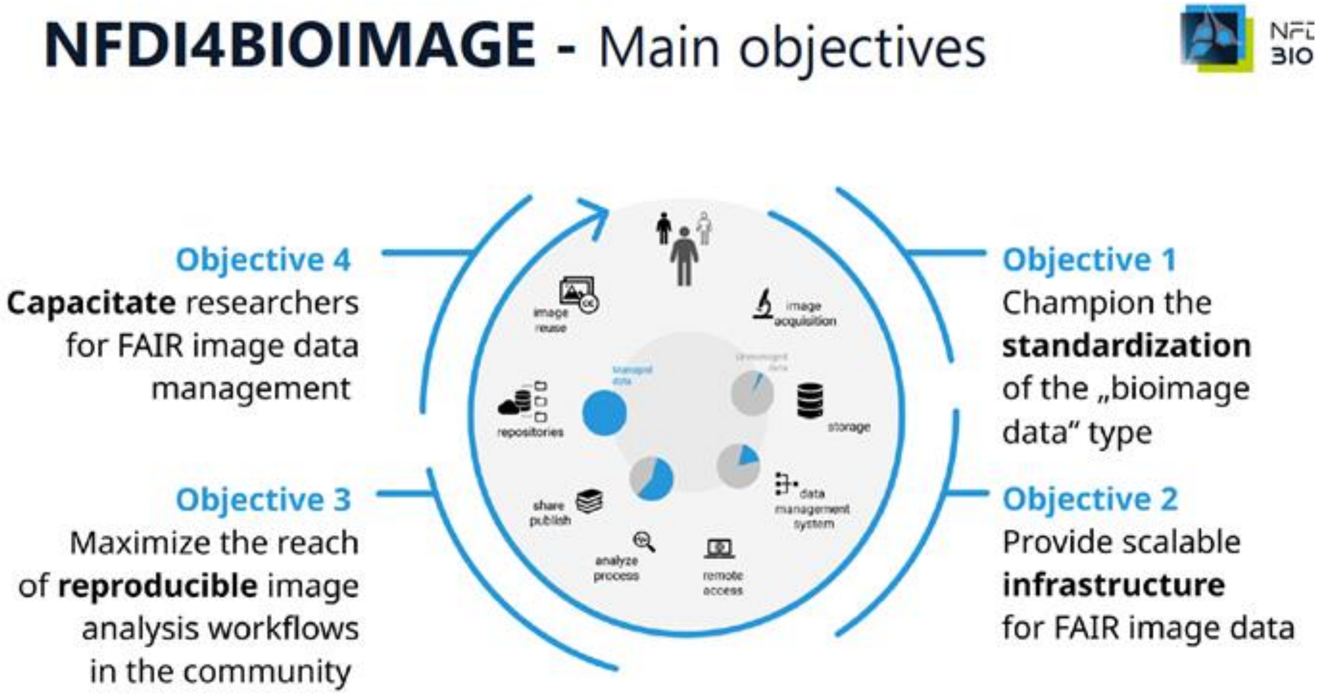
NFDI4BIOIMAGE - a consortium of the National Research Data Infrastructure



6 Task Area teams

Weidtkamp-Peters, S., Boissonnet, T., & Schmidt, C. (2025, July 3). [Webinar] A journey to FAIR bioimage data. Zenodo. https://doi.org/10.5281/zenodo.15796252

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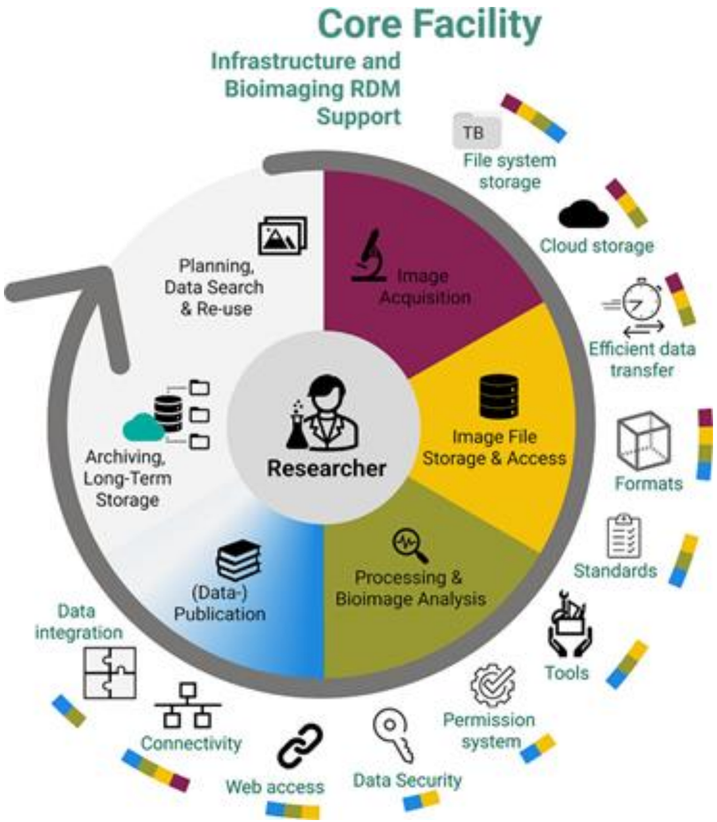
A practical guide to bioimaging research data management in core facilities



Schmidt, C., Boissonnet, T., Dohle, J., Bernhardt, K., Ferrando-May, E., Wernet, T., Nitschke, R., Kunis, S., & Weidtkamp-Peters, S. (2024). A practical guide to bioimaging research data management in core facilities. Journal of Microscopy , 294, 350–371.
https://doi.org/10.1111/jmi.13317

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A practical guide to bioimaging research data management in core facilities

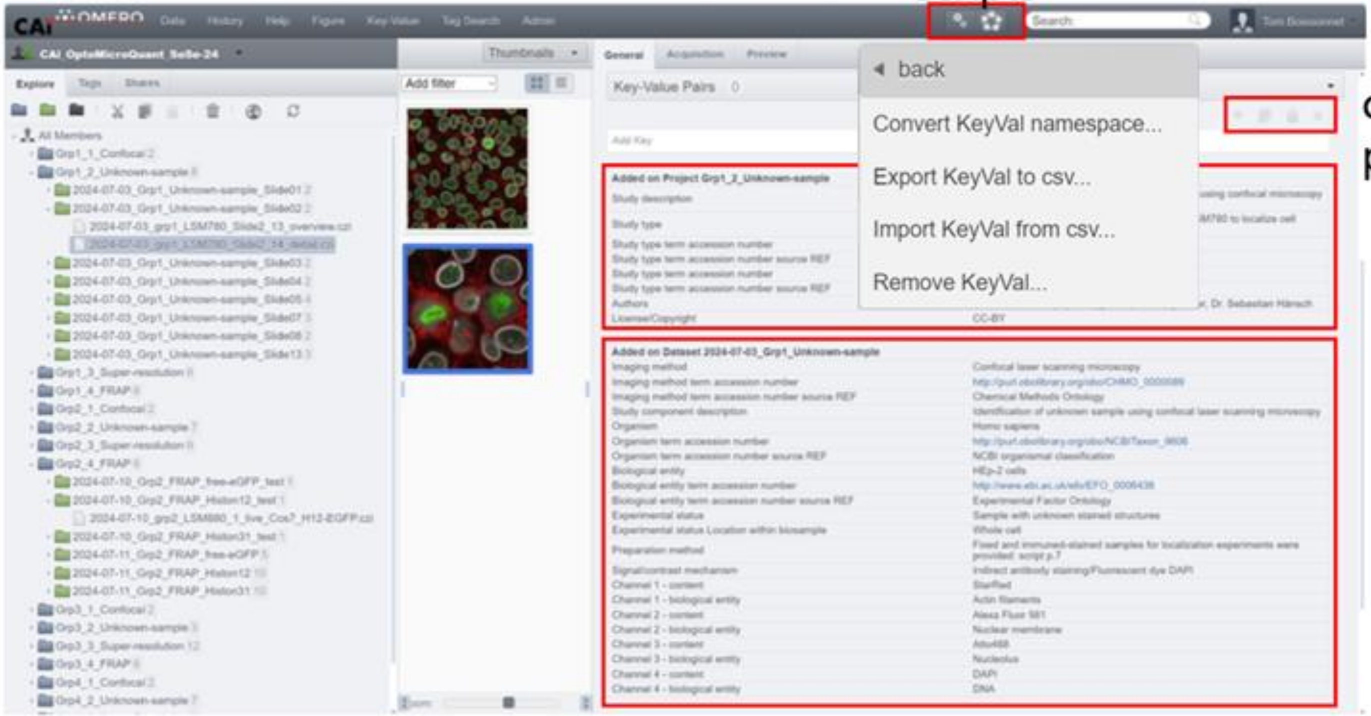
TABLE 1 Common issues and solutions regarding bioimage data management planning.

Common issue	Proposed solution
Underestimating the complexity of bioimage data handling	Consulting and taking advice from core facility staff. Using DMP templates that have been approved or provided by core facilities.
Reusing existing data is not considered	Data search and review in public repositories and institutional archives can become part of experiment planning. Reusing data from repositories can help users familiarise themselves with the data types and test workflows before acquiring their own data. Suitable archives can be found on re3data.org or on FAIRsharing.org. Bioimaging-specific archives include, for example, the BioImage Archive (BIA) and the Image Data Resource (IDR). Institutional archives may be another data source.
Lack of adaptation strategies for changing demands throughout the experiment	Writing a data management plan and using it as a living document to adapt where required, and record changes.
Suboptimal instrument choice	User training and guidance on instrument choice and setup, as well as recommendations on sample preparation. Consulting the core facility and taking advice on instrument and software choice.
Image analysis and data processing are not planned prior to the acquisition	Consulting with core facility staff, image analysts and data stewards before starting a new project. Searching example data and protocols that have established the desired procedure previously to test image analysis a priori. Planning suitable data structures and metadata for analysis and running a pilot experiment before adjusting and scaling up.
The data volumes, storage space and data transfer requirements are not considered before the acquisition	Gathering comprehensive information about typical image sizes of the core facility's instruments. Highlight limits to memory load and data transfer with large files. Estimating expected data volumes and required storage and ensuring resource availability.
The importance of metadata collection is overlooked	Preparing a metadata annotation checklist for the data even before acquisition, for example, based on REMBI. Optionally, enforce annotation as part of a usage policy.
Lack of a strategy for the publication of image data	Considering data publication, candidate public repositories, and reviewing submission guidelines. Defining authorship and licensing criteria with collaboration partners early in the project.

Schmidt, C., Boissonnet, T., Dohle, J., Bernhardt, K., Ferrando-May, E., Wernet, T., Nitschke, R., Kunis, S., & Weidtkamp-Peters, S. (2024). A practical guide to bioimaging research data management in core facilities. Journal of Microscopy , 294, 350–371. https://doi.org/10.1111/jmi.13317

Working with images — key-value pairs

scripts



copy
paste

Recommended Metadata for Biological Images, Sarkans et al, Nature Methods 2021

Weidtkamp-Peters, S., Boissonnet, T., & Schmidt, C. (2025, July 3). [Webinar] A journey to FAIR bioimage data. Zenodo. https://doi.org/10.5281/zenodo.15796252

Release Date: 18 June 2025 • Modified: 18 June 2025 • Views: 2

[Cite] {JSON} →PageTab↓ HTTP FTP Globus

Fluorescence lifetime separation of fluorophores in C. elegans embryos and hermaphrodites

Cornelia Wetzker [1], Marcelo Leomil Zoccoler [1], Svetlana Iarovenko [2], Chukwuebuka William Okafornta [1], Anja Nobst [1], Hella Hartmann [1], Thomas Müller-Reichert [1], Robert Haase [3], Gunar Fabig [1]

[1] Technische Universität Dresden [2] Institute of Molecular Biotechnology of the Austrian Academy of Sciences, Vienna [3] Leipzig University

Accession S-BIAD1967

DOI 10.6019/S-BIAD1967

ReleaseDate 2025-06-18

Description Fluorescence lifetime imaging microscopy (FLIM) translates the duration of excited states of fluorophores into lifetime information as additional source of contrast in images of biological samples. This offers the possibility to separate fluorophores particularly beneficial in case of similar excitation spectra. Here, we demonstrate the distinction of fluorescent molecules based on FLIM phasor analysis, called lifetime separation, in live-cell imaging using open-source software for analysis. We showcase two applications using Caenorhabditis elegans as a model system. First, we separated the highly spectrally overlapping fluorophores mCherry and mKate2 to distinctively track tagged proteins in six-dimensional datasets to investigate cell division in the developing early embryo. Second, we separated fluorescence of tagged proteins of interest from masking natural autofluorescence in adult hermaphrodites. For FLIM data handling and workflow implementation, we developed the open-source plugin napari-FLIM-phasor-plotter to implement conversion, visualization, analysis and reuse of FLIM data of different formats. Our work thus advances technical applications and bioimage data management and analysis in FLIM microscopy for life science research.

Keywords FLIM, live-cell imaging, phasor, napari

Acknowledgements Imaging was performed at the Light Microscopy Facility at the Center for Molecular and Cellular Bioengineering (CMCB-LMF) of TU Dresden. We wish to acknowledge the support of the CMCB-LMF team, particularly Ellen Geibelt for technical support and maintenance of the microscope, and Silke Tulok (Core Facility Cellular Imaging at the Faculty of Medicine at TU Dresden) for support in bioimage data management using OMERO. We thank the Bio-Image Analysis Technology Development team of Physics of Life at TU Dresden (BiA-PoL) for provision of infrastructure and fruitful discussions, Tom Boissonnet (Heinrich Heine Universität Düsseldorf, I3D:bio) for support with metadata annotation and Josh Moore (German BioImaging - Gesellschaft für Mikroskopie und Bildanalyse e.V.) for feedback on the resulting manuscript.

Funding statement C.W. is funded by the German consortium NFDI4BIOIMAGE (Deutsche Forschungsgemeinschaft, grant number NFDI 46/1, project number 501864659). Research in the Müller-Reichert lab is supported by grants

Data files

Show 5 entries Search:

Name
TMR17_2_sptw_summed_intensity_with_segme
TMR17_3_sptw_summed_intensity_with_segme
TMR31_1_sptw_summed_intensity_with_segme
TMR31_2_sptw_summed_intensity_with_segme
TMR31_3_sptw_summed_intensity_with_segme

EMBL-EBI

Release Date: 18 June 2025 • Modified: 18 June 202

Fluorescence lifetime s hermaphrodites

Cornelia Wetzker [1], Marcelo Leomil Zoccoler [1], Svetl
, Thomas Müller-Reichert [1], Robert Haase [1], Gunar Fabi

[1] Technische Universität Dresden [2] Institute of Molecular Bio

Accession S-BIAD1967

DOI 10.6019/S-BIAD1967

ReleaseDate 2025-06-18

Description Fluorescence lifetime imaging m
fluorophores into lifetime information as addit
possibility to separate fluorophores particularl
demonstrate the distinction of fluorescent mol
live-cell imaging using open-source software f
elegans as a model system. First, we separate
to distinctively track tagged proteins in six-dim
embryo. Second, we separated fluorescence o
adult hermaphrodites. For FLIM data handling
napari-FLIM-phasor-plotter to implement con
formats. Our work thus advances technical ap
microscopy for life science research.

Keywords FLIM, live-cell imaging, phasor, i

Acknowledgements Imaging was performed a
Bioengineering (CMCB-LMF) of TU Dresden. W
particularly Ellen Geibelt for technical support
Cellular Imaging at the Faculty of Medicine at
We thank the Bio-Image Analysis Technology I
provision of infrastructure and fruitful discussi
for support with metadata annotation and Josl
Bildanalyse e.V.) for feedback on the resulting

Funding statement C.W. is funded by the Germ
grant number NFDI 46/1, project number 501

Release Date: 12 September 2025 • Modified: 19 September 2025

[Cite] {JSON} →PageTab HTTP FTP Globus

Interdisciplinary Study on Drug-Induced-Phospholipidosis of Repurposing Libraries through Machine Learning and Experimental Evaluation in Different Cell Lines

Maria Kuzikov [1], Adelinn Kalman [2], Jeanette Reinshagen [1], Johanna Huchting [1], Kun Qian [3], Hanna Axelsson [3], Marianna Tampere [2], Päivi Östling [2], Brinton Seashore-Ludlow [2,3], Yojana Gadiya [4,1], Philip Gribbon [1], Andrea Zaliani [1], Jens Wendt [5,6]

[1] Fraunhofer Institute for Translational Medicine and Pharmacology [2] Department of Oncology-Pathology, Karolinska Institutet, Science for Life Laboratory, Solna, Sweden [3] Department of Medical Biochemistry and Biophysics, Karolinska Institutet, Science for Life Laboratory, Solna, Sweden [4] Bonn Aachen International Center for Information Technology [5] University of Münster [6] NFDI4Bioimage

Accession S-BIAD2282

DOI 10.6019/S-BIAD2282

ReleaseDate 2025-09-12

Description Phospholipidosis (PLD), a cellular adverse effect that is, among others, caused by numerous cationic amphiphilic drugs. Interest is raised within pharma discovery to predict this phenomenon, as it can impact the outcome of phenotypic cellular screens and significantly delay drug development processes. The development of accurate and validated machine learning models for predicting drug-induced PLD across different cell lines and research centers could provide a valuable early application tool for the pharmaceutical industry, potentially accelerating drug discovery and reducing the risk of late-stage failures. We report here the assembly, curation, testing and modeling of one of the largest datasets of repurposed drugs (5000+) tested for PLD induction on different cell lines. A machine-learning classification method was developed and validated to predict whether molecules are prone to induce PLD effects when applied in cell-based screens.

This constitutes the accompanying publication of the acquired image data.

Keywords Phospholipidosis, HCS, Machine learning, Drug repurposing

Acknowledgements We thank Jens Wendt (NFDI4BIOIMAGE, DFG project ID 501864659) for help with metadata annotation and upload to the BioImage Archive repository.

Funding statement This work and the authors were primarily funded by the EU-Horizon HLTH Remedi4ALL (EU 101057442).

Funding

Horizon Europe: 101057442

Vero-E6 Kidney cells from African green monkey

Data files

Show 5 entries Search:

Name	Size	Section	sourc
ESP0030315.zarr.zip	12.21 GB	Study Component	
ESP0030316.zarr.zip	12.19 GB	Study Component	
ESP0030317.zarr.zip	12.08 GB	Study Component	
ESP0030460.zarr.zip	12.83 GB	Study Component	
ESP0030461.zarr.zip	12.75 GB	Study Component	

Showing 26 to 30 of 32 entries

Previous 1 2 3 4 5 6 7 Next

Download all files

ORCID: Data claiming

You can sign-in to ORCID to claim your data

Remember me on this computer

Similar Studies

- PolyA tail segmentation improves the stability of the template DNA and increases the translatability of in vitro transcribed mRNA. [S-BIAD2347]

EMO BON: European Marine Omics Biodiversity Observation Network

https://www.ebi.ac.uk/training/events/mastering-data-management-go-fair-go-open-go-smart-emo-bon-and-trec-showcases/

Centralised data management via GitHub

CENTRALISED data space to deal with DISTRIBUTED data

- Allowing each data team to use formats familiar to them
- But adding interoperability to that via metadata, standards, templates, formatting

Why GITHUB?

- GitHub offers version control, automated actions, mini-webpages, issue tracking, data (re)formatting, etc
- Workflows to reformat data as required for external publishers
- For large files: need to be archived somewhere else, but via LFS can be dealt with as if are on GitHub

DATA PACKAGING via Research Object Crate (RO-Crate)

- Packaging research data with their metadata
- Who, what, where, when, why, how and internal/external data linking

VLIZ

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DOI: 10.6019/TOL.DataManagement-EMOBON-TREC-w.2025.00001.1

End-user best practices for data organisation

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Folder structure tips

- Use folders to separate raw from processed/analysed data
- It is alway useful to add a README.md file to guide people on the contents and structure of that folder and subfolder(s)
- Organize your files in a logical structure that reflect the relationships and hierarchy between these files
- Use file naming conventions that will make it easier to identify and retrieve the data

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File naming tips

- Try and use meaningful names related to the data
- Provide a version of the file
  - Use a minimum of 2 digits
  - Avoid using _final, _final_final
  - Don’t use initials to version a file
  - Use version control e.g. GitHub or Gitlab
- Try and avoid using personal names
  - e.g. mut1_confocal_GFP_Melina
- Add date at the beginning if you want to sort out chronologically
  - Follow the YYYY-MM-DD format
- Avoid using special characters or accents
  - e.g. ‘+’ will cause problems in a BIA submission
- Try and avoid using white spaces to separate words.
  - Best to use underscores, or camelCase
- Avoid extra long names
- If there are known abbreviations in the field, use them

Good data management practices

- Folder and file structures are good for data organisation but it is not data management.
- Folder structures and file names are **NOT** your metadata storage.

  - Filenames have a character limit, you can’t put all metadata in a filename.
  - Folder structures and filenames are not accountable and changes are not trackable. They don’t provide provenance.
  - What happens if you or someone else move the data or accidentally rename the data?
  - What happens if you forget what the abbreviation you put in meant?

Storage and sharing strategies and tools

Hot, Warm or Cold Storages: Which one to choose?

- **Hot Storage:** Refers to data that is frequently used and accessed. The problem does not go away with Cloud storage.

- **Cold Storage:** is the opposite; this is data that you want to keep (probably off site and on slower equipment), but you rarely need to access. Less expensive to store than hot data. Cold storage could sit off-site on a service like [Amazon Glacier]

- **Warm Storage:** Something in the middle. Data stored close enough for easy access, but with fewer financial hits than your hot storage data e.g Network-attached Storage or a remote file server e.g. SharePoint.

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	Hot	Warm	Cold
Access Speed	Fast	Slower	Slowest
Access Frequency	Regular	Irregular	Rare
Storage Media	Standard Hard drive, easy access cloud storage	Network Attached storage (NAS), remote file server	Archive Cloud storage (e.g. Amazon Glacier), Tape
Cost	Higher	Medium	low
Ideal User	Everyone	Users with lots of data	Users with lots of data

Aspects to consider for storage strategy

- **Scalability**
- **Data access frequency**
- **Data retention requirements**
- **Budget**
- **Data security**

SPEAK TO YOUR IT. DO NOT DO IT YOURSELF

Sharing data

- Public repositories e.g. BIA, IDR, EMPIAR. See https://rdmkit.elixir-europe.org/bioimaging_data
- Host your own OMERO instance and publish data using public account
- Share your data as OME-Zarr - best works with hot storage (fast cloud)

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Sharing data - transfer methods and tools

- Ftp - command line/programmatic, or apps like FileZilla, Cyberduck
- Globus - basic local transfers via free personal account; server/HPC integration and managed sharing require institutional setup
- Tools requiring institutional setup and payment, such as Aspera
- Hot storage/cloud s3
  - Http - you don't need to download or transfer the data to access and analyse or view it.
  - AWS Console
  - command line tools (aws s3, rclone etc.)
  - Applications like CyberDuck

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Data formats - Standardisation and interoperability

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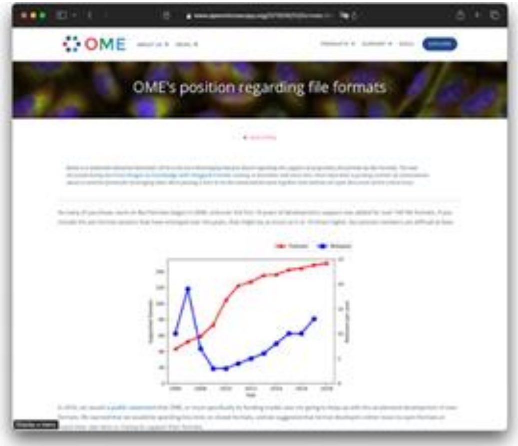
BIO-FORMATS

Benefit: Enable Users to get the work done

Challenge: Encourage the development of more formats

Strategy: Work on common format



"Put simply the formats landscape has scaled beyond a manageable level (2019)"

https://www.openmicroscopy.org/2019/06/25/formats.html

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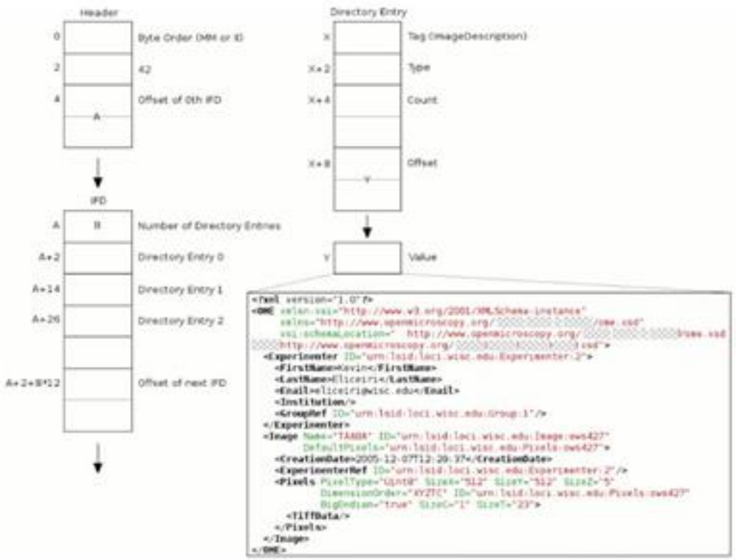
OME-TIFF Common Exchange Format



https://ome-model.readthedocs.io/en/stable/ome-tiff/specification.html

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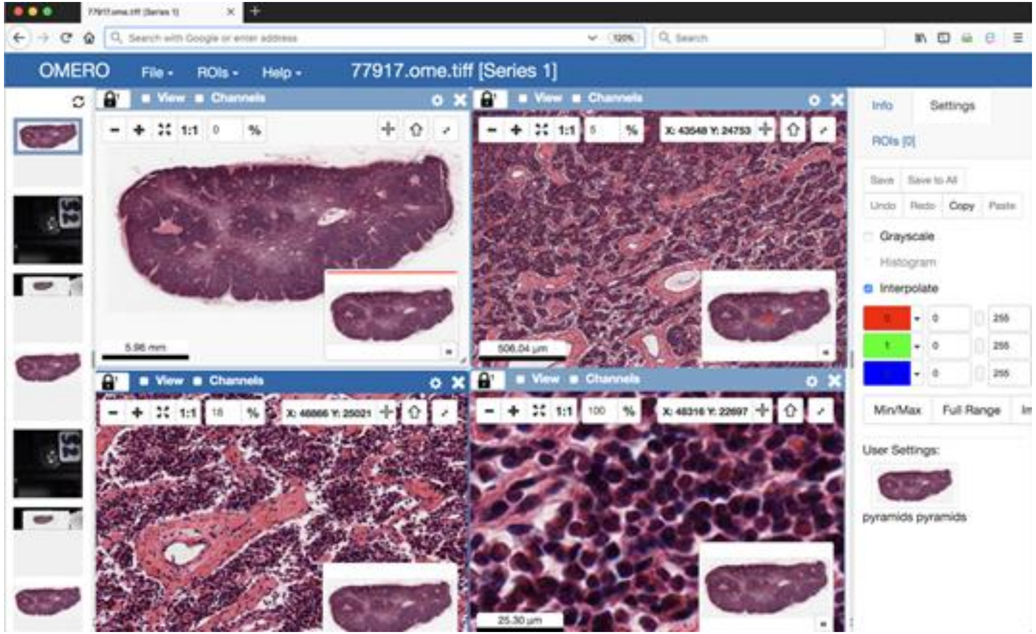
OME-TIFF (Pyramidal) Common Exchange Format

Benefit: Pre-calculated multi-resolution

Challenge: Limited annotation options



https://www.openmicroscopy.org/2018/11/29/ometiffpyramid.html

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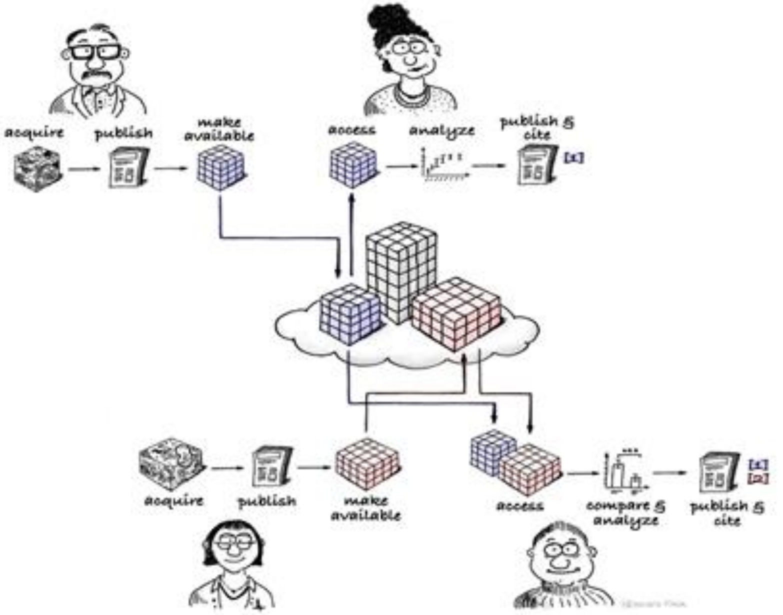
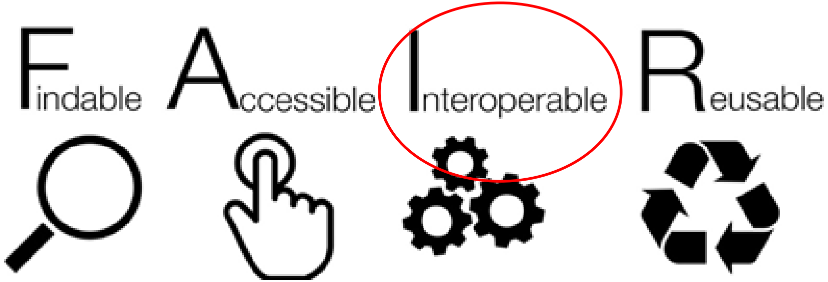
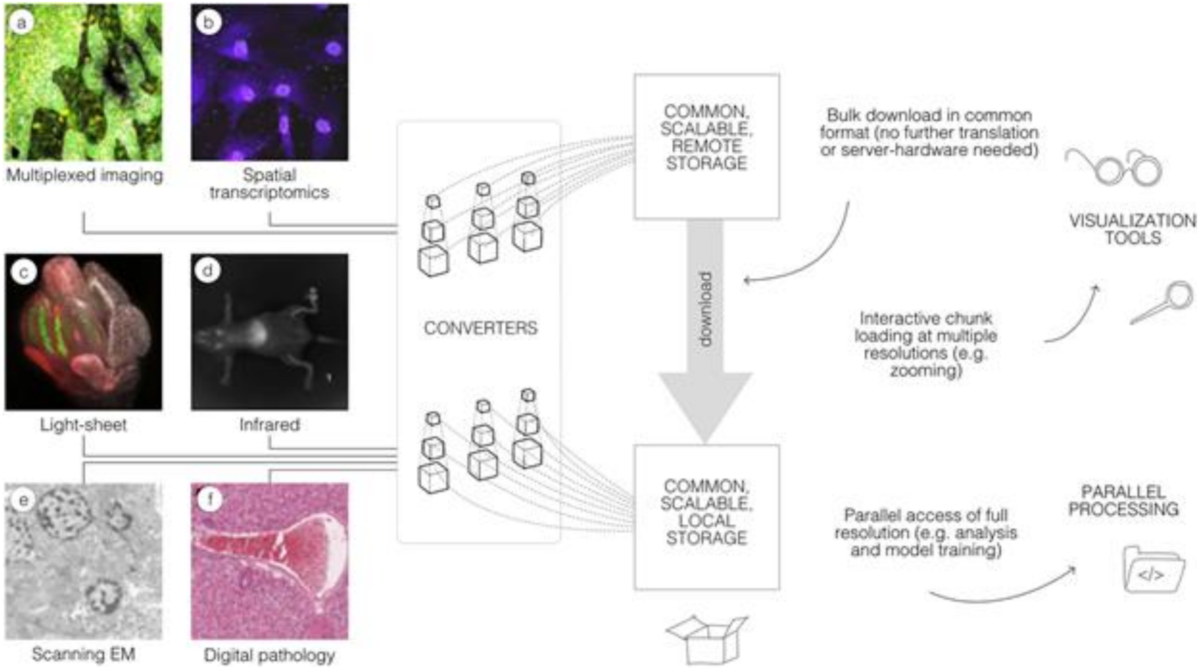
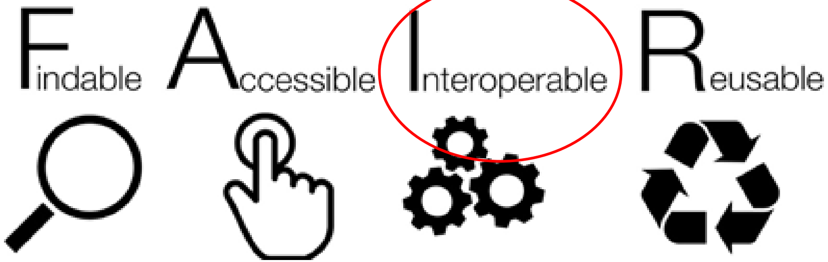
Challenge: FAIR principles even for largest images

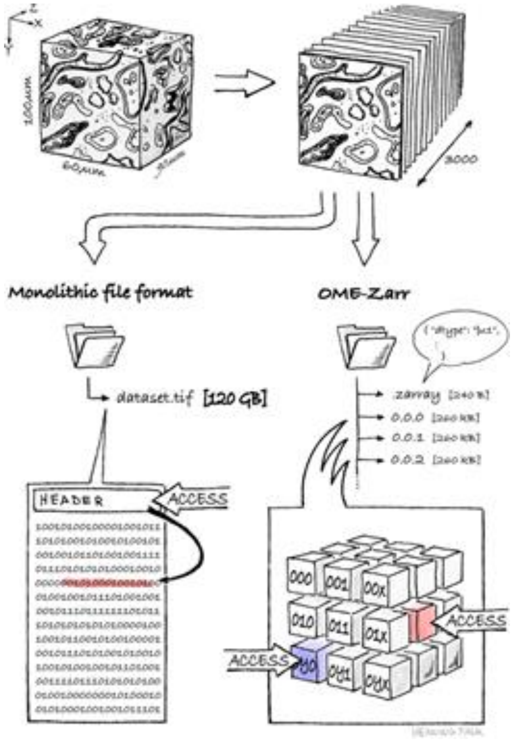






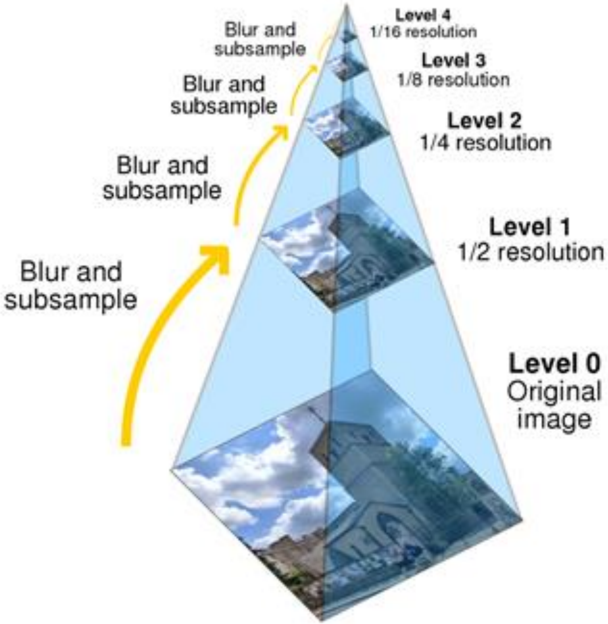
OME-Zarr: a cloud-optimized bioimaging file format with international community support, Moore et al. 2023





Next generation file format:

chunking + pyramids



Why would I use OME-Zarr?

1. **Chunking**: “Zarr” files are stored in independently-accessible blocks.

1. **OME-Zarr**: “Zarr” with embedded standardized metadata.

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When should I not use OME-Zarr?

1. Files are not big and you are working locally.

1. If data is lossy compressed e.g. SVS => Huge Increase in the number of files.

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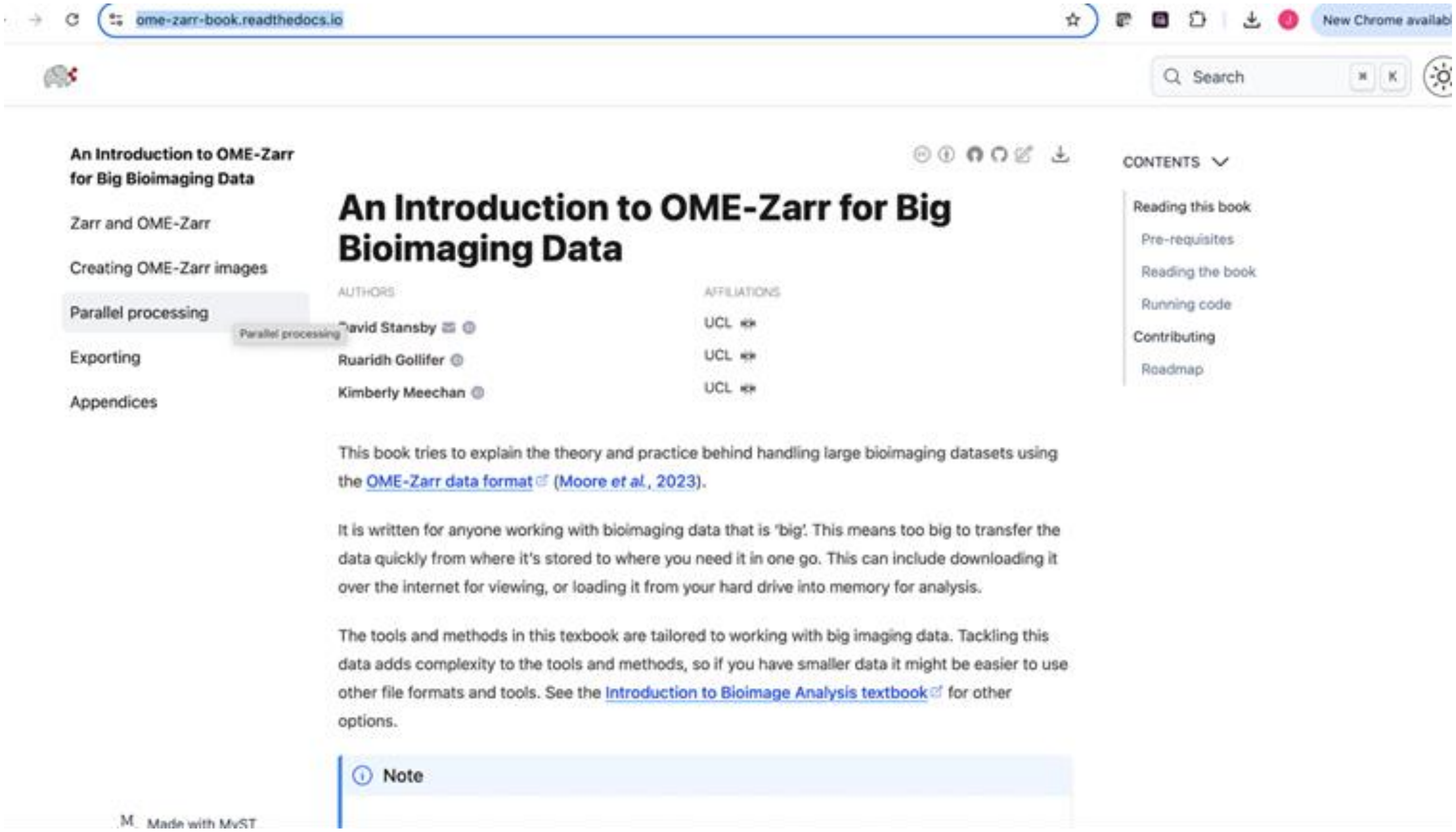
When do know more about it?



https://ome-zarr-book.readthedocs.io/

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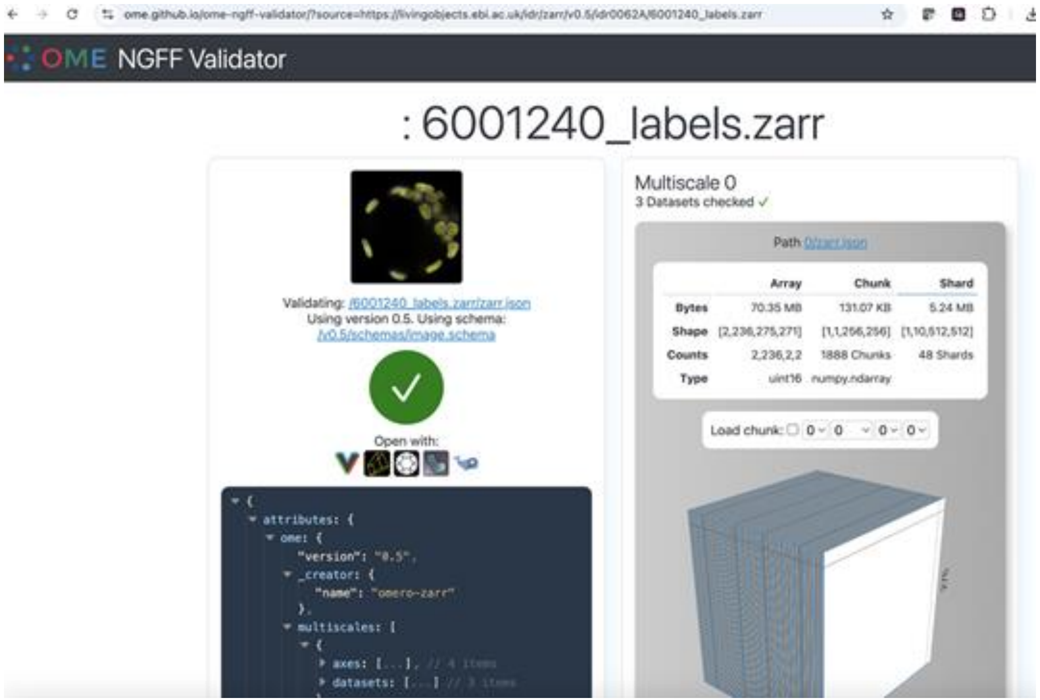
How does it look?



https://ome.github.io/ome-ngff-validator/?source=https://livingobjects.ebi.ac.uk/idr/zarr/v0.5/idr0062A/6001240_labels.zarr

Zarr converter with UI

Name	Link	Description
NGFF-Converter		A desktop application for conversion of bioimage formats into OME-Zarr or OME-TIFF.

https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-converters-with-a-ui

Zarr converters

Name	Link	Description
BatchConvert		A Nextflow based command-line tool that wraps bioformats2raw for parallelised conversion of image data collections to OME-Zarr.
bioformats2raw		Java application to convert image file formats, including .mrxs, to an intermediate Zarr structure compatible with the OME-Zarr specification.
BioIO Conversion		CLI & scripting tool for easily converting images to OME-Zarr. Requires bioio-ome-zarr
EuBI-Bridge		A tool for distributed conversion of microscopic image collections into the OME-Zarr format.
ITKIOOMEZarrNGFF		An ITK external module for IO of images stored in OME-Zarr file format.
multiscale-spatial-image		Generate a multiscale, chunked, multi-dimensional spatial image data structure that can serialized to OME-Zarr.
Nextflow (nf-omezarr)		A Nextflow pipeline for converting directories of images to OME-Zarr using bioformats2raw
stack-to-chunk		A Python library to convert stacks of 2D images to OME-Zarr with minimal memory use and maximum concurrency.
stack_to_multiscale_ngff		A tool for converting multi-terabyte stacks of images into a multiscale OME-Zarr.

https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-converters

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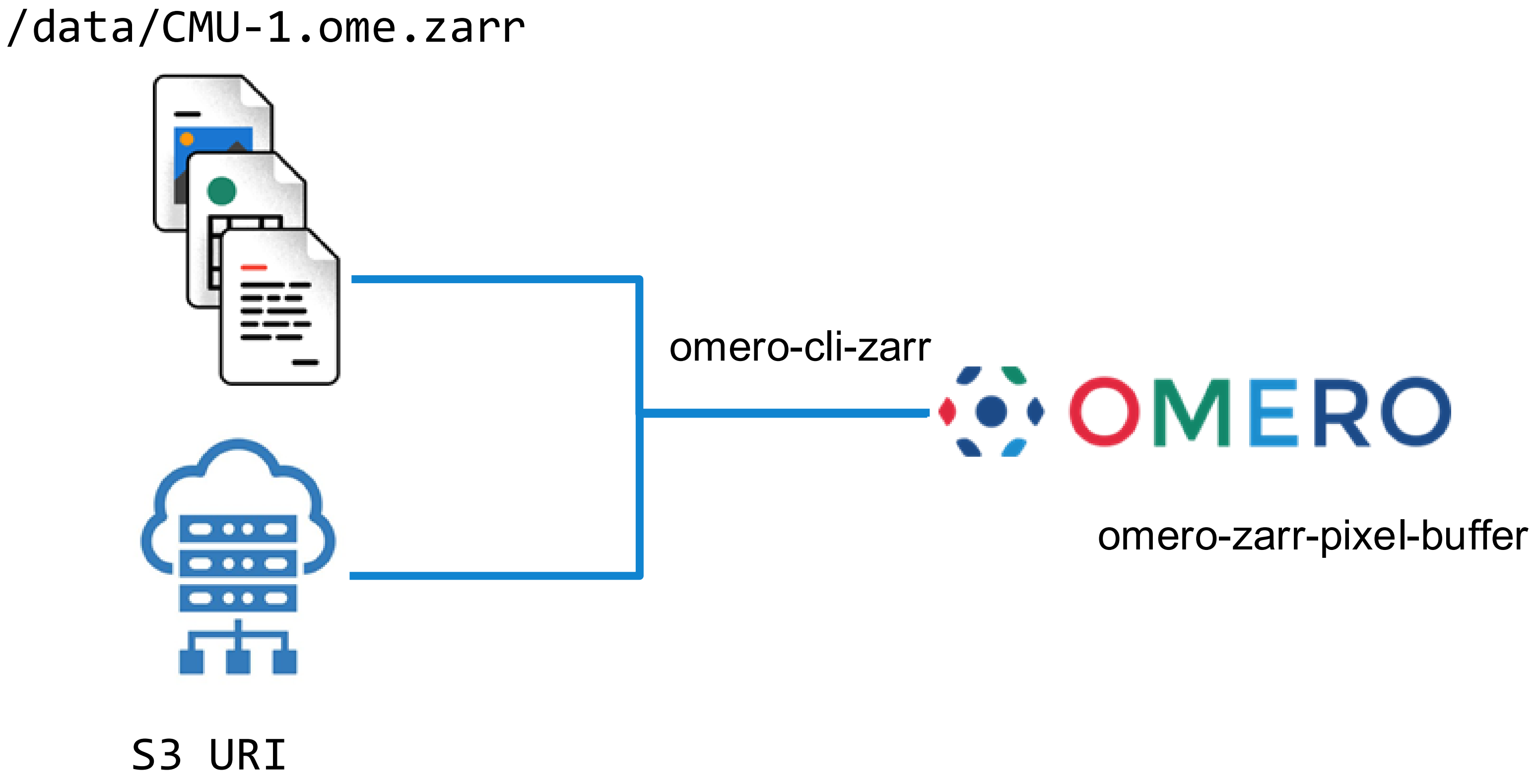
Zarr Viewers

Name	Link	Description
AGAVE		Desktop application for viewing multichannel volume data powered by your GPU
FIJI (MoBIE / BigDataViewer)		FIJI plug-in for exploring and sharing big multi-modal image and associated tabular data
FIJI (n5-ij)		FIJI plug-in for loading and saving image data to OME-Zarr and other formats supported by the N5 API
ITKWidgets		Python tool for interactively viewing images (ex. in Jupyter)
Microscopy Nodes		Blender add-on for visualizing high-dimensional microscopy data
napari		napari plug-in for viewing Zarr
Neuroglancer		A browser-based volume viewer
Viv (Avivator, Vizarr, Vitessce, ...)		A toolkit for interactive visualization of high-resolution, multiplexed bioimaging datasets. The viv toolkit is used by the Avivator, Vizarr and Vitessce image viewers, among others
Vol-E		A browser-based volume viewer
WEBKNOSSOS		An open-source tool for annotating and exploring large 3D image datasets

https://ngff.openmicroscopy.org/resources/tools/index.html#zarr-viewers

OMERO & OME-Zarr: "in-place" import



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Example: Access possibilities of images

image from idr0044, McDole et al.	Download	IDR API Access	OME-Zarr Access via S3
Load image subregion, e.g., single chunk or tile	No, only per file	Yes	Yes
Lazy loading	No	No	Yes. Use Dask collections: da.from_zarr(endpoint_url)
Easily analyze in parallel	No, depends on file format which may require a translation library.	Difficult due to the transfer protocol used (zeroc-ice)	Yes. Use Dask schedulers: dask.delayed(analyze)(t, c, z)

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Exercises

- Read Data in IDR using Python API
  https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_Image_From_IDR.ipynb

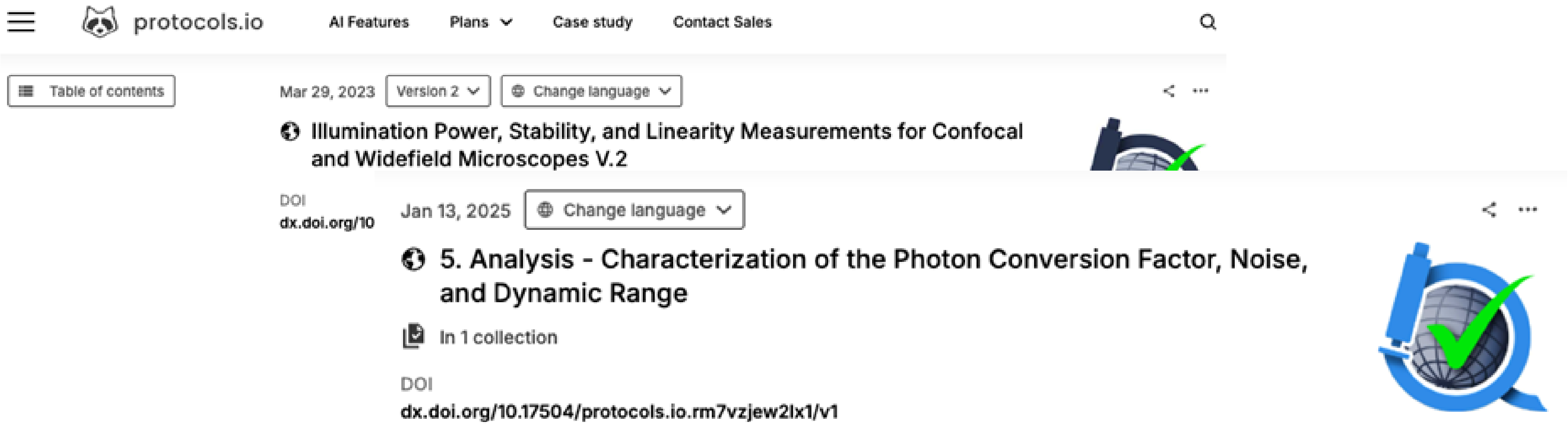
- Explore the multi-resolution
  https://github.com/BioImagingUK/Training/blob/main/notebooks/Multi_Resolution.ipynb

- Read OME-Zarr data using Python
  https://github.com/BioImagingUK/Training/blob/main/notebooks/Read_OME_Zarr_Image.ipynb

Data preservation: what to preserve?

Quality Control checks

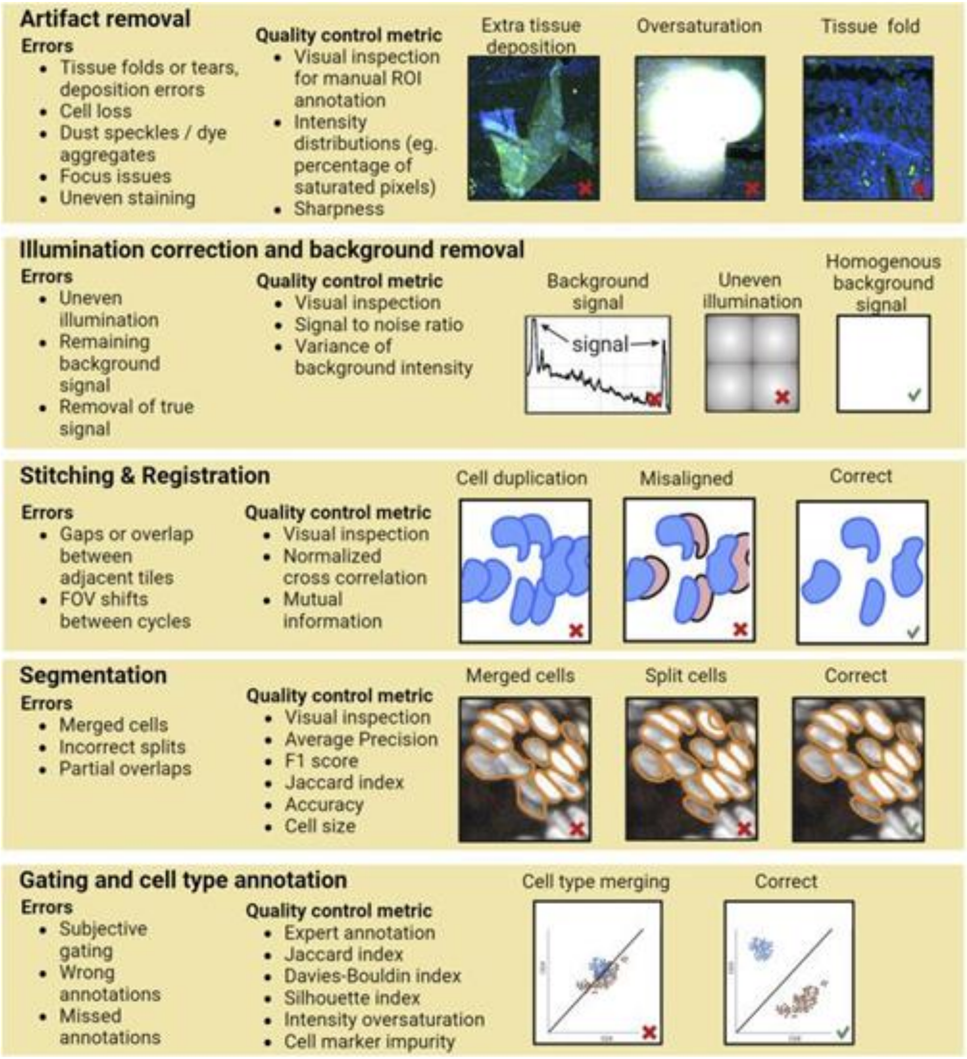
- QUAREP-LiMi: Quality Assessment and Reproducibility for Instruments & Images in Light Microscopy
  https://quarep.org/
  - **comprehensive set of community-agreed guidelines, protocols, automation procedures and other resources to improve quality control, quality assurance, and instrument/method calibration.**



Quality Control checks

- Corrupt data (machine read)
- Intensity distribution, e.g. percentage of saturated pixels, percentage of blank pixels
- Focus
- md5 checks after transfers or use transfer protocols that support data integrity

➔ Automated checks at the instrument or primary storage.



Vierdag, W. A. M., & Saka, S. K. (2024). A perspective on FAIR quality control in multiplexed imaging data processing. Frontiers in bioinformatics, 4, 1336257. https://doi.org/10.3389/fbinf.2024.1336257

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What to preserve

- Preserve raw data (keep it untouched)
- Can preserve processed data (or raw data plus processing and analysis workflows)
- Analysis workflows - workflowhub.eu

Data Retention Policies and Data Embargoes

Data retention policies vary depending on a funder or repository perspective.

Funder (UKRI):

- Different policies depending on the research council.
- BBSRc and MRC say primary data should be retained for 10 years after the project ended.
- BBSRc say data should be shared as soon as possible after publication, and preferably no more than 3 years after the data was generated
- MRC also should be retained for at least 20 years when it's population health and clinical data
- EPSRC also mention 10 years, but metadata should be published after 12 months of the data being generated

Public Image Repository:

- Data can be embargoed for a period after submission.

Licensing and Data Retention

Licenses used by the archives

Creative Commons (CC)

https://creativecommons.org/share-your-work/cclicenses/

CC 0: You can reuse the data with no restrictions

CC BY: You can reuse the data with no restrictions other than give proper credit to the creator

There are other more restrictive licenses

Open-source Software Licenses

Do what you want, as long as you include attribution/copyright notice

Examples:

MIT: Very short, minimal requirements. Just include the license and attribution.

Apache 2.0: Like MIT, but also provides an explicit **patent grant**

BSD (2-Clause, 3-Clause): Similar to MIT, with slight differences in attribution requirements.

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What do public repositories offer

PLOS BIOLOGY

OPEN ACCESS

COMMUNITY PAGE

The Open Knowledge Foundation: Open Data Means Better Science

Jennifer C. Molloy

Published: December 6, 2011 • https://doi.org/10.1371/journal.pbio.1001195

Article | Authors | Metrics | Comments | Media Coverage

Patterns of database citation in articles and patents indicate long-term scientific and industry value of biological data resources

David Bousfield,[1,2] Johanna McEntyre,[3] Sameer Velankar,[3] George Papadatos,[3] Alex Bateman,[3] Guy Cochrane,[3] Jee-Hyub Kim,[3] Florian Graef,[3] Vid Vartak,[3] Blaise Alako,[3] and Niklas Blomberg[a,1]

Author information ▸ Article notes ▸ Copyright and License information ▸

Increased visibility, citations and collaboration

> PLoS One. 2007 Mar 21;2(3):e308. doi: 10.1371/journal.pone.0000308.

Sharing detailed research data is associated with increased citation rate

Heather A Piwowar [1], Roger S Day, Douglas B Fridsma

Published: 12 March 2015

The open access advantage considering citation, article usage and social media attention

Xianwen Wang, Chen Liu, Wenli Mao & Zhichao Fang

Scientometrics **103**, 555–564 (2015) | Cite this article

5672 Accesses | **119** Citations | **368** Altmetric | Metrics

nature > career feature > article

CAREER FEATURE | 13 May 2019

Data sharing and how it can benefit your scientific career

Open science can lead to greater collaboration, increased confidence in findings and goodwill between researchers.

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Funder and Journal requirement

What to include in your application

Contents

- Introduction
- Case for support
- Data management plan
- Diagrammatic work plan
- Justification of resources

- methods for data sharing – planned mechanisms for making these data available, for example through deposition in existing public databases or on request, including access mechanisms

UKRI Biotechnology and Biological Sciences Research Council

Data availability

Please provide a Data Availability statement in the Methods section under "Data Availability"; detailed guidance can be found in our data availability and data citations policy. Certain data types must be deposited in an appropriate public structured data depository (details are available here), and the accession number(s) provided in the manuscript. Full access is required at publication. Should full access to data be required for peer review, authors must provide it.

Editorial | Published: 17 June 2025

Light microscopy reporting for reproducibility

Nature Cell Biology 27, 877 (2025) | Cite this article

We announce a cross-journal pilot at Nature Portfolio journals with a goal of implementing standardized light and fluorescence microscopy reporting to improve methodological description and aid in reproducibility efforts.

In the interest of fostering reuse of light and fluorescence microscopy data, we also encourage authors to deposit their microscopy datasets in public repositories, as is routinely expected in other fields, such as genomics and proteomics, which has expanded both the community and the value of data. For deposition, we recommend the Image Data Resource, the BioStudies database, the BioImage Archive or figshare, which is integrated in our submission system.

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Long term data management is hard

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Lots of data!

When you're looking for a dataset, use BIA, IDR and EMPIAR first, instead of google or sometimes even papers.

BioImage Archive: https://www.ebi.ac.uk/bioimage-archive/
https://beta.bioimagearchive.org/

IDR: https://idr.openmicroscopy.org/

EMPIAR: https://www.ebi.ac.uk/empiar/

Exercise (5 minutes)

Find data on your research topic

- https://beta.bioimagearchive.org/

- https://idr.openmicroscopy.org/

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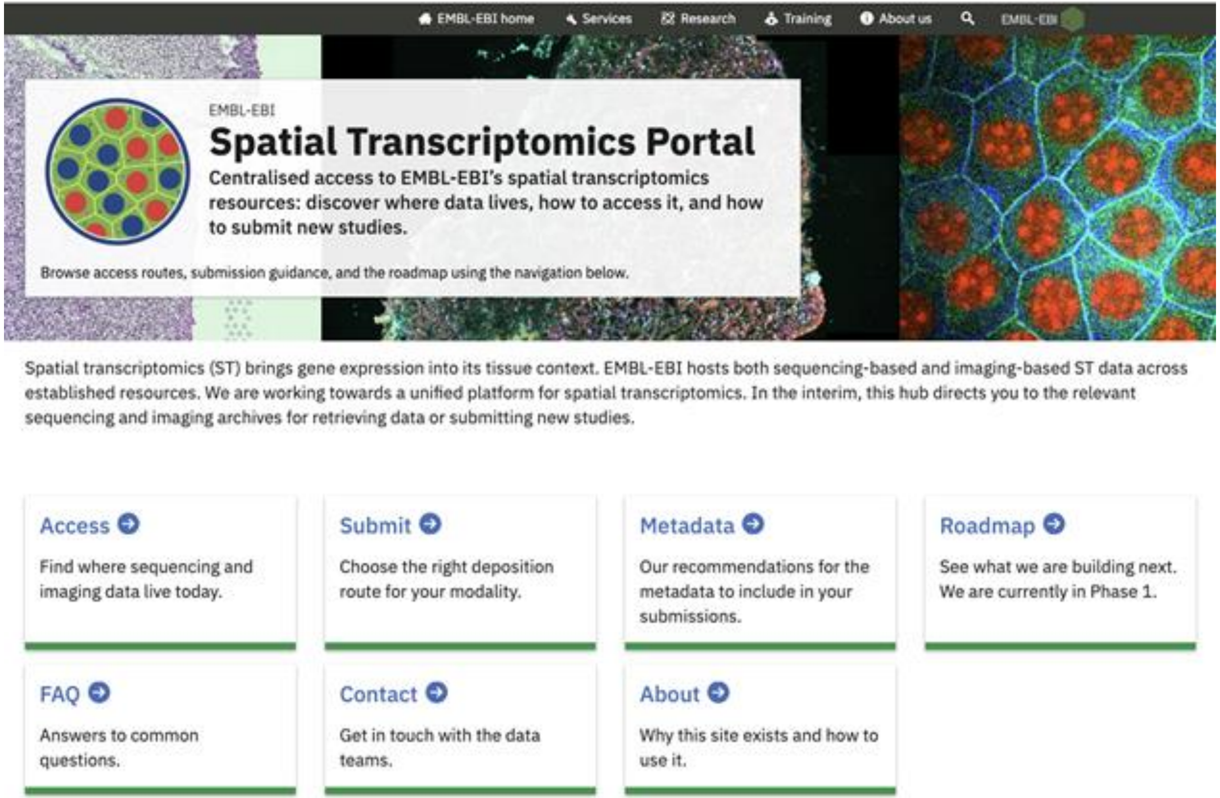
Spatial Transcriptomics Portal



https://www.ebi.ac.uk/spatial-transcriptomics/

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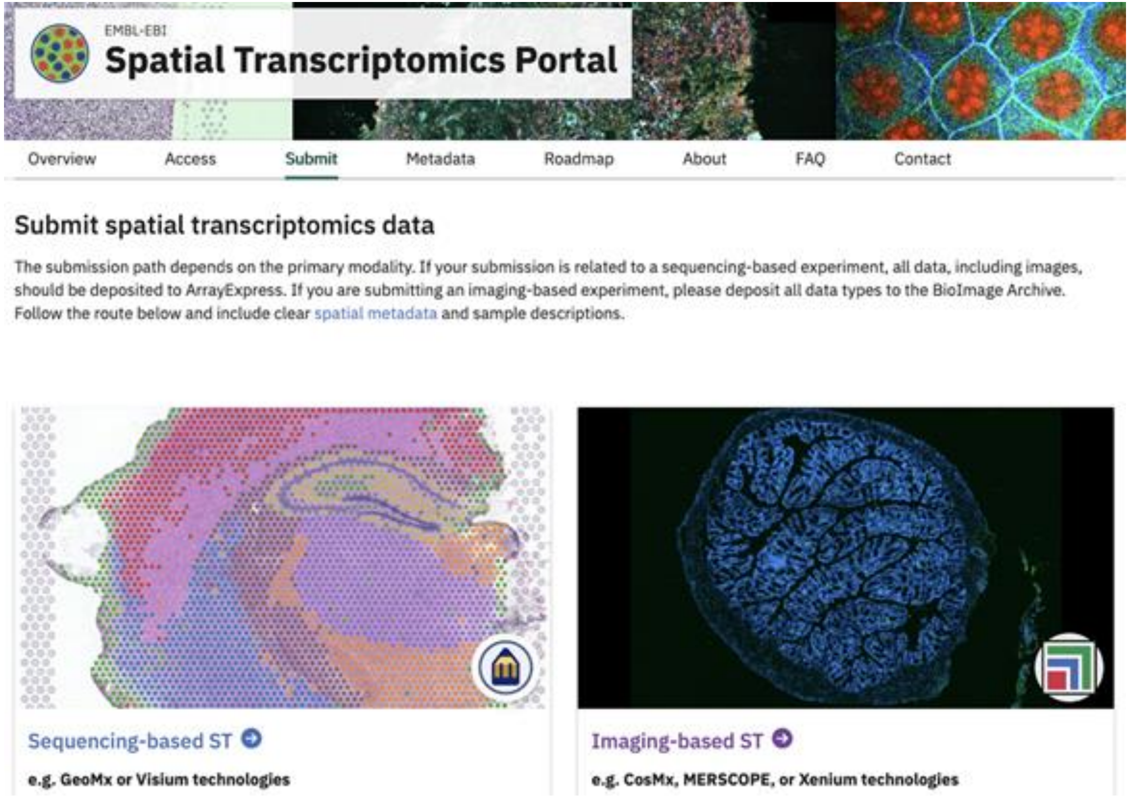
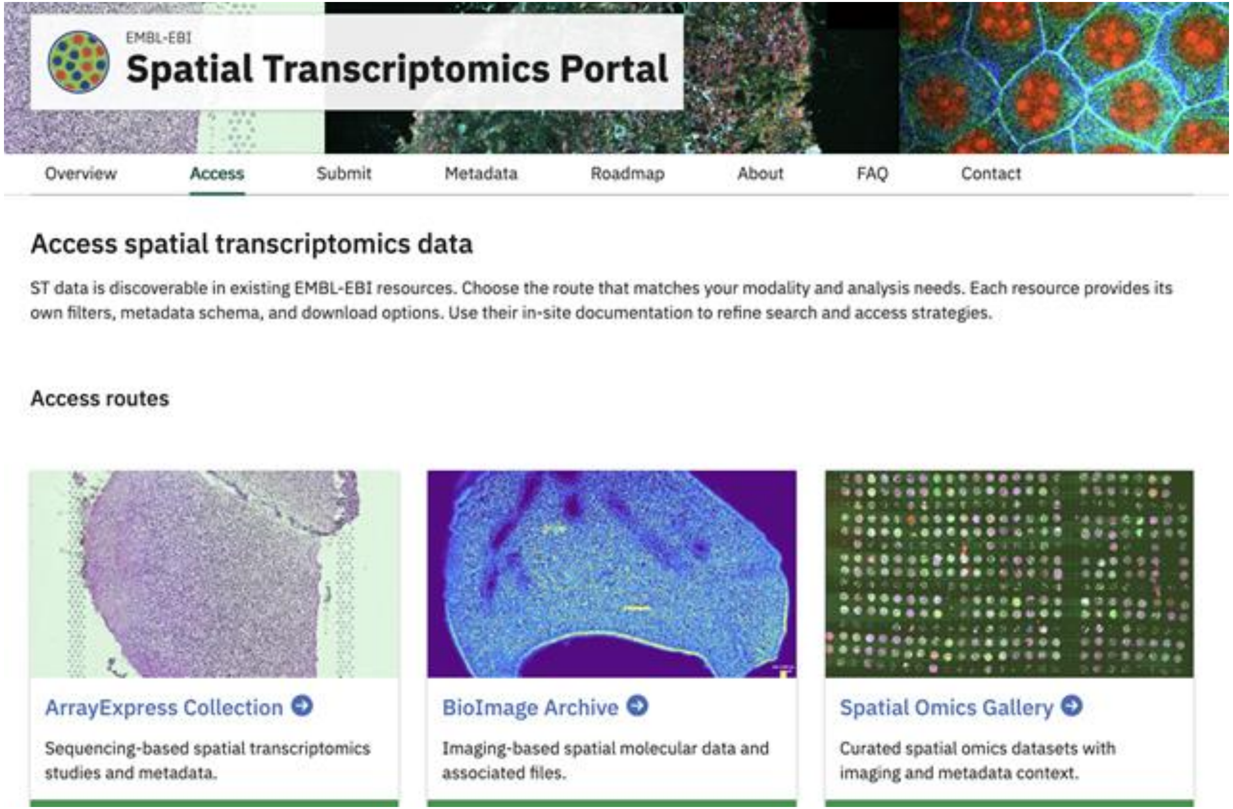
Spatial Transcriptomics Portal



Spatial Transcriptomics Portal



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Spatial Transcriptomics Portal

Metadata

While guidance on how to submit through Annotare or the BioImage Archive is available in the respective documentation pages, we are working to establish a shared metadata standard for spatial transcriptomics experiments.

Below are our recommendations for the metadata to include in your submissions in order to facilitate interpretation and reuse of your data.

Sample

Specimen

Experimental

Analytical

File types imaging based

File types sequencing based

Sample

Field	Definition	Example	Imaging-based ST	Sequencing-based ST
Sample ID	A unique identifier assigned to a biological specimen used in the imaging experiment	liver_01	required	required
Slide ID	A unique identifier for the glass slide or cartridge on which the tissue section was mounted and processed	BSMC_01	required	required
Position on slide	Describe where the capture area is located on the slide. This is required if the slide contains multiple capture areas, for example for visium	Area A1; number 1	optional	optional
Sample preservation method	The method used to preserve the tissue prior to processing (e.g., fresh frozen, FFPE)	fresh frozen	required	required
Organism name	The scientific name of the source organism (e.g., Homo sapiens, Mus musculus)	Homo sapiens (NCBITaxon:9606)	required	required
Time from collection to preservation	Elapsed time between tissue excision/collection and preservation, reported in hours	1h	optional	optional
Section	The thickness of the tissue section	5 µm	required	required

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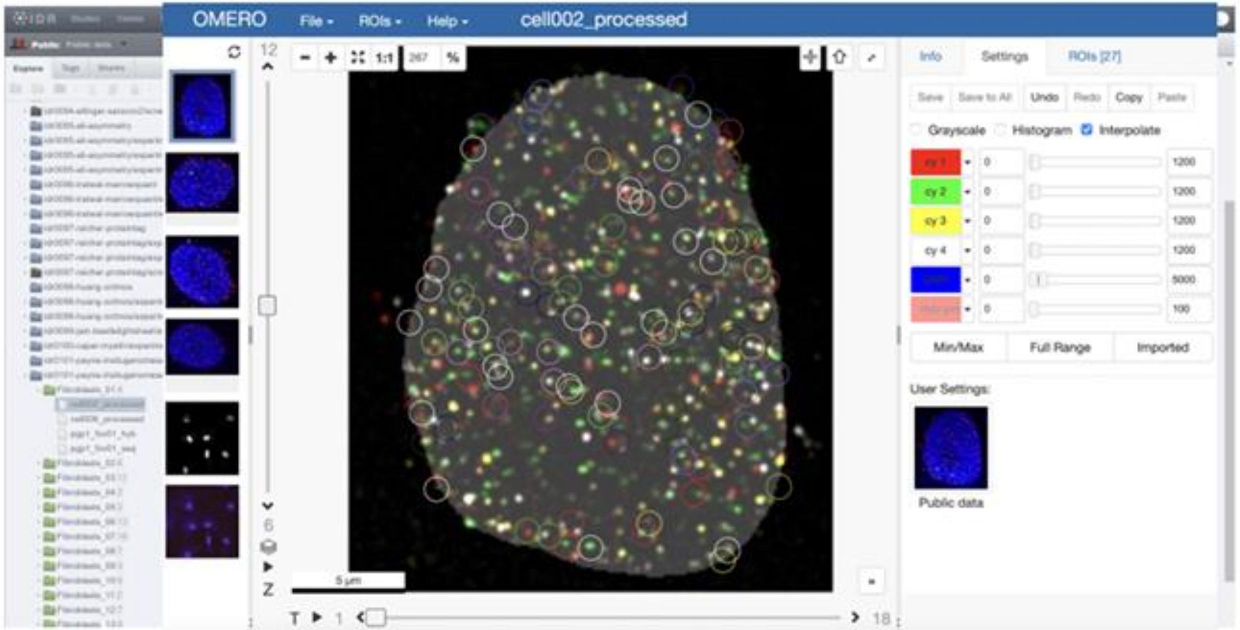
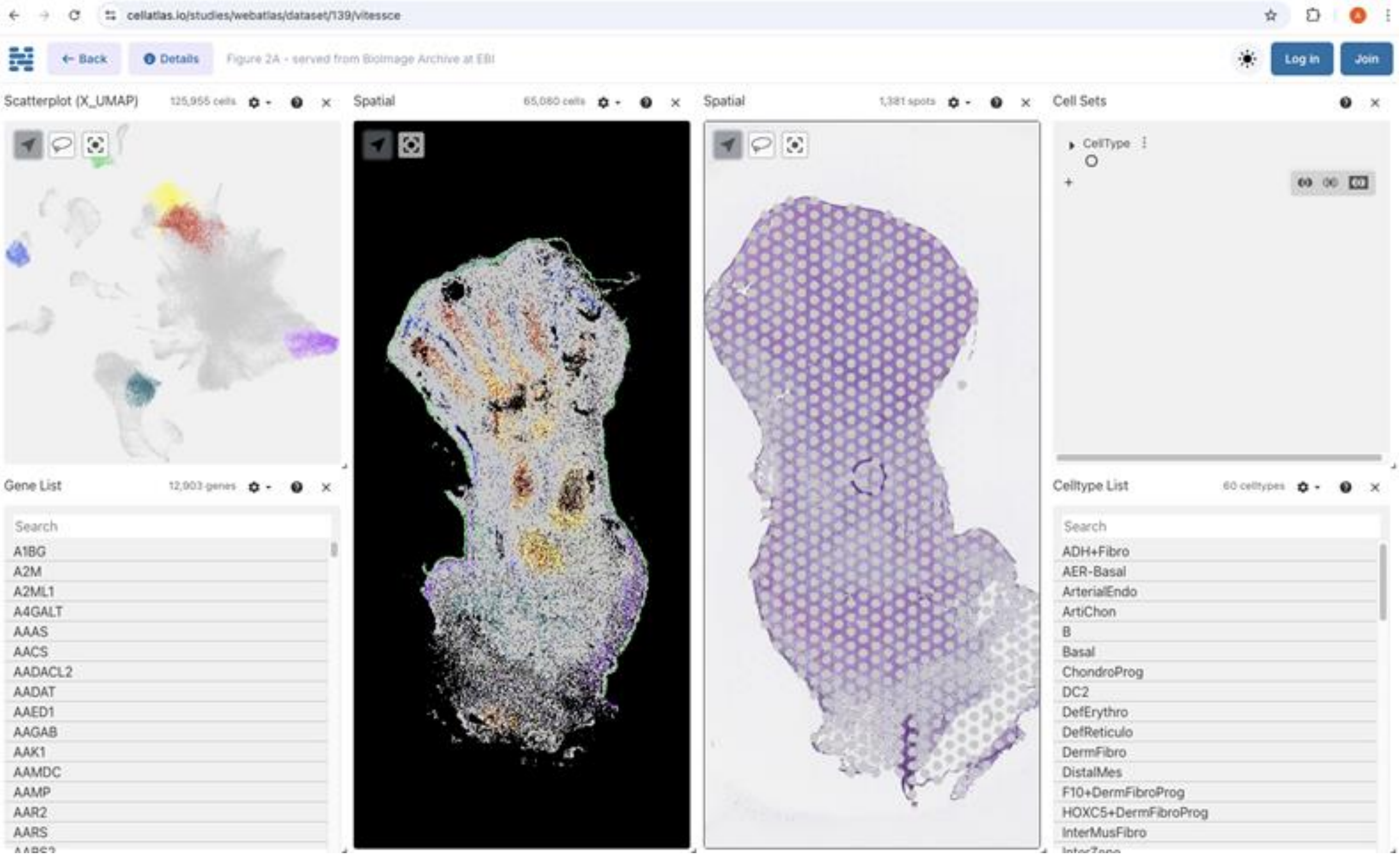
In Situ Sequencing - Linking Multimodal Metadata



idr0101, Payne et al *Science* DOI:10.17867/10000169

Connect Data stored in different resources using the Nucleome Browser
https://www.nature.com/articles/s41592-022-01559-3

WebAtlas integrated scRNA-seq, ISS and Visium dataset of the human lower limb



Li, T., Horsfall, D., Basurto-Lozada, D. *et al.* WebAtlas pipeline for integrated single-cell and spatial transcriptomic data. *Nat Methods* 22, 3–5 (2025). https://doi.org/10.1038/s41592-024-02371-x S-BIAD822, S-BIAD887

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Re-use cases

Unsupervised and Source-Free Ranking of Biomedical Segmentation Models

Talks et al (2024). https://arxiv.org/pdf/2503.00450

S-BIAD895


ZeroCostDL4Mic - Stardist example training and test dataset

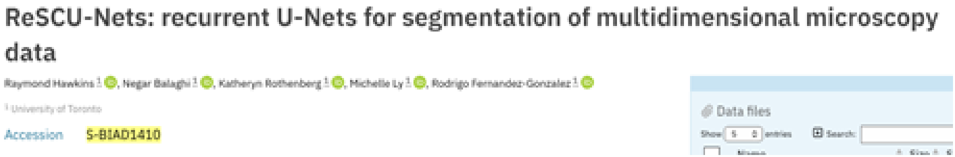
Johanna Jukkala [1], Craig Russell [2]

[1, 2] European Molecular Biology Laboratory, European Bioinformatics Institute

Accession S-BIAD895

https://www.ebi.ac.uk/biostudies/BioImages/studies/S-BIAD895?query=s-biad895

S-BIAD1410


ReSCU-Nets: recurrent U-Nets for segmentation of multidimensional microscopy data

Raymond Hawkins [1], Negar Balaghi [1], Katheryn Rothenberg [1], Michelle Ly [1], Rodrigo Fernandez-Gonzalez [1]

[1] University of Toronto

Accession S-BIAD1410

Data files

https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD1410?query=s-biad1410

Used to train instance segmentation datasets

S-BIAD634


An annotated fluorescence image dataset for training nuclear segmentation methods

Sabine Taschner-Mandl [1], Inge M. Ambros [1], Peter F. Ambros [1], Klaus Beiske [2], Allan Hanbury [3], Wolfgang Doerr [4], Tamara Weiss [1], Maria Berneder [1], Magdalena Ambros [1], Eva Bozsaky [1], + 2 more

[1] Children's Cancer Research Institute [2] Oslo University Hospital [3] TU Wien [4] TRAB-Applied and Translational Radiobiology, Department of Radiation Oncology, Medical University of Vienna [5] European Bioinformatics Institute (EMBL-EBI)

Accession S-BIAD634

Data files

https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD634

Used as an instance target dataset

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Re-use cases

Unsupervised and Source-Free Ranking of Biomedical Segmentation Models

Talks et al (2024). https://arxiv.org/pdf/2503.00450

S-BIAD895

ZeroCostDL4Mic - Stardist

Johanna Jukkala [1], Craig Russell [2]

[1, 2] European Molecular Biology Laboratory, European Bioinformatics

Accession S-BIAD895

https://www.ebi.ac.uk/biostudies

S-BIAD1410

ReSCU-Nets: recurrent U-Nets for s
data

Raymond Hawkins [1], Negar Balaghi [1], Katheryn Rothenberg [1], Michelle Ly [1], Rodrigo Fernandez-Gonzalez [1]

[1] University of Toronto

Accession S-BIAD1410

Data files

Show 5 entries Search:

https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD1410?query=s-biad1410

to train instance
entation datasets

Take home message

- The **original authors did not perform the reuse**
- It illustrates the **diverse reuse potential** of annotated biomedical image datasets

S-BIAD634

An annotated fluorescence image dataset for training nuclear segmentation methods

Sabine Taschner-Mandl [1], Inge M. Ambros [1], Peter F. Ambros [1], Klaus Beiske [2], Allan Hanbury [3], Wolfgang Doerr [4], Tamara Weiss [5], Maria Berneder [1], Magdalena Ambros [1], Eva Bozsaky [1], + 2 more

[1] Children's Cancer Research Institute [2] Oslo University Hospital [3] TU Wien [4] TRAB-Applied and Translational Radiobiology, Department of Radiation Oncology, Medical University of Vienna [5] European Bioinformatics Institute (EMBL-EBI)

Accession S-BIAD634

Data files

Show 5 entries Search:

Name Size

https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD634

Used as an instance target dataset

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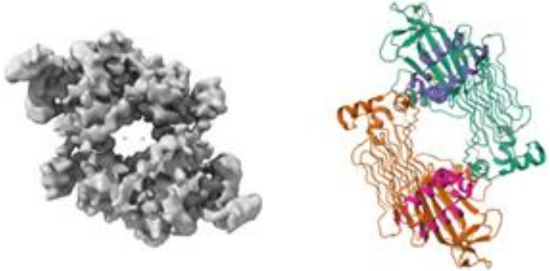
Re-use cases

Reanalysis of cryo-EM data reveals ALK-cytokine assemblies with both 2:1 and 2:2 stoichiometries

Felix et al., *PLoS Biology* (2025) https://pubmed.ncbi.nlm.nih.gov/40208865/

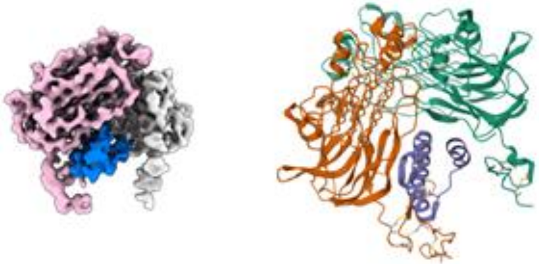
Independent re-processing of raw EMPIAR data

- Used the **uncleaned particle stack** from EMPIAR-10930
- Avoided particle binning used in the original workflow



The re-analysis revealed that:

- ~53% of particles in the original dataset correspond to a **previously unresolved 2:1 ALK:ALKAL2 stoichiometry**
- Preferred particle orientations had obscured this assembly in the original analysis
- A **new 3.2 Å cryo-EM structure** of the **2:1 ALKTG-EGFL-ALKAL2 complex** was reconstructed



Re-use cases

Reanalysis of cryo-EM data reveals ALK-cytokine assemblies with both 2:1 and 2:2 stoichiometries

Felix et al., *PLoS Biolog*

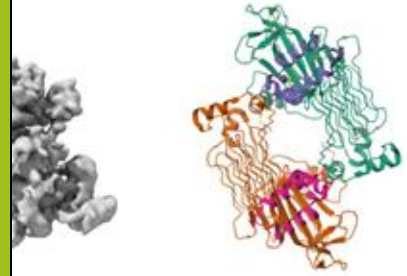
Independent re-proc

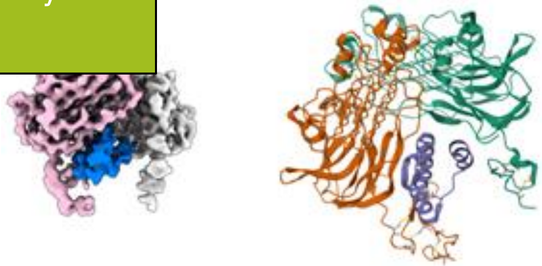
- Used the **uncl**
- Avoided parti

The re-analysis reveal

Take home message

- The **original authors did not perform the reuse**
- The **same raw dataset** supported **multiple, biologically distinct interpretations**
- Methodological advances made *years later* unlocked new structural information
- Without public deposition of raw data, this discovery would not have been possible





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- Preferred particle orientations had obscured this assembly in the original analysis
- A **new 3.2 Å cryo-EM structure** of the **2:1 ALKTG-EGFL-ALKAL2 complex** was reconstructed

Submission to the public image repositories

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BioImage Archive

A free, publicly available resource for life sciences images from any imaging modality, at any scale

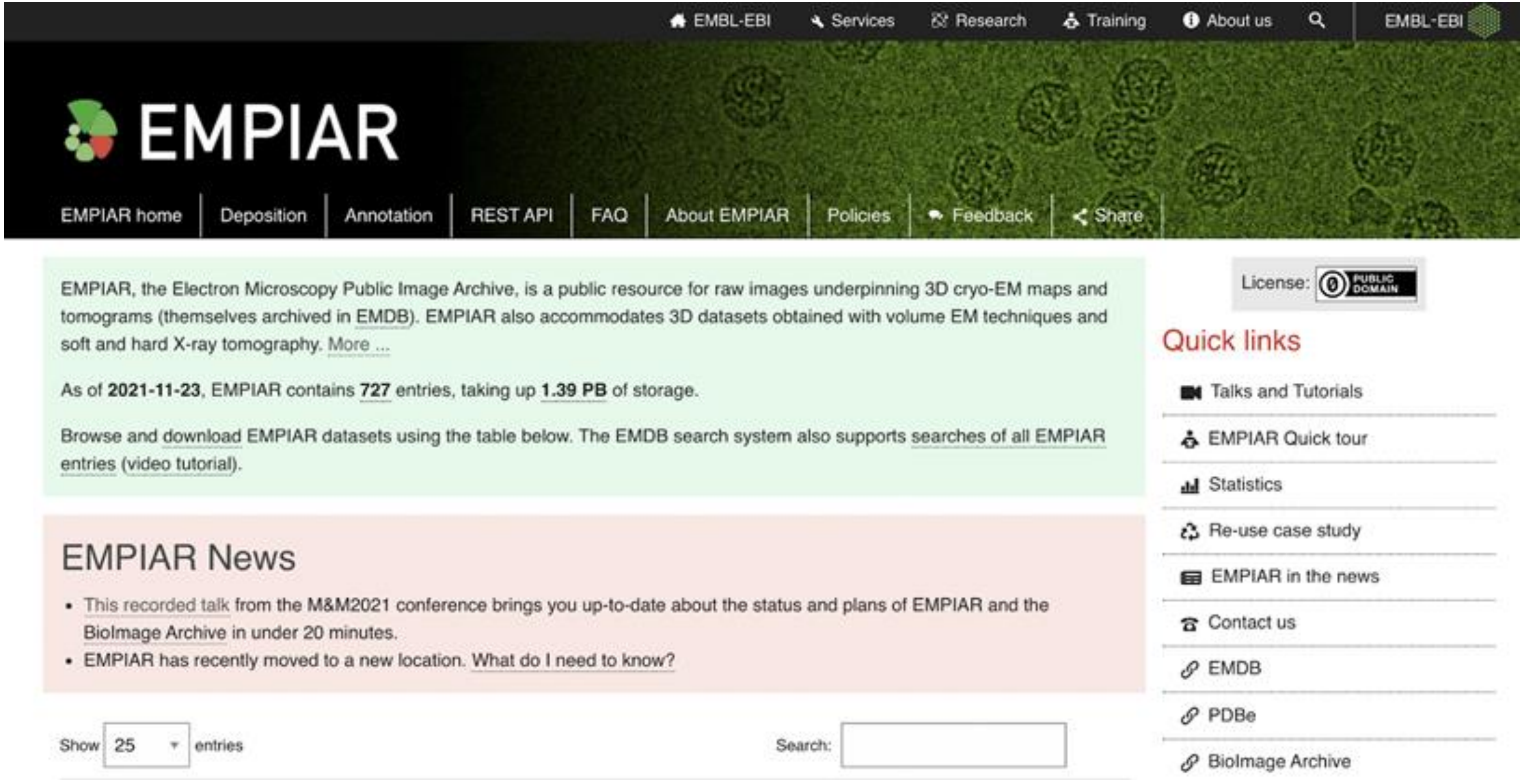
Data are either **associated with a peer-reviewed publication, or of value beyond** a single experiment

Out of scope:

- Datasets linked to an identifiable individual
- Datasets with a primary use in medical treatment (e.g. diagnostic imaging)

https://www.ebi.ac.uk/bioimage-archive New website (beta version): https://beta.bioimagearchive.org/

EMPIAR - Electron Microscopy Public Image Archive



elixir Deposition Database

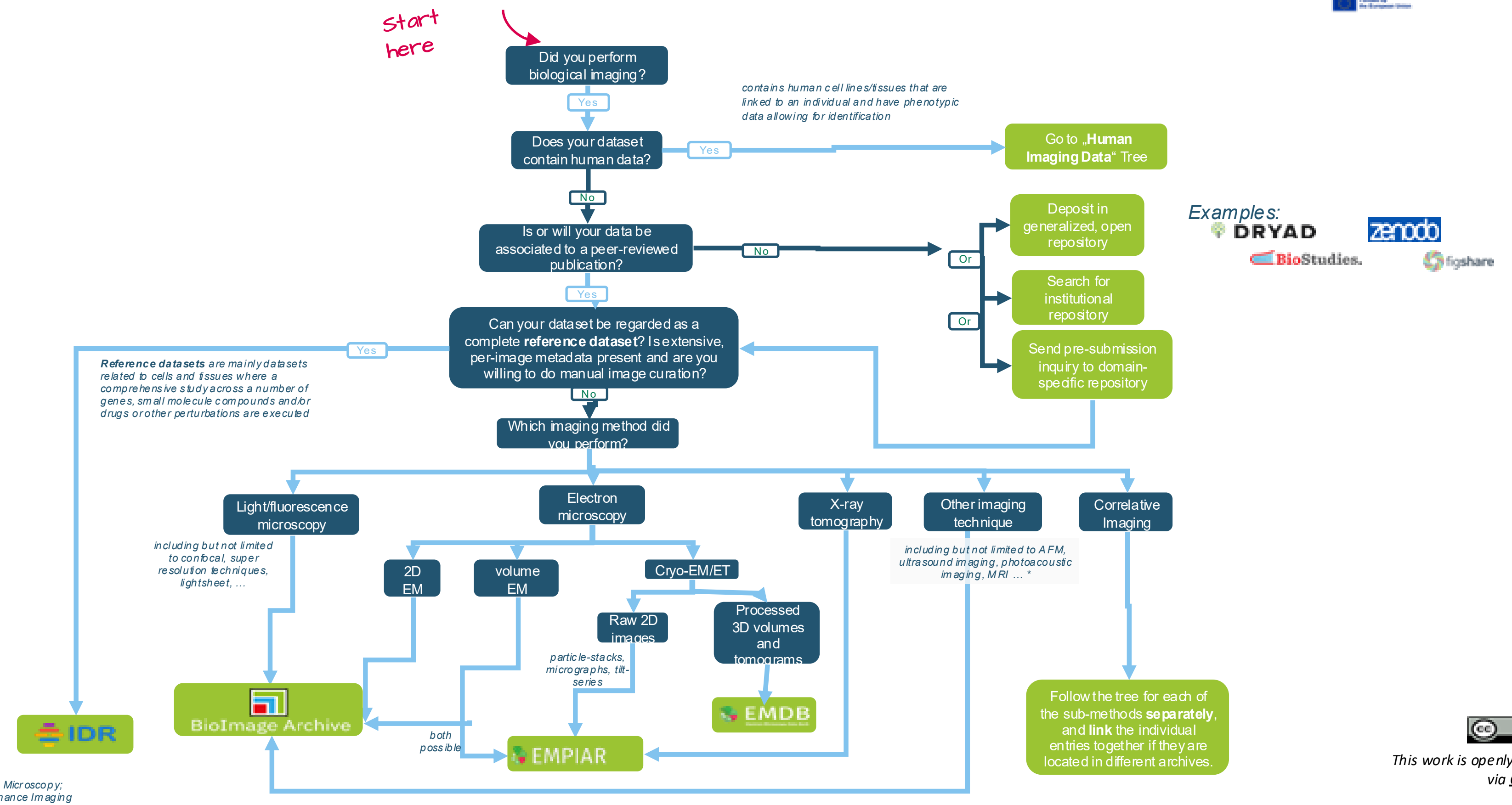
EMBL-EBI

https://www.ebi.ac.uk/empiar/

EURO-BIOIMAGING Guidance

Where do I deposit my open bioimaging data?

Made possible by EVOLVE



* **AFM:** Atomic Force Microscopy; **MRI:** Magnetic Resonance Imaging

Kemmer, I., Romdhane, F., & Euro-BioImaging ERIC. (2026). Image Repository Decision Tree - Where do I deposit my imaging data. Zenodo. https://doi.org/10.5281/zenodo.18378649

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Submitting a study to the BIA

1. Create an account in Biostudies (https://www.ebi.ac.uk/biostudies/submissions/signin)
2. Upload your data: web interface, FTP, Aspera, (Globus)
3. Create a draft and fill all the metadata fields that are mandatory or necessary for your submission
4. See how best to organize your submission:
   a. One study component or more?
   b. Do you have annotations?
   c. Spatial-transcriptomics
1. You will need one file list per study or annotations component
2. Submit when you are ready
3. Release date is your choice

More detailed information on the BIA website (with a short 2 min video) and the BIA online tutorial

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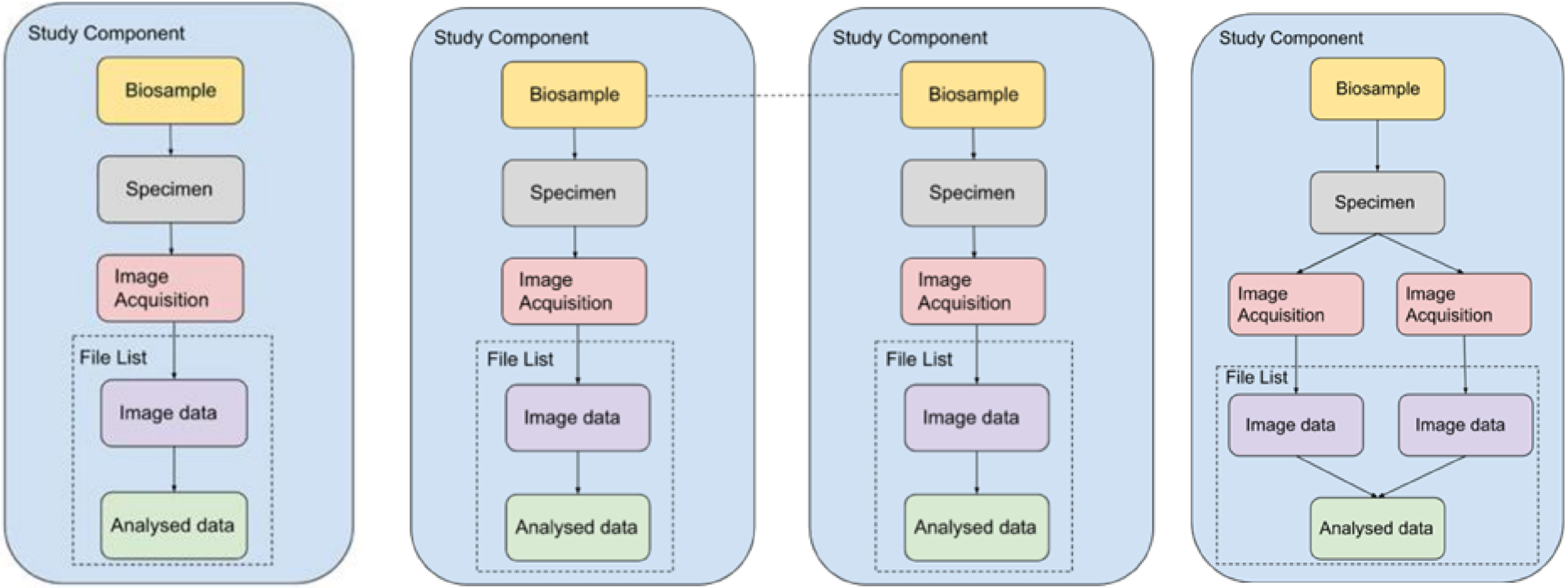
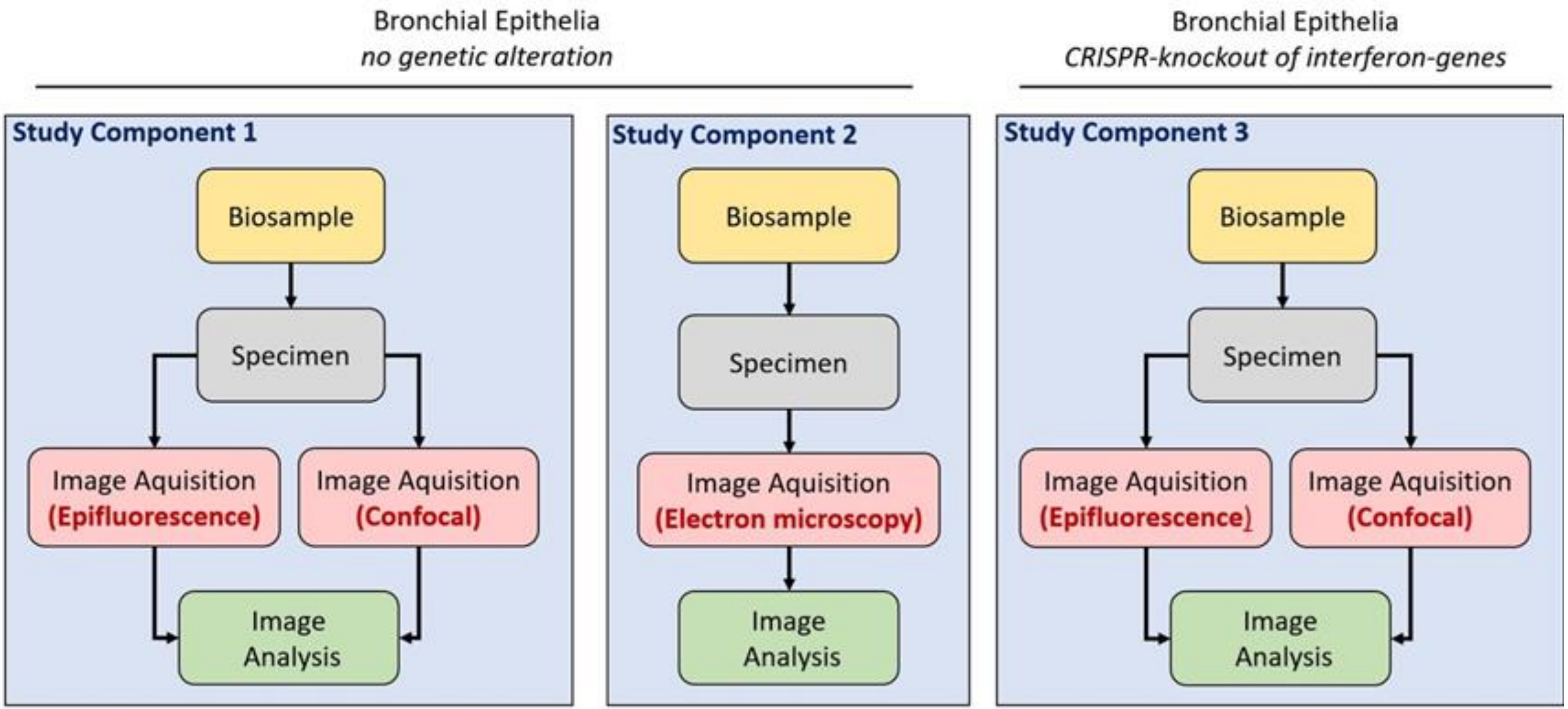
REMBI - Study Component Examples



REMBI - Real Life Examples of Study Components



Isabel Kemmer

https://github.com/Euro-BioImaging/FAIR101-Workshop-on-data-deposition-into-BioImage-Archive/blob/main/README.md

REMBI - Real Life Examples of Study Components

https://www.ebi.ac.uk/biostudies/bioimages/studies/S-BIAD677

Transfer methods available at the BIA

Submission size	Recommended upload method
If less than 50GB total size and Less than 20GB per individual file	Submission tool upload
If up to 1TB total size	FTP
If larger than 1TB	

Please note that we cannot accept single files larger than 2TB.

We also offer Globus by request

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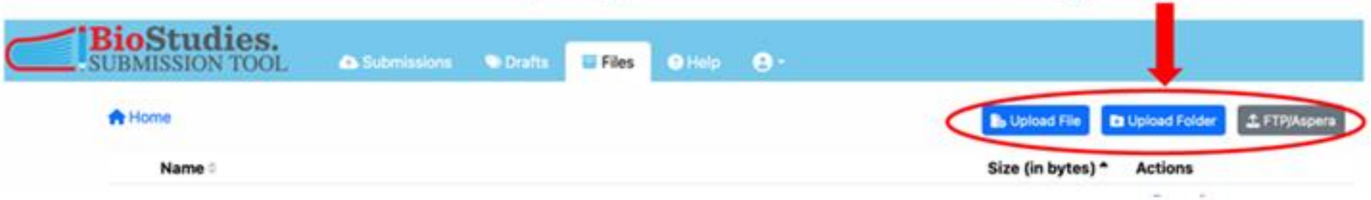
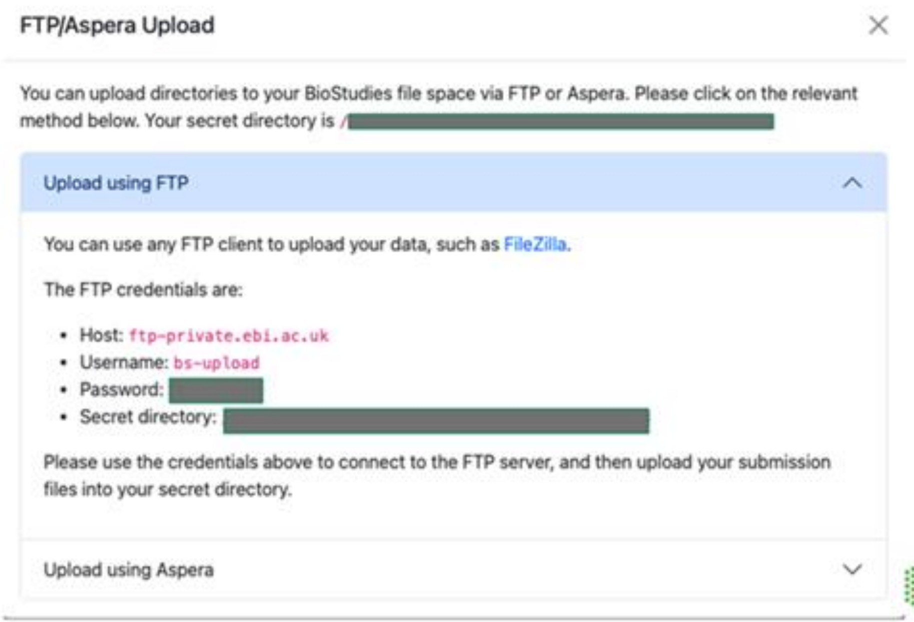
Submission Tool

Upload your data via the web interface or get the credentials for FTP/Aspera



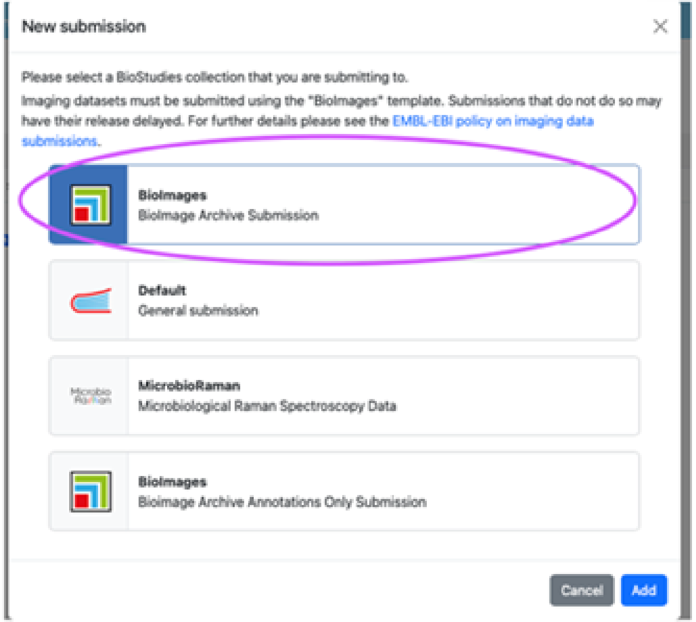


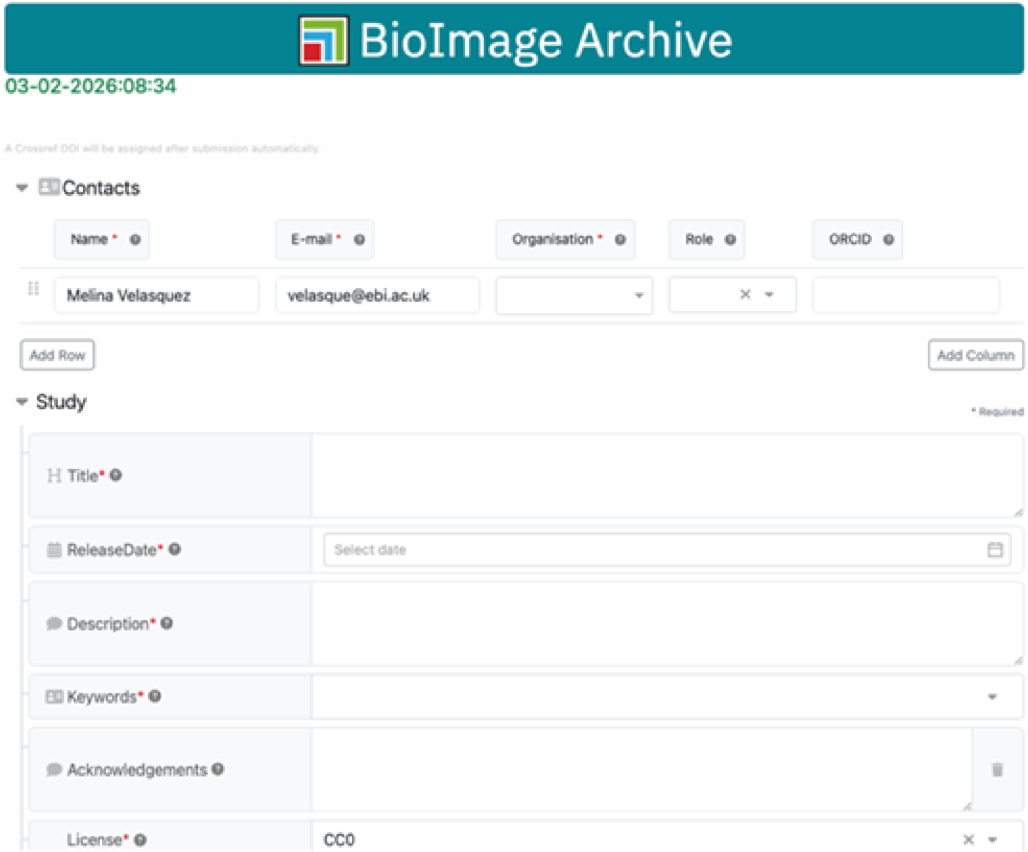
Submission Tool

Click here to create a draft

Submission Tool - Biostudies

Submit





Filelist

- List of all the files in your study component(s) and/or annotations component
- File lists are tabular data, either in tsv or Excel (.xlsx) format.
- The first column of the header has to be the word “Files”.
- If submitting annotations, the annotations file list should also include a “source_image” column.
- It has to have one file per line.
- Do not leave blank lines.
- Filenames are case-sensitive.
- File path separator must be forward slash “/”.
- You can assign metadata to an image level

Let’s look at the file list of [S-BIAD29]

Exercise

BIA live submission

Mock dataset

https://bit.ly/3VeOYJ3

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Exercise

BIA live submission

1. Create an account https://wwwdev.ebi.ac.uk/biostudies/submissions/#/signup
2. Prepare the file list
3. Upload the demo data https://bit.ly/3VeOYJ3
4. Create a draft and fill all the metadata fields that are mandatory or necessary for your submission
5. Select the file list
6. Submit

Making changes after submission

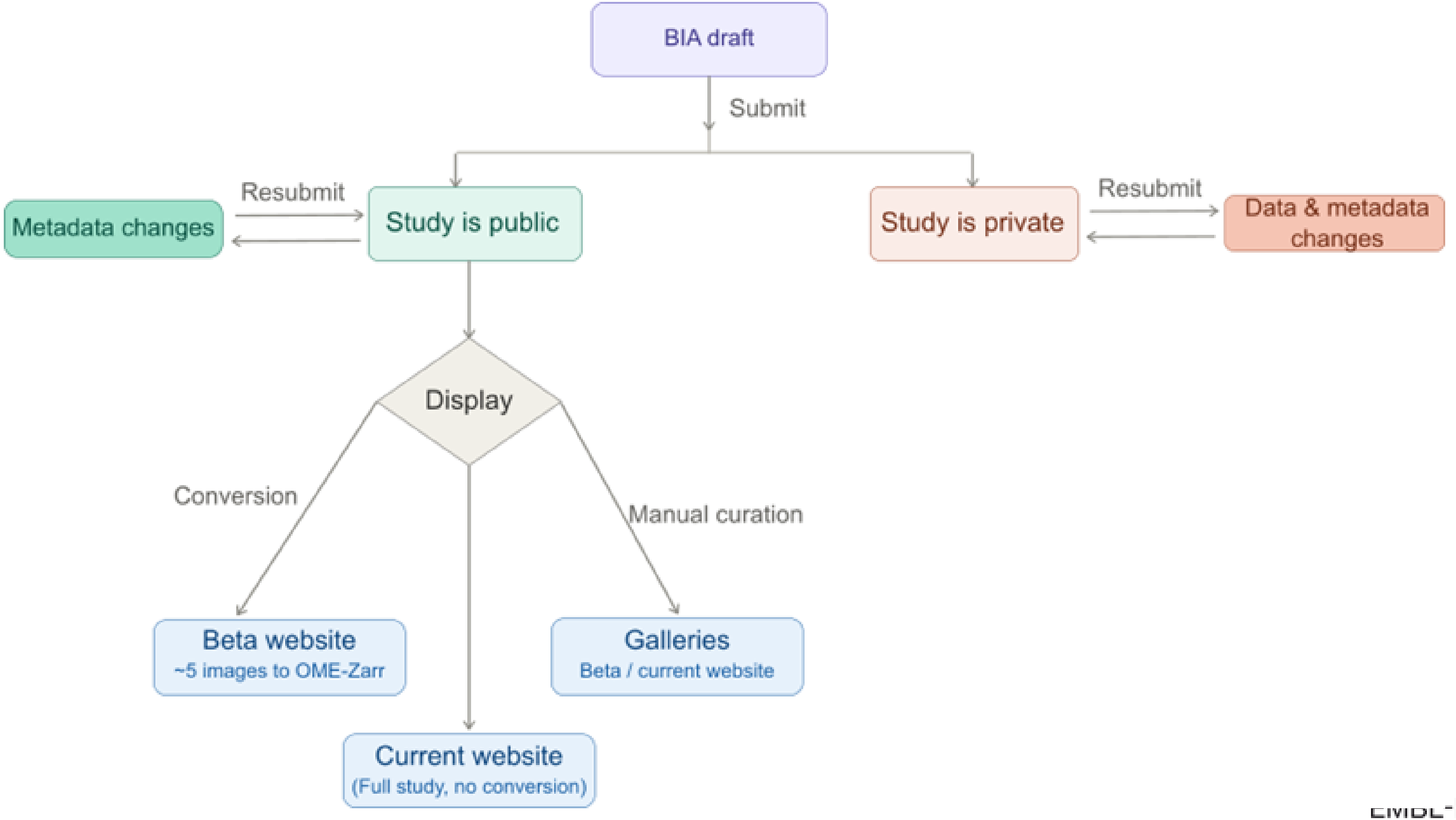
If the submission is still private, you can:

- Change metadata and data
- Add more data
- Change release date
- Delete the whole submission

If the submission is public:

- Change metadata
- Add more data
- You can’t modify/delete data

Submission pipeline overview



Take home message

- ➔ **Plan and include data management human and infrastructure** needs in your plan and funding application
- ➔ Start **collecting metadata** early on, at the instrument if possible and systematically **throughout the data lifecycle**
- ➔ Metadata standardisation: Use **REMBI** and ontologies, controlled vocabularies
- ➔ Data standardisation: **Bio-Formats and OME-NGFF (OME-Zarr)**
- ➔ If you’re publishing, submit data to **BIA, (IDR) or EMPIAR** (if not patient identifiable) or another FAIR public bioimage repository

Opportunities coming up

- Fellowship call mid-2026
- Pathfinder call late-2026

- DisCouRSE flexible fund 05/06/2026

- Fund call and Events https://casdar.ac.uk/funding/

CaSDaR
Careers and Skills for Data-driven Research

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Moving forward

- Github repo for training materials: https://github.com/BioImagingUK/Training
    - We will add this slide deck
- Community Slack channel for knowledge and information sharing
- Feedback for this training: https://forms.gle/ZBFYSTNDQt9G8Hvb7

Join slack #

Give feedback

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